Gene Regulatory Network-Guided Causal Generative Modeling of Single-Cell Transcriptomics

A thesis submitted to McGill University in partial fulfillment of the requirements for the degree of Master of Science.

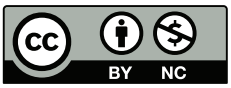
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No part of this thesis was created using generative artificial intelligence.

Abstract

Résumé

"You don't see yet, Genry, why we perfected and practice Foretelling?"

"No—"

"To exhibit the perfect uselessness of knowing the answer to the wrong question."

— Ursula K. Le Guin, *The Left Hand of Darkness* (1969)

To my parents, this thesis is dedicated to you for being my support through it all.

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List of Abbreviations

AE	Autoencoder
AEP	Affinity evaluation and prediction
AI	Artificial intelligence
ANN	Artificial neural network
AP	Average precision
API	Application programming interface
AP-MS	Affinity purification, mass spectroscopy
ASA	Accessible surface area
ASGD	Averaged stochastic gradient descent
ASL	Asymmetric loss
ATP	Adenosine triphosphate
AUPR	Area under the precision-recall curve
AWD-LSTM	ASGD Weight-Dropped LSTM
AUROC	Area under the receiver operating characteristic
BERT	Bidirectional encoder representations from transformers
BFDR	Bayesian false-discover rate
BiFC	Bimolecular fluorescence complementation
BioGRID	Biological general repository for interaction datasets
BioID-MS	Proximity-dependent biotin identification, mass spectrometry
BLAST	Basic local alignment search tool
BLOSUM	Blocks substitution matrix

BRET Bioluminescence resonance energy transfer

CD-HIT Cluster database at high identity with tolerance

CLI Command-line interface

CNN Convolutional neural networks

coFrac-MS, CF-MS Co-fractionation, mass spectrometry

CO-IP Co-immunopercipitation

CPU Central processing unit

DIP Database of interacting proteins

DNA Deoxyribonucleic acid

DNN Deep neural network

D-SCRIPT Deep sequence contact residue interaction prediction transfer

ESM Evolutionary scale model

FLOP Floating point operation

FRET Förster resonance energy transfer, or fluorescence resonance energy transfer

GELU Gaussian error linear unit

GO Gene ontology

GPT Generative pre-trained transformer

GPU Graphics processing unit

GRU Gated recurrent unit

HER2 Human epidermal growth factor receptor 2

HIPPIE Human integrated protein–protein interaction reference

HSSP Homology-derived secondary structure of proteins

HuRI Human reference interactome

INTREPPPID Incorporating triplet error for predicting PPIs using deep learning

IMEX International molecular exchange consortium

JSD Just sustainability design

KEGG Kyoto encyclopedia of genes and genomes

KISS Kinase substrate sensor

*k*NN	*k*-nearest neighbours
LLM	Large language model
LSTM	Long short-term memory
MaMTH	Mammalian-membrane two-hybrid
MAPPIT	Mammalian protein-protein interaction trap
MCC	Matthew's correlation coefficient
MIST	Molecular interaction search tool
MLM	Masked langauge model
MMseqs	Many-against-Many sequence searching
MOE	Mixture of experts
MP-BERT	MindSpore ProteinBERT
MSRP	Manufacturer's suggested retail price
MYTH	Membrane yeast two-hybrid
NCBI	National center for biotechnology information (USA)
NDCG	Normalized discounted cumulative gain
NHK	Nippon Hōsō Kyōkai
NLP	Natural language processing
NT-ASGD	Non-monotonically triggered averaged stochastic gradient descent
OMA	Orthologous matrix
OOD	Out of distribution
PAM	Point accepted mutation
PCA	Principle component analysis
PDB	Protein data bank
PEPPI	Pipeline for the extraction of predicted protein-protein interactions
PFC	Position-wise fully-connected
PIPE	Protein-protein interaction prediction engine
PIPR	Protein–Protein Interaction Prediction Based on Siamese Residual RCNN
pLLM, pLM	Protein (large) language model
PPI	Protein-protien interaction

Prose	Protein sequence embeddings
psi-blast	Position-specific iterated blast
rapppid	Regularized automatic prediction of protein–protein interactions
rcnn	Recurrent convolutional neural network
rna	Ribonucleic acid
rnn	Recurrent neural network
sae	Stacked autoencoder
saint	Significance analysis of interactome
sd	Standard deviation
sds-page	Sodium dodecyl sulfate–polyacrylamide gel electrophoresis
sgd	Stochastic gradient descent
SimCLR	Simple framework for contrastive learning of visual representations
skempi	Structural database of kinetics and energetics of mutant protein interactions
snc	Skill-normalized concordance
sota	Simple framework for contrastive learning of visual representations
sprint	Scoring protein interactions
ssl	Self-supervised learning
string	Search tool for recurring instances of neighbouring genes
svm	Support vector machine
swa	Stochastic weight averaging
tdp	Thermal design power
tpca	Thermal proximity co-aggregation
tuna	Transformer-based uncertainty aware model for ppi prediction
umap	Uniform manifold approximation and projection
usd	United States dollar
vram	Video random-access memory
y2h	Yeast two-hybrid
xl-ms	Cross-linking, mass spectrometry

Chapter 1

Introduction

Motivation

Of all the macromolecules one might find in the cells of living organisms, proteins are the class which are overwhelmingly responsible for the execution of the cell's functions, operating within and beyond its borders. Proteins are at the heart of vital cell functions like metabolism, DNA repair, cell polarity, and scaffolding.

To execute these myriad functions, proteins often participate in sometimes complex chains of chemical and physicochemical reactions with other proteins more broadly referred to as "protein-protein interactions" or PPIs. Understanding which proteins interact with other proteins is vital to our knowledge of *how* cells carry-out their many functions as well as the ways in which they fail in the case of disease.

In fact, cancers and other diseases which are caused by protein mutations often result in a change in protein binding affinity, altering how those proteins interact with others [1]. Infection and diseases caused by viruses like HIV and SARS-CoV-2 are also carried out by interactions between host and viral proteins [2, 3]. What's more, treatments for these diseases sometimes come in the form of so-called "biologic" therapies like small peptides, antibiotics, or mRNA vaccines, which in turn carry out their functions through the same fundamental processes as PPIs do.

While the study of PPIs has immeasurably shaped our understanding of biology, the methods by which we've traditionally come to characterize them require immense investments of time, labour, and capital. These methods, which include yeast two-hybrid (Y2H), affinity chromatography, and mass spectroscopy, are primarily conducted at the bench with the aid of specialized material, equipment, and expertise [4, 5, 6].

Consuming a fraction of the labour and capital resources, computational methods for identi-

fying putative PPIs are a compelling methodology for complimenting the challenges posed by traditional experimental methods. A single Y2H experiment, for instance, is conducted over multiple days [7, 8]. Contrastingly, the interactions between a given protein and the entire human proteome can be conducted in approximately three minutes using the INTREPPPID method through the online PPI.bio portal [9]. This is to make no mention of the discrepancies in capital investment between these two examples, with Y2H requiring expensive specialist equipment and materials, while the RAPPPID & INTREPPPID models runs on commodity computer hardware [10, 9].

Building upon decades of painstaking research, sizable databases of PPIs have been compiled across a growing number of organisms [11, 12]. These databases, in turn, have greatly enabled methods for identifying putative PPIs using data analysis and mining. Combined with the explosion of advances in the field of deep learning in recent years, it is increasingly viable for designers of PPI inference methods to leverage neural networks to increase benchmark accuracies.

Despite best efforts, accurate PPI inference that generalizes to all use-cases remains regrettably out of reach. Surprisingly prevalent and persistent challenges in the nature of training datasets and model architecture frustrate efforts to train PPI inference models which perform well on out-of-distribution examples [13, 14, 15]. While these hurdles have long been known to researchers, the challenges remain relevant and affect recent methods [15].

This thesis contains a collection of three studies aimed at improving PPI inference performance on out-of-distribution data, as well as characterizing new challenges on the horizon that threaten the generalization of tomorrow's PPI inference models when trained on datasets with maximum sequence identity thresholds ranging from 50% to 90%.

Thesis Organization & Contributions

This thesis is organized into four additional chapters whose main technical contributions are:

- **Chapter 2** – This thesis begins with a comprehensive review of the biological and computational literature which inform this work.
- **Chapter 3** – This chapter introduces RAPPPID, a simple model for PPI inference which relies on judicious applications of regularization to achieve state-of-the-art performance at the time of publication. This study also highlights that contemporaneous models fail to generalize to proteins outside the training set, and introduces tokenization typically applied to natural language to amino acids.

- **Chapter 4** – Expanding on the work done on RAPPPID, Chapter 4 reports on an investigation into leveraging orthology to further generalize PPI inference to unseen proteins belonging to evolutionarily-distant organisms. The resulting method, INTREPPPID, outperforms RAPPPID and other contemporaneous models when trained on human PPIs, and subsequently evaluated on other organisms. To do so, a novel "orthologous locality" loss term is defined to reshape the protein embedding latent space, "pushing" orthologous proteins close to one another and "pulling" non-orthologous proteins apart.

- **Chapter 5** – In which we reveal a before-uncharacterized source of data leakage in PPI inference models which incorporate pre-trained protein large language models (PLLMs). Furthermore, we demonstrate how careful attention to the creation of training datasets can result in PLLMs which are both performant and do not leak data. Finally, failures to generalize across PLLM-based and non-PLLM-based are described.

Publications & Author Contributions

- **Chapter 3**

  Szymborski, Joseph, and Amin Emad. 2022. "RAPPPID: Towards Generalizable Protein Interaction Prediction with AWD-LSTM Twin Networks." *Bioinformatics* 38 (16): 3958–67.
  DOI: 10.1093/bioinformatics/btac429.

  A.E. and J.S. conceived the study, designed the project and the algorithm, and wrote the article. J.S. implemented the pipeline and performed the statistical analyses of the results. All authors read and approved the final article.

- **Chapter 4**

  Szymborski, Joseph, and Amin Emad. 2024. "INTREPPPID—an Orthologue-Informed Quintuplet Network for Cross-Species Prediction of Protein–Protein Interaction." *Briefings in Bioinformatics* 25 (5): bbae405. DOI: 10.1093/bib/bbae405.

  A.E. and J.S. contributed to project conceptualization, methodology as well as writing, editing, and reviewing. A.E. contributed to supervision and funding acquisition. J.S. contributed to software, data curation, investigation and visualization. All authors read and approved the final article.

- **Chapter 5**

  The manuscript in Chapter 5 is under-review and is currently available as a pre-print on the bioRxiv online pre-print repository.

  Szymborski, Joseph, and Amin Emad. 2025. "A Flaw in Using Pre-Trained pLLMs in Protein-Protein Interaction Inference Models." *bioRxiv*. DOI: 10.1101/2025.04.21.649858.

  A.E. and J.S. contributed to project conceptualization, methodology as well as writing, editing, and reviewing. A.E. contributed to supervision and funding acquisition. J.S. contributed to software, data curation, investigation and visualization. All authors read and approved the final article.

Chapter 2

Background Material and Literature Review

The studies of which this thesis is composed are multi-disciplinary in scope and nature. They encompass a wide range of fields as varied as molecular biology, biocuration, machine learning, and deep neural networks. This chapter serves as a gentle but comprehensive introduction to the relevant aspects of these fields to help guide the reader in their understanding of the chapters that follow it.

The Biological Basis of Protein-Protein Interactions

Proteins are the subject, verb, and object of a dizzying number of molecular functions that make-up life. This chapter section discusses those functions, the scientific phenomena that causes them to occur, and the sometimes grim outcomes of their dysregulation.

Biological Processes, Pathways, and Networks

The number of interactions between two proteins ("binary interactions") in humans has been estimated at around 600 000, while the literature is estimated to contain upward of 110 000 (or ≈ 18%) of those interactions [16, 17]. As there are upwards of 20 000 human proteins with over 400 million possible combinations, the proportions of proteins that interact with one another (*i.e.*, members of the "interactome") are relatively few. This makes for a rather sparse network of interactions. Despite this, the proportion of non-interacting proteins which appear in the literature is far smaller than that of those which do interact. A mere 2 171 non-interacting pairs found in literature are present in the Negatome database, an effort by researchers at the

Technical University of Munich to collect evidence of non-interacting protein pairs [18]. If the aforementioned estimates are to be believed, approximately one thousandth of one percent of non-interacting protein pairs have been verified experimentally. An additional layer of complexity atop this network of interactions is that the sequence, speed, and environment in which protein interactions occur greatly impact their downstream effects. The large and complex network formed by these interactions has led to comparisons to a hairball [19, 20].

The structural organization, metabolism, proliferation, DNA repair, and death of a cell are just a sample of the very many functions for which PPIs are primarily responsible for. The functions are carried out by an often-diverging chain of interactions that are typically referred to as "pathways". The latest version of Reactome, a knowledgebase which catalogues peer-reviewed reactions and pathways, identifies 2 799 pathways in *Homo sapiens* broadly sorted into 29 categories reflecting their function and role within the cell, ranging from signal transduction to extracellular matrix organization (Figure 2.1) [21].

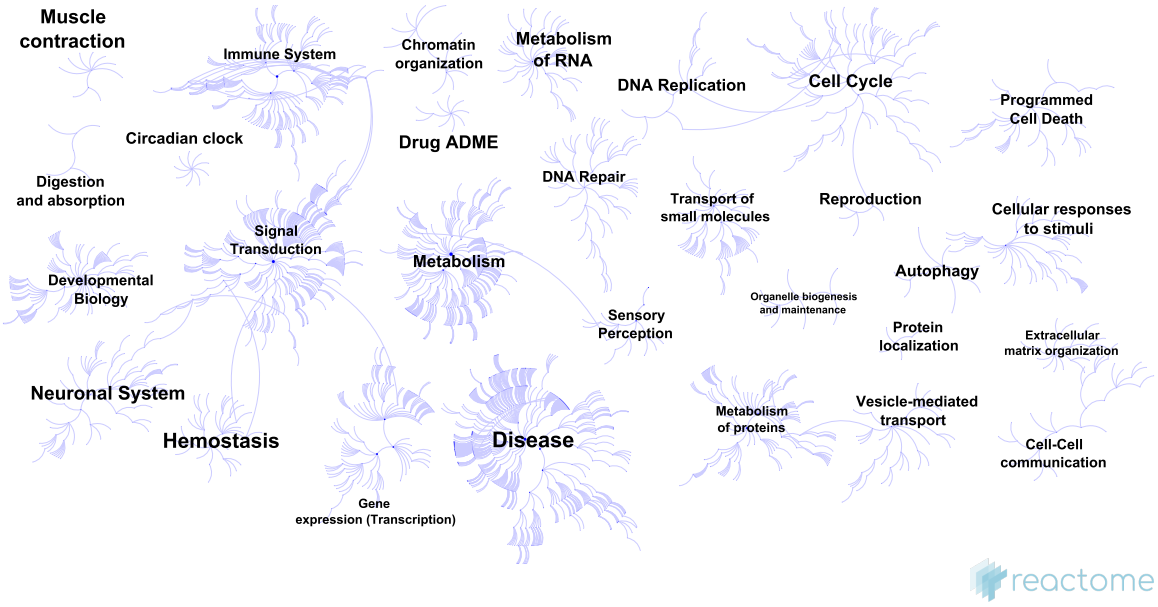


Figure 2.1: A high-level schematic of the human pathways in the Reactome knowledgebase. Each blue curved line segment represent a protein interaction within a broader pathway. Pathways are sorted in 29 broad categories that reflect their function and role. © 2025 Reactome / CC-BY 4.0.

These pathways are composed of interactions which can involve pairs of discrete proteins or many proteins acting in concert as a complex [22]. While these complexes may be composed of proteins which are different to one another in both sequence and structure ("heter-oligomeric" complexes), those comprise entirely of the same protein are also common ("homo-oligomeric"). In addition to other proteins, proteins interact with other macromolecules like ribonucleic acids (RNA), deoxyribonucleic acids (DNA), and even glycans to perform these various functions. These interactions, however, are less understood and, to our current knowledge, fewer in number [23, 24]. Proteins also regularly interact with small molecules like adenosine triphosphate (ATP)

and ions like those of the metals zinc and calcium [25, 26, 27]. For the purposes of this thesis, we will restrict our scope to binary interactions between proteins.

Physiochemical Fundamentals of Protein Interactions

Protein interactions, at their most fundamental level, are chemical reactions. These reactions can alter the structural or chemical properties of the interacting proteins or other macromolecules. Additionally, enzymes are a class of protein which intercede to catalyze other biochemical reactions.

Proteins within a cell travel through its different liquid phases via a random process known as Brownian diffusion [28, 29]. This undirected movement results in random collisions between proteins which then have an opportunity to physically and/or chemically interact with each other.

The strength with which two or more proteins interact is referred to as their "binding affinity" with respect to another protein. One way of quantifying such an interaction is its dissociation constant (K_D), which measures the concentration of ligand at which half of the protein is bound[1] [30]. For an equilibrium between a protein P and a ligand L

$$P + L \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} PL$$

where k_{on} and k_{off} reflects the forward and reverse rate of reaction, the value of K_D is defined as

$$K_D = \frac{k_{off}}{k_{on}} = \frac{[P][L]}{[PL]}$$

[31].

Protein interactions vary greatly with regard to their reaction kinetics, with some protein interactions being transient and short-lived (*i.e.*, high K_D value), while others are long-lived stable interactions (*i.e.*, small K_D value).

Some kinase reactions are short-lived. One such example is given by the Mitogen-activated protein kinase (*MAPK*) cascade, whose involvement in cell proliferation, differentiation, stress, and apoptosis set the stage for its role in, among other things, metabolic disease and cancers [32, 33, 34]. Specifically, the isoforms of the mitogen-activated protein kinase kinase (*MEK1/2*) and the isoforms of the extracellular signal-regulated kinase (*ERK1/2*) bind very quickly (k_{on} =

[1] This definition only holds for binary interactions involving a single binding site, an assumption held throughout this thesis.

3.23×10^8 $M^{-1}s^{-1}$) and dissociate relatively quickly (k_{off} = 18.7 s^{-1}) [35]. In comparison, Biotin and Avidin form very long lasting reactions, with a comparable association rate to *MEK–ERK* (k_{on} = 7×10^7 $M^{-1}s^{-1}$) and a much slower dissociation rate ($k_{off} \approx 10^{-7}s^{-1}$) [36]. This is reflected in the difference of their dissociation constants. The *MEK–ERK* reaction has a dissociation constant of 5×10^{-8} M while the Biotin–Avidin constant is much smaller at 7×10^{-15} M.

Proteins most commonly selectively bind their ligands to some degree, however some "promiscuous" proteins are capable of almost indiscriminately binding to other proteins [37]. This promiscuity, however, is often evolutionarily balanced by an increase in protein instability [37].

The binding affinity and specificity of a protein to another protein is largely determined by their three-dimensional structure and chemistry — their "molecular complementarity" [38]. Our understanding of how a protein's structure factors into its function was put into focus when researchers solved the first protein structures, that of myoglobin in 1958 and hæmoglobin in 1960 [39, 40, 41]. In the early 1960s, shortly after the first protein structures were resolved, the role of amino acid sequences in the structure of a protein was largely crystallized through work done by Christian B. Anfinsen and his colleagues. Anfinsen for his part would go on to win the 1972 Nobel Prize in Chemistry, having established "...the connection between the amino acid sequence and the biologically active conformation" [42, 43]. Anfinsen advanced a "thermodynamic hypothesis" which states "...that the native conformation is determined by the totality of inter-atomic interactions and hence by the amino acid sequence, in a given environment" [43]. This would later be commonly referred to by many as "Anfinsen's dogma", by way of comparison to Crick's "central dogma of molecular biology" [44, 45, 46].

Together, these works uncovered a triune of protein biochemistry, whereby a protein's sequence determines its structure and its structure in turn determines its function. As is the case with most statements of fact with wide-sweeping implications, this is a gross oversimplification and hides within it many pernicious exceptions and conditions [47, 48, 44, 45]. Nonetheless, it shall serve as a useful guide for our discussion of the basis of protein interactions.

Protein Interactions and Pathologies

Having established that proteins are essential for many essential functions of the cell, it requires little imagination to arrive at the conclusion that their dysfunction can have grave implications for organismal health. Our understanding of how protein interaction networks and how diseases alter their fabric has lead to many therapies and treatments. For instance, network analysis on an organism's interactome can be useful in identifying which proteins are "indispensable" to normal cell function and therefore whose dysfunction is more likely to cause disease and illness [49, 50]. This can be done by analyzing the effects of removing entire proteins from the protein

interaction graph [49], or by observing the result of a protein which no longer interacts with one of its usual partners due to *e.g.*, mutations which alter one of multiple binding sites [51]. The latter are known as "edgetic" perturbations, and give us particular insight into genetic diseases which cause small changes in proteins rather than the drastic sort which eliminates all function or expression of the protein [51].

Analyses of the "edgetic" changes are particularly relevant in the context of cancers which are largely driven by germline and somatic mutations which alter protein function [52]. Network analysis of the binding partners of mutant drivers of breast cancer, such as *TP53* and *PTEN*, helped identify new proteins previously unknown to be implicated in the onset of cancer [53]. Similarly, network analysis has helped us understand some of the mechanisms by which hereditary diseases like Huntingtin's disease progress, implicating novel interactors of the mutant *HTT* protein [54]. The new discoveries can then help identify new targets for the development of small-molecule therapies to restore wild-type function [55, 56].

The clinical utility of protein interaction networks aren't limited to cancers and hereditary diseases. Early networks of between human proteins and the SARS-CoV-2 virus, the cause of Coronavirus disease 2019 (COVID-19), helped quickly identify druggable protein targets months after a worldwide pandemic was announced [3].

The study of protein-protein interaction networks, therefore, is one of the rare fields of study in the biological sciences whose pursuit readily advances both basic and translational research.

Experimental Approaches to Discovering Protein-Protein Interactions

A huge number of PPI interactions have been characterized to date, with over 20 billion interactions across more than a thousand organisms being catalogued in version 12 of the STRING database [11]. The scale of the studies whose findings populate databases like STRING are increasing as well, with as many as 53 000 PPIs being identified in a single study.

Protein-protein interaction have been discovered using a wide variety of methods. Many methods involve identifying the unknown partners ("prey" proteins) of a known target protein ("bait" protein). Each method presents its own unique advantages and challenges. These limitations, including high costs, biases and lack of sensitivity, present as obstacles to uncovering the complete network of proteins which interact [5, 6].

This section serves as an overview of common methods for discovering and verifying PPIs.

Yeast Two-Hybrid and Related Methods

The Y2H system is one of the oldest and best studied methods for experimentally discovering and validating PPIs, originating from work by Fields and Song 36 years ago [57, 4]. The Y2H method involves fusing the binding domain of a transcription factor to the bait protein and that same transcription factors activation domain to the prey proteins [57]. When the two proteins interact, they effectively reconstitute the function of the transcription factor, which results in the expression of some desired reporter gene. In their original work, Fields and Song used the *Gal4-UAS* system, which allowed for screening by the ability of cells to grow on galactose or for the blue colour they produce when grown on indicator plates [57].

(MYTH) is a variation of the Y2H system which adapts the method to the study of membrane proteins [58].

Many related methods for studying and observing PPIs involve fusing domains to subjects of study such that bait-prey interactions reconstitute functional systems whose activity is directly or indirectly observable [5]. Examples include MaMTH and BiFC, whose interactions cause the reconstitution of ubiquitin and a fluorescent complex, respectively [59, 60].

The *JAK/STAT3* signal transduction pathway has also proved useful in identifying PPIs experimentally. Both the MAPPPIT and KISS methods involve fusing kinase and/or receptor domains which, upon prey-bait interactions, causes activation of the *JAK/STAT3* signalling pathway and the downstream expression of a reporter gene [61, 62].

Affinity-based Methods

Affinity methods involve the physical capture, through the use of antibodies, of target proteins which have bound proteins.

Co-immunoprecipitation (CO-IP) is a useful technique for validating hypothesized PPIs. Primary antibodies which bind the target protein are first incubated with cell lysate, often from cells which over-express the protein of interest [63]. Secondary antibodies, conjugated to agarose or magnetic beads, are used to isolate the target protein [63]. Following the separation of the lysate's constituents on an SDS-PAGE gel, the presence of the proteins hypothesized to interact is confirmed by Western blot [63].

Whereas CO-IP is suitable for validating suspected interaction partners, affinity purification mass-spectroscopy (AP-MS) is a method which can be used for the discovery of new interaction partners. Baits are fused with short peptide sequences known as affinity tags (*e.g.*, FLAG or HA) [64]. Purification is then conducted much in the same way as it is for CO-IP, but primary

antibodies here target the affinity tag [64]. The most significant difference is that, rather than identifying interacting partners using a Western blot, AP-MS relies on mass spectrometry. Mass spectrometry allows for the identification of partners without making assumptions as to the proteins which have interacted with the bait, as is the case with CO-IP [64].

Both AP-MS and CO-IP require that protein interactions survive cell lysis which, depending on the strength of the interaction, may not be the case. BIOID-MS is a method whereby a promiscous biotin ligase (*e.g.*, BirA) is fused to a bait protein, resulting in the biotinylation of interactors [65]. The biotinylated proteins can then be isolated using avidin/streptavidin affinity columns and the partners can be identified by mass spectrometry [65].

Energy Transfer Methods

The methods for PPI discovery and validation here thus far have not been suitable for real-time observation of protein interaction dynamics. Förster resonance energy transfer, or sometimes fluorescence resonance energy transfer, (FRET) is capable of spatial and temporal resolution of PPIs in living cells [66, 5].

FRET shares some similarity with BIFC, but rather than fusing fragments of one fluorescent protein, fluorophores are fused to both proteins. One of the fluorophores (the "donor") emits at a wavelength which excites the other (the "acceptor") [66, 5]. The acceptor only emits when the donor is near enough to excite it, unlike with BIFC where the reconstitution of the fluorescent complex is permanent [5].

Bioluminescence resonance energy transfer (BRET) is a similar method which helps reduce background noise and photobleaching which are obstacles caused by FRET's reliance on external illumination [66]. Rather than fusing two fluorophores, BRET experiments replace the "donor" fluorophore with luciferase, eliminating the need for external illumination to excite the "donor" [66].

Other methods

Cross-linking mass spectrometry (XL-MS) is a method involves forming covalent bonds between nearby protein residues using a chemical cross-linker. The cross-linked proteins are then identified using mass spectrometry. XL-MS is both high-throughput and allows us to identify protein complexes. Cross-linking, however, can often have low-efficiency and are slow, limiting the interactions that can be detected using XL-MS [6].

Many of the methods here involve high-impact manipulations involving greatly altering the

natural environments of cells or proteins, which can distort or mask certain reactions. coFrac-MS is a method for identifying possible protein complexes while avoiding any such harsh conditions at the cost of measuring functional association via co-expression/co-localization rather than direct PPIs [6]. This is achieved by fractionating cell lysates under native conditions and identifying the protein constituents of fractions by mass spectrometry [6].

Thermal proximity co-aggregation (TPCA) is a method that exploits the finding that proteins which interact co-aggregate when thermally denatured [67]. By comparing the melting curves of proteins, interaction partners are inferred [67]. This method can successfully identify protein complexes, is sensitive to weakly bound proteins and can be performed *in situ*.

Computational & Statistical Models of Protein-Protein Interactions

Methods for computationally inferring PPIs span a wide range, with the field largely moving from statistical methods to machine learning methods and finally to deep learning models (Figure 2.2). I have found that, much as musical artists resist being assigned genre labels, so too do models of PPI inference. Nevertheless, an effort was made to thematically group models into broad categories on the basis of the methodologies they employ. Viewing these groups over the passage of time gives us useful insight into the trends and discoveries that have shaped the field of PPI inference.

The list of studies reviewed here is not exhaustive, but rather a representative cross-section of the methodologies that have been used for PPI inference. They are presented here with the aim that they might help situate the novel approaches presented later in this thesis within the broader field.

Non-Machine Learning Methods for PPI Inference

Interaction Site Prediction

The earliest methods for PPI inference largely centred identifying on identifying similarities across proteins and networks belonging to other species, and interpolating interactions between them on that basis [68, 69, 70]. The earliest study of this nature of which I am aware also is arguably the simplest; Kini and Evans described a method for predicting interaction sites in 1996 by noting the prevalence of proline residues which flank such regions [68]. This method was used to identify the existence of a polymerization site in the epitope of fibrinogen which

they found to be bound by a specific monocolonal antibody [68]. A further generalization of the identification of interaction sites on the basis of the composition of amino acid residues and their three-dimensional structure closely followed in the work of Jones and Thornton the following year [70]. Jones and Thornton devised a scoring system for surface patches of the

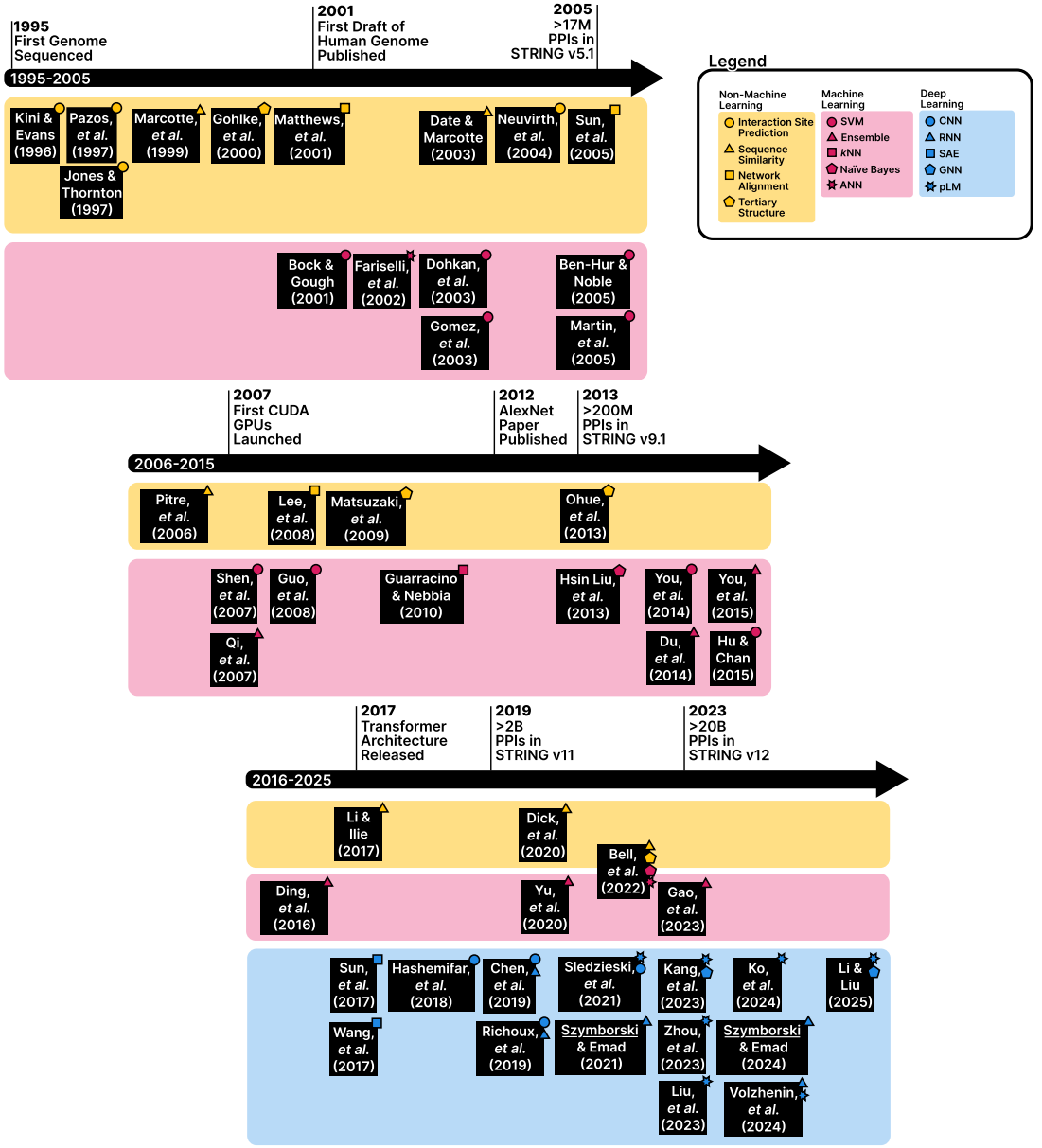


Figure 2.2: A historical timeline of PPI inference methodologies. A timeline of papers representative of the last 30 years of advances in PPI inference research, with particular focus to the methodologies employed. Each black box represents a study, and is placed along a timeline across three decades beginning from 1995. Coloured rounded boxes group the timeline according to broad methodological themes: non-machine learning (yellow), machine learning (red), deep learning (blue). Emblems at the right of each black box indicate the methodology studied in each publication. Emphasis has been placed on publications of which I am an author.

resolved three-dimensional structure of proteins which, among other factors, accounted for its solvation potential, hydrophobicity, accessible surface area (ASA), and planarity [70]. Seven years later, a study would revisit the analysis of surface properties of proteins which might help identify interactions sites by characterizing differences between the interfaces of bound and unbound protein complexes [71]. Another early method by Pazos *et al.* published in 1997 uses the correlation of mutations between proteins to identify putative interaction sites [69].

Sequence Similarity

The similarity between protein sequences has been demonstrated to have relationship to their relative structure and function for over 30 years [72]. This relationship has been leveraged to infer interactions between proteins on the basis of their similarity to each other, or to other homologous proteins which interact. One of the earliest such methods was devised in 1999 by Marcotte *et al.* who describe a "Rosetta Stone" method, which uncovers interacting proteins which appear as fused singular proteins in other species [73]. This finding serves as the seed for later work expanding and generalizing these ideas by Date and Marcotte in 2004 [74]. Specifically, Date and Marcotte constructed what they termed "phylogenic profile" vectors for each protein for which PPIs were to be constructed. Each element of these 57-dimension $\mathbf{p}$ vectors corresponds to a reference organism j. The value of each element $p_{i,j} = \max\left[-\log\left(E_{i,j}\right)^{-1}, 1\right]$, where $E_{i,j}$ is the lowest expectation value (E-value) found between the protein in question and the the proteome of organism j, as measured by the BLAST software [75, 74]. The mutual information between the $\mathbf{p}$ vectors of pairs of proteins correlate with their functional similarity [74]. Functional similarity was determined using the KEGG knowledgebase [76]. Notably, Date and Marcotte's method infers functional associations between proteins, sometime termed "functional interactions", which are not to be confused with physical interactions.

Sequence similarity methods remain the most common variety of non-machine learning models that are developed in recent years. These methods infer interactions on the basis of having previously observed other pairs of proteins with similar amino acid sequences which themselves interact with one another. A pioneering example of these methods is Pitre, *et al.*'s PIPE method [77].

PIPE begins by sliding overlapping windows over two proteins (protein A and B) whose interaction status is being evaluated. A search for proteins whose sequence contains similar regions to the overlapping windows over protein A is conducted. Subsequently, proteins which interact with the proteins found in the search are catalogued in a list R. Finally, a search is conducted for similarities between the windows from the sequence of protein B and those in R to construct a final score matrix. Each element $s_{i,j}$ of the score matrix corresponds to instances where the jth window from protein B matched a member of R who was added to the list due to its interaction

with a protein who shared sequence similarity to the ith window from protein A. An *ad hoc* visual analysis of the scoring matrix is then performed, whereby strong peaks in regions of the scoring matrix are indications of possible interactions.

The original PIPE algorithm was very computationally demanding, with its authors citing an approximate 1 000 hours of computational time required to conduct the evaluations that appear in the original publication [77]. The computational performance and sensitivity of the PIPE algorithm has since been iterated upon and improved over the course of 14 years [78, 79, 80, 81]. Li and Ilie developed SPRINT, a method reminiscent of PIPE in its theory, and boldly claimed in their 2017 publication that that theirs was "...the only sequence-based program that can effectively predict the entire human interactome: it requires between 15 and 100 [minutes], depending on the dataset" [82].

Bell *et al.* would later describe PEPPI, a pipeline which integrates multiple signals to infer PPIs, including protein sequence and stucture similarity.

Tertiary Structure Methods

The non-machine learning methods discussed thus far have largely operated on the basis of the amino acid sequences of proteins. Other methods rely on the tertiary structure of proteins to screen for pairs which are likely to interact.

Tsukamoto *et al.* introduced their Affinity Evaluation and Prediction (AEP) aglorithm in 2009, whereby the three-dimensional structure of proteins are first voxelized [83]. The complementarity of these structures are then computed and used to infer interaction. Matsuzaki *et al.* outlined another such method in the same year, but this time using a protein docking software (ZDOCK) to assess complementarity [84, 85]. Both these methods require exhaustive searches and are very computationally costly.

MEGADOCK, a method describe in a 2013 publication by Ohue *et al.*, was conceived to address these computational performance concerns [86]. Ohue *et al.* claimed a 7.5-fold improvement in docking calculation speed all while maintaining "an acceptable level of accuracy".

Machine Learning Methods for PPI Inference

From the years 1995 to 2000, PPI inference methods were primarily predicated on expert analyses and heuristics. These, often partially manual, methods are time consuming and often have limited accuracy. This is perhaps partly responsible for the limited scope of the applied examples of these publications, likely further exacerbated by the accessibility and scope of the contemporaneous

databases available. Work by Bock and Gough is an early, if not the first, example of machine learning deployed in the then-nascent field of PPI inference [87]. Bock and Gough use the Support Vector Machine (SVM) algorithm, which would become one of the most common machine learning approaches for inferring PPIs before deep learning would become prevalent [87].

Support Vector Machines

The SVM is a very well-studied machine learning algorithm whose underlying concepts arose in the 1960s and which would eventually enjoy a great deal of popularity in the early-to-mid 2000s across a number of fields, not least of which being computational biology [88, 89, 90]. SVMs operate by maximizing the distance between a hyperplane and the nearest data point on either of its sides, resulting in what's termed as the "maximum-margin hyperplane" [90]. While SVMs were originally characterized a linear classifier, they can be easily adapted to non-linearly separable classification tasks by applying a kernel which projects the data into a higher dimension where such linear maximum-margin hyperplanes are more likely to exist [91, 90].

Many of the studies which sought to apply the SVM algorithm to the PPI inference task concern themselves primarily with feature engineering and devising appropriate kernels. In the same year, Gomez *et al.* and Dohkan *et al.* used SVMs by representing proteins as a "bag of domains" as well as incorporating amino acid composition features [92, 93]. Representing variable-length amino acid sequences as fixed-length vectors is a subject which has received much focus, as they are required by the SVM algorithm as well as other machine learning approaches [94, 95, 96]. Martin *et al.* used features inspired by chemical signatures to represent variable-length amino acid sequences into fixed vector representations [96].

All applications of SVMs to the binary PPI inference task must conceive of kernels which allow for the pairwise comparison of proteins. That is to say that methods must devise of a kernel $\tilde{k}\left[(A,B),(X,Y)\right] \rightarrow \mathbb{R}$ which calculates the similarity of the pair of proteins A and B with the pair X and Y. Ben-Hur and Noble term this variety of kernel a "pairwise kernel" [97]. A pairwise kernel often makes use of a second "sequence" kernel which computes the similarity between two protein sequences [97]. Martin *et al.*, for instance, use the sequence kernel

$$k(A,B) = s(A) \cdot s(B)$$

where k is the sequence kernel and s is the aforementioned signature function which projects protein sequences into fixed vectors. Martin *et al.* go on to define the pairwise kernel as

$$\tilde{k}\left[(A,B),(X,Y)\right] = k(A,B)k(X,Y)$$

where $\tilde{k}$ is the pairwise kernel [96]. Ben Hur and Noble concurrently and independently described the same kernels and termed the sequence kernel a "spectrum kernel" [97].

Shen *et al.* use a so-called "conjoint triad" of residues as features which take into consideration the neighbourhood of each residue and their physiochemical properties. A sequence kernel which computes the distance between these residue feature vectors was used and can be expressed, for two proteins A and B, as

$$k(A,B) = ||A - B||^2.$$

This sequence kernel is then incorporated into a symmetric pairwise kernel of the form

$$\tilde{k}\left[(A,B),(X,Y)\right] = \min\left[k(A,X) + k(B,Y),\ k(A,Y) + k(B,X)\right].$$

Supervised machine learning methods like SVM require large representative training sets to perform optimally, and as a result it is desirable to scale these methods to handle such datasets. You *et al.* explore a method to improve scalability by parallelizing a SVM model for PPI inference using the MapReduce algorithm [98, 99].

Ensemble Methods

Ensemble methods combine multiple weaker predictors to form a single, more performant one. An early example of such a method is provided by Qi *et al.* who describe a Mixture-of-Experts (MOE) method composed of four experts [100]. The first expert (P) was trained on high-throughput PPI experiments, the second expert (E) trained on gene expression data, the third expert (S) was trained on sequence-based features like homology, and the fourth and final expert (F) were trained on functional annotations. Bell *et al.*'s PEPPI method can be viewed as an ensemble method, as it classifies the interaction status of pairs of proteins by computing the consensus of four independent models which (1) perform look-ups in existing PPI databases, searches for (2) sequence and (3) structure interologues, and finally (4) a shallow neural network which operates on amino acid sequence triads [101].

The Random Forest algorithm is very well represented among studies which use ensemble methods [102, 103, 104, 105, 106, 107]. The Random Forest models consist of multiple shallow decision trees whose output are averaged to assign a final inferred label [102]. In contrast to Qi

et al.'s MOE model, the features upon which sub-models are trained are chosen at random in Random Forest models rather than being selected manually [102]. Additionally, it is typical for Random Forest models to have many more sub-models than the four found in Qi *et al.*'s MOE model and Bell *et al.*'s PEPPI.

Other Methods

While the most common traditional machine learning algorithms used for inferring PPI models are SVMs and Random Forest, other algorithms are also sometimes used. Guarracino and Nebbia created a PPI inference model using the k-nearest neighbours (kNN) algorithm [108, 109]. For this method, Guarracino and Nebbia computed amino acid composition feature vectors for each protein in a interacting or non-interacting pair, concatenated those vectors. To infer the interaction status of a new pair of proteins x, the L^1 (Manhattan) distance between the concatenated amino acid feature vectors of x and those in the training set is measured to find its k nearest neighbours [110]. If the majority of these neighbours are features of interacting pairs, the members of x are deemed to interact, and *vice versa*.

Another algorithm sometimes used for inferring PPI interactions is the naïve Bayes classifier [111, 112]. Hsin Liu *et al.* use the naïve Bayes classifier to combine 91 likelihood ratios which reflect the divergence in co-evolution between proteins [112]. Bell *et al.* similarly rely on a naïve Bayes classifier to ascertain a consensus between the five independent models they train.

In 2002, Fariselli *et al.* published a work which made the rare use of an artificial neural network (ANN) for PPI inference. It would be another 15 years before neural networks would become common place for such a task [113].

Deep Learning Methods for PPI Inference

While implementations of ANNs have been around since the late 1950s in some form or another, it would not be until approximately five decades later in the 2010s when networks with very many layers would see widespread use [114]. Much of the groundwork for these "deep" neural networks (DNNs) was laid between the early 1980s and the mid 1990s, with the development and implementation of concepts such as Hopfields networks, backpropagation, convolutional neural networks (CNNs), and recurrent neural networks (RNNs) [115, 116, 117, 118, 119, 120].

The term "deep learning" as applied to neural networks would be coined at the turn of the millennium but widespread adoption of deep neural networks came on the back of multiple simultaneous developments — including general-purpose computing on graphics processing units (GPUs), improvements to neural network implementations, and the advancement of big

data algorithms [121, 120]. A watershed moment for deep learning occurred in 2012 when Krizhevsky, *et al.* published AlexNet, a network for image classification that marked the first time a neural network won the ImageNet competition [122]. Computer vision and would be the field that arguably benefitted the most from the flurry of DNN research that followed AlexNet until self-attention and the transformer architecture would revolutionize the field of natural language processing (NLP) [123].

The adoption of DNN architectures to the task of PPI inference largely follow the trends of the broader field, with readily available examples of such models based on stacked autoencoders, RNNs, CNNs, protein language models and graph neural networks.

Stacked Autoencoders

In 2017, Sun *et al.* claimed in an article published in BMC Bioinformatics:

> To our knowledge, this research is the first to apply a deep-learning algorithm to sequence-based PPI prediction, and the results demonstrate its potential in this field. [124]

Sun *et al.*'s first-of-kind deep PPI inference method would adopt the stacked autoencoder (SAE) architecture [124, 125]. A similar SAE method for PPI inference by Wang *et al.* would be submitted for publication in Molecular BioSystems 23 days later [126]. These methods would foreshadow a burgeoning field of DNN-based PPI inference methods in the years to come.

SAEs are a neural network architecture that involve chaining the outputs and inputs of a series of autoencoders [125]. Autoencoders (AEs) are in turn neural networks which are composed of two sub-networks: an encoder and a decoder. The encoder is a network which projects an input into some smaller dimension, and a decoder approximates the encoder's inverse function. That is to say,

$$g : \mathbb{R}^n \rightarrow \mathbb{R}^m \ , \ g' : \mathbb{R}^m \rightarrow \mathbb{R}^n$$

where g is an encoder, g' is a decoder, and $n >> m$ [125]. Given some input $x \in \mathbb{R}^n$, the encoder computes a latent vector $z \in \mathbb{R}^m$. The decoder, being optimized to invert the encoding given the latent vector, means that

$$g'(z) = \hat{x} \approx x$$

where $\hat{x}$ is the reconstructed input encoded by latent vector z [125].

While important to the initial development of DNN-based PPI inference, the SAE architecture

remained uncommon outside these two studies, being primarily superseded in the next few years by RNNs and CNNs.

Convolutional Neural Networks

Retellings of the invention of the CNN often begin with Kunihiko Fukushima who published the idea while working at the NHK Broadcasting Science Research Laboratories in 1980 [118, 120]. Using the name "neocognitron" for the concept — an iteration of his earlier "cognitron" — Fukushima created a feature detector network which took inspiration from research by David Hunter Hubel and Torsten Nils Wiesel who conducted research on the cells of the visual neural cortex of mammals and their relation to pattern recognition [127, 128, 129, 120]. The two layer types used in the neocognitron, "S-layers" and "C-layers", resemble convolutional and pooling layers, respectively — both commonly used neural network layers used for computer vision tasks [118].

Nearly 20 years later, LeCun *et al.* would describe how fully-connected layers could be constrained to be convolutional, thereby describing the CNNs we know today [130, 131]. Briefly, CNNs apply convolutions to input tensors using a learnable kernel. CNN's are accordingly well suited to computer vision tasks. When convolutions are only applied in one-dimension, however, they lend themselves to sequences of features [132].

An early example of CNN's being applied to the PPI inference task is Hashemifar *et al.*'s DPPI method [133]. CNN's are, in the context of PPI inference methods however, are commonly used in conjunction with RNNs.

Recurrent Neural Networks

In 1990, Jeffrey L. Elman of the University of California, San Diego published a study entitled "Finding Structure in Time" in the journal Cognitive Science. This seminal article provides one of the earliest descriptions of an RNN architecture, later to be known eponymously as an Elman network [119]. RNNs were conceived with the goal of better modelling time-series data using neural networks. RNNs add a time-based inductive bias to the architecture of neural networks, where the locality of elements in the time series is made relevant. Incremental improvements on RNN architecture would ultimately cumulate into the Long Short-Term Memory (LSTM) network seven years later [134]. Devised by Sepp Hochreiter and Jörgen Schmidhuber, the LSTM gated architecture addressed many of the problems plaguing older RNN constructions by improving gradient flow and training stability as well as performance on tasks that require long-range dependencies [134]. The following year Hochreiter would publish a simplified construction similar to the LSTM architecture which he named the Gated Recurrent Unit (GRU).

RNNs are useful for any number of domains with a sequence of elements, be that a string of letter or words, as is the case for the field of NLP, a series of closing stock prices over the last quarter, as might be the case for quantitative finance, or a sequence of amino acid residues found in a protein — a scenario which is most relevant to this thesis.

Compared to Elman or Jordan networks, LSTMs and GRUs greatly reduce the risk of vanishing gradients due to long time-series. Despite this, LSTMs and GRUs still often suffer from training instability when trained on protein sequences due to their long length.

My methods, RAPPPID and INTREPPPID, both rely on LSTM encoders to learn latent vector representations of amino acid sequences. In both cases, to mitigate the training stability issue, amino acid sequences are first tokenized using the Unigram algorithm resulting in common groupings of common amino acids to be considered as one time-step when encoding the RNN [135, 136].

Richoux *et al.*'s work and Chen *et al.*'s PIPR both use a combination of RNN and CNN layers [137, 138]. The CNN layers, followed by pooling layers, serve to greatly reduce the number of time-steps, thereby increasing training stability of the downstream LSTM (in the case of Richoux *et al.*) and GRU (in the case of PIPR) layers.

Protein Language Models

Some two decades after Claude E. Shannon described a method for estimating the information entropy of the words that appear in printed English in 1950, Thomas M. Cover and Roger C. King would elaborate the idea by describing a "gambling" method for inferring the next letter of an English phrase [139, 140]. This is achieved by placing "sequential bets on the next symbol of text" using information about the underlying information entropy distribution [140]. Cover and King's method would, however, rely on human subject to "...work interactively at a computer terminal for a period of approximately five hours..." to generate predictions of a text sample approximately 250 words long, for an average of 50 words per hour [140]. It would be another quarter century before Bengio *et al.* would propose to model language neural networks using a similar method [141]. Bengio *et al.* began by modelling language as the "...conditional probability of the next word given all the previous ones...", given by

$$\hat{P}(W) = \prod_{t=1}^{T} \hat{P}\left(w_t | w_1, \ldots, w_{t-1}\right),$$

where w_t is the tth word in the sequence $W = w_1, \ldots, w_T$ [141]. Concisely, Bengio *et al.* summarize their proposed method for modelling language using the following numbered list of steps:

1. associate with each word in the vocabulary a distributed *word feature vector* (a real-valued vector in $\mathbb{R}^m$),
2. express the joint *probability function* of word sequences in terms of the feature vectors of these words in the sequence, and
3. learn simultaneously the *word feature vectors* and the parameters of that *probability function*.[141]

While research on neural language models, particularly using RNNs, continued in the years following Bengio *et al.*'s introduction of the concept, it would be the introduction of self-attention and the transformer architecture that would result in huge gains in predictive performance and reignite a field [142, 123]. While originally trained on a neural machine translation task, transformers would later be employed to model language in a 2018 white paper by OpenAI, inc. introducing a method named GPT. GPT models language in a similar manner to that proposed by Bengio *et al.*, conditioning the probability of the next member of the sequence given all previous ones. The following year, researchers at Google's AI language group would release BERT, a language model that diverges from Bengio *et al.*'s "causal" approach [143]. Instead, BERT infers masked regions of sequences given *all* other tokens in the sequence. This variety of language model is referred to as a "masked language model" (MLM).

Rather than sequences of natural language, protein language models (pLLMs) apply the idea of neural language models to sequences of amino acids of which proteins are made-up. Early seminal work in modelling protein sequences by next-residue prediction include the UniRep and Prose method [144, 145, 146]. Both methods are based on LSTMs, as was common for contemporaneous language models [142]. The stated utility of these methods is primarily the generation of useful vector representation of proteins. As discussed at length in section 2, machine learning methods thus far have relied on various methods to generate fixed-length features for protein sequence with mixed-results. Prior to pLLMs, these methods often relied on heuristics that effectively produced a hash or signature of proteins, or learned embeddings which were often limited due to compromises intrinsic to the RNN or CNN architectures.

In the years that would follow the release of GPT and BERT, attention-based language models — of natural language and protein sequences alike — would balloon in parameter size (Figure 2.3). This would lead to the *terme de l'art* "large language model" or LLM in reference to the scale of the number of parameters. Much like the *deep* in "deep learning" or *big* of "big data", *large* is a relative qualifier whose definition in absolute terms is murky at best and rapidly changing with developments in the field.

A variety of attention-based pLLM models would be released by a number of groups in rapid succession between 2022 and today [147, 148, 149, 150, 151, 152, 153, 154]. When used as part of PPI inference models, these vector representations of proteins generated by these pLLMs

have resulted in great improvements in reported predictive performance. One of the earliest methods to use pLLM embeddings for PPI inference is the D-SCRIPT method, which made use of the ProSE language [155, 146]. D-SCRIPT computes a "broadcast matrix" from the protein embeddings of the proteins whose interaction status is being inferred. Using convolutional and pooling layers, a contact map between the two proteins is inferred as well as the probability of interaction. Kang *et al.*'s AFTGAN method uses embeddings from the ESM-1B pLLM to generate features for proteins in a graph of PPI interactions [156]. A graphical attention network is then used to learn PPI-specific features, before a fully-connected layer is used to infer interaction status [156]. Geometric learning is once more paired with pLLM's in work conducted by Li and Liu and their MuToN method. Unlike AFTGAN, MuToN uses attention-based message passing to learn structural features from the three-dimensional structure of the protein in question. MuToN uses a pre-trained pLLM to obtain embeddings of each residue in the protein as part of the features used to train the message passing network.

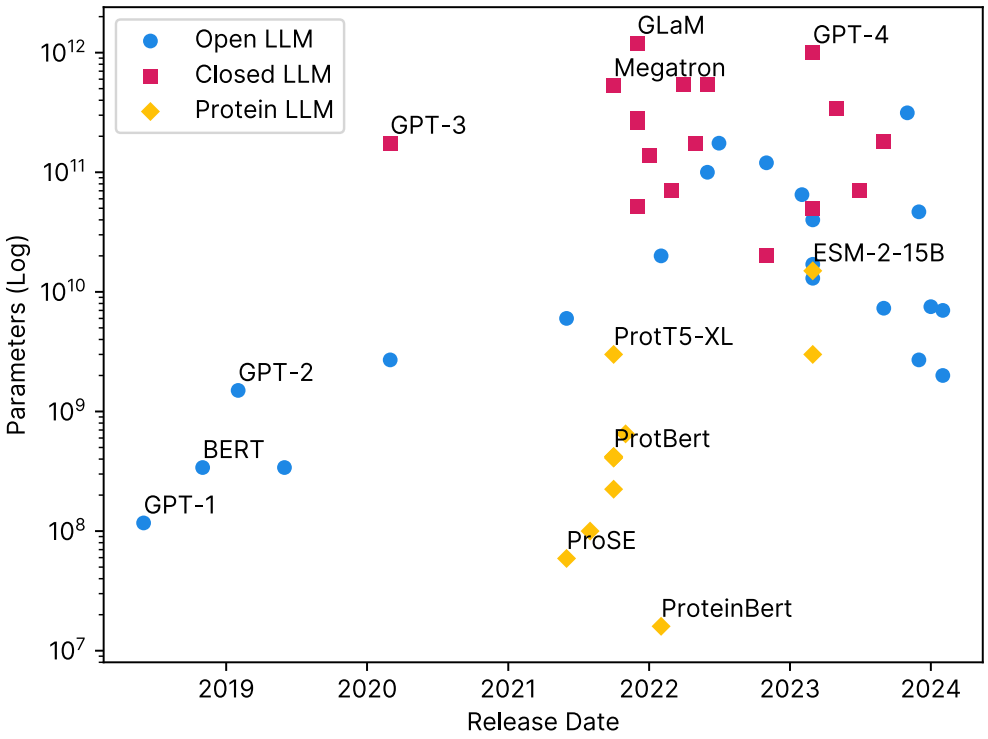


Figure 2.3: Number of parameters in language models over time. The number of model parameters (y-axis, log scale) found in LLMs as a function of their release date (x-axis). Round, blue markers indicate natural language models with open-source architectures. Red, square markers indicate natural language models whose architecture is closed-source. Parameter counts in these cases are difficult to verify and are either reported by the model's authors or otherwise estimated. Yellow, diamond markers report parameter counts of protein language models. The names of some notable models are included above their markers. Models included in this figure were released between 2018 and 2024.

Other pLLM-based methods use more conventional approaches to PPI inference. For instance, Zhou *et al.* simply compute the Hadamard product of protein vectors generated by ESM-1B and input the result to a four-layer fully-connected neural network which infers the interaction status [157]. Liu *et al.* input both protein sequences directly into a language model, delimited by a special "separation" token [151]. Unlike the other pLLM-based PPI inference models here, however, Liu *et al.* train their own pLLM (MP-BERT) from scratch. Ko *et al.* developed TUNA, which uses features from ESM2-150M as input to a twin transformer-based neural network [158]. A final neural Gaussian process layer is used to predict uncertainty and provide interval estimates [158, 159].

The Issue of Generalization in PPI Inference Models

In a 2012 letter to *Nature Methods*, Park and Marcotte sounded the alarm about a pervasive source of data leakage among methods which purport to make accurate inferences on paired biological data such protein-protein or protein-drug interactions [13]. Data leakage occurs when data or information from the testing set appears in the training set [160]. The implication being that the cross-validation being performed is invalidated by the lack of independence between the training and testing datasets [161, 162].

This oversight in the preparation of evaluation datasets had caused methods to appear as they had made accurate predictions of the interaction status of protein pairs, but instead had gravely overfitted to the training set and were incapable of generalizing to real-world use-cases. So severe was this masking that models which were evaluated under the flawed scheme which appeared to achieve AUROC scores as high as 0.82±0.01 were in reality scoring 0.58±0.03 when evaluated on a corrected dataset [13]. The collapse of metrics to near-random levels was consistent across seven models tested, all representative of methodologies common to the time.

As described in Section 2, significant changes in the adopted methodologies for PPI inference between when Park and Marcotte published their warning in 2012 and today. The methods tested had largely been superseded by new ones based on deep learning (Section 2), each purporting to having achieved ever more accurate testing scores. A study in 2024 found, however, that leading methods failed to correct for data leakage in precisely the same way Park and Marcotte first described [15]. When re-evaluated on a corrected dataset, the familiar collapse of testing metrics to near-random levels is replicated once more.

Notable exceptions among methods of this era are RAPPPID and INTREPPPID, which were developed with a focus on generalization and minimizing data leakage during evaluation [10, 9]. Both methods far out-performed their contemporaries on datasets which control for data leakage as result.

Around the same time as pLLM adoption for PPI inference grew, so did awareness of the challenges that Park and Marcotte first described. This resulted in these methods increasingly adopting stricter evaluation datasets which control for data leakage [158, 157]. The adoption of pre-trained pLLMs, however, presented a new potential source of data leakage. pLLMs commonly used for PPI inference are typically trained on datasets that are exhaustive and are representative of all known protein sequences. The extent and nature of the data leakage caused by the incorporation of pre-trained pLLMs in PPI inference models is explored in length by a study of which I am an author and whose text is reproduced in Chapter **??** [163]. In that same study, I demonstrate that current PPI inference models, including those which do not incorporate pLLMs, fail to generalize in multiple practical real-world application. When evaluated on a dataset of protein interactions between proteins which originate from the SARS-CoV-2 virus and human proteins, all PPI inference models tested failed to make accurate predictions, obtaining scores which are consistent with random classifiers [163]. Likewise, when faced with pairs comprised of sequences with small mutations, the inferred probability of interaction computed by models had no correlation with their true changes in binding affinity [163]. Together, these results indicate that many challenges remain for PPI inference, particularly with respect to their ability to generalize.

Background and Context for "*GRouNdGAN: GRN-guided simulation of single-cell RNA-seq data using causal generative adversarial networks*"

In Chapter 2, I explored both the landscape of PPI inference methodology (Section 2), as well as their persistent and pervasive problem of these methods failing to generalize to out-of-distribution testing samples (Section 2). It is in this context that, in 2021, my supervisor and I submitted for publication RAPPPID. Prior to RAPPPID few PPI inference models, and no deep learning techniques to my knowledge, were evaluated on datasets which appropriately controlled for data-leakage. Indeed, when we tested existing PPI inference models at the beginning of RAPPPID's development, it became clear that models (particularly those using deep-learning methods) were still failing to generalize as Park and Marcotte observed in 2012 and Bernett *et al.* would go on to observe in 2024 [13, 15].

RAPPPID is a PPI inference model which was created to answer the question: could conventional deep PPI inference models be modified to overcome their failure to generalize if a focus on regularization was applied?

Inspired by then-recent studies, we began developing a RNN-based PPI inference model which incorporated LSTM-specific regularization techniques outlined by Merity *et al.* in 2017 [164]. Merity *et al.*'s AWD-LSTM is a carefully tuned approach to regularizing and optimizing RNNs which take into consideration how traditional approaches sometimes are inappropriate for RNNs.

As introduced in Subsection 2, a common problem encountered when applying RNN architectures is that of vanishing gradients. RNN variants like LSTM and GRU, which help to lessen the negative effects of large sequences on training stability, succumb under a number of time-steps which is even still below the average protein sequence length. While other RNN-based PPI inference methods address this limitation through the use of CNNs to greatly reduce the length of the sequence, we identified a few problems with this approach.

Firstly, by initially filtering sequences through the use of CNNs and pooling layers, the resolution of the time-steps is greatly reduced. By reducing the resolution, the LSTM's ability to model long-term dependencies is rendered useless, thereby squandering an opportunity to model the non-linear nature of PPI contact surfaces in protein sequences. Secondly, the addition of convolutional layers added to a very constrained parameter budget. A very small number of PPI training pairs and a dataset which controls for data leakage makes overfitting both likely and easy to detect. For this reason, the number of parameters that make up the RAPPPID was carefully selected.

Ultimately, use of the Unigram tokenization algorithm via the Sentencepiece library allowed us to increase training stability while never bringing the resolution lower than three amino acids, grouped according to their common co-occurence [136]. Additionally, rather than contributing to overfitting, this tokenization approach had a regularization effect due to a process called "sub-word regularization".

The final result is a model which, achieved then-state-of-the-art accuracy on out-of-distribution evaluation datasets [10]. The regularization, optimization, and training modifications as applied to a fairly conventional deep PPI inference architecture are responsible for RAPPPID's performance, as ablation experiments show [10].

RAPPPID is proof positive that, despite a literature replete with PPI models that fail to generalize, it is possible to improve performance on out-of-distribution evaluation sets of simple architecture through thoughtful application of regularization and modification to the training process.

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Chapter 3

GRouNdGAN: GRN-guided simulation of single-cell RNA-seq data using causal generative adversarial networks

Abstract

We introduce GRouNdGAN, a gene regulatory network (GRN)-guided reference-based causal implicit generative model for simulating single-cell RNA-seq data, in silico perturbation experiments, and benchmarking GRN inference methods. Through the imposition of a user-defined GRN in its architecture, GRouNdGAN simulates steady-state and transient-state single-cell datasets where genes are causally expressed under the control of their regulating transcription factors (TFs). Training on six experimental reference datasets, we show that our model captures non-linear TF-gene dependencies and preserves gene identities, cell trajectories, pseudo-time ordering, and technical and biological noise, with no user manipulation and only implicit parameterization. GRouNdGAN can synthesize cells under new conditions to perform in silico TF knockout experiments. Benchmarking various GRN inference algorithms reveals that GRouNdGAN effectively bridges the existing gap between simulated and biological data benchmarks of GRN inference algorithms, providing gold standard ground truth GRNs and realistic cells corresponding to the biological system of interest.

Introduction

Unraveling gene regulatory interactions, often represented as a gene regulatory network (GRN), plays a crucial role in studying biological processes under different conditions[1,2,3], simulating knockdown and knockout experiments[4,5], and identifying therapeutic drug targets[6,7]. Many al-

gorithms have been proposed for GRN reconstruction using bulk or single-cell RNA sequencing data (scRNA-seq), alone[8,9,10,11,12,13,14] or with other modalities[15,16,17,18]. While these advances have provided great biological insights, evaluating the performance of GRN Inference algorithms remains challenging[19,20] due to the lack of reliable ground truth for the biological processes under study. Existing evaluation approaches often resort to curated databases[21,22,23]. However, the regulatory interactions in these databases are aggregated from a wide range of datasets and are not specific to a biological system, making them not ideal benchmarks due to context specificity of gene regulation. Another strategy is to verify regulatory interactions by conducting perturbation experiments on the system under study[20]. However, this approach is tedious, lengthy, and expensive. Another approach is to employ scRNA-seq simulators. Despite great progress in this domain, most simulators lack the essential properties for this task, such as the preservation of gene identities and simulation based on a user-provided ground truth GRN. For example, scGAN[24], cscGAN[24], scDESIGN2[25], and SPARSIM[26] capture inter-gene correlations in their datasets. However, since they do not explicitly impose a known GRN (that can act as the ground truth), they are not suitable for benchmarking GRN inference methods.

A small subset of simulators (e.g., BoolODE[19], SERGIO[4], and GeneNetWeaver (GNW)[27]) explicitly incorporate GRNs capturing transcription factor (TF)-gene dynamics. GNW was used to generate benchmarks for Dialogue for Reverse Engineering Assessment and Methods (DREAM) challenges[27,28,29]. However, being a bulk RNA-seq tool, its simulated datasets do not replicate the distribution of experimental scRNA-seq datasets nor exhibit their statistical properties, despite attempts to adapt it for this purpose by externally inducing dropout events[10,30]. SERGIO and BoolODE were designed to simulate scRNA-seq data using stochastic differential equations (SDEs)[4,19] and have been used to benchmark a variety of GRN inference methods. However, often a mismatch between the benchmarking results based on experimental and their simulated data have been reported, which may be attributed to differences between the simulated and experimental datasets. For example, in the BEELINE study (using BoolODE), some of the top ranking methods on simulated benchmarks reported near random performance on curated and experimental benchmarks[19]. Both simulators enable the user to simulate more realistic datasets by carefully selecting the values of the SDE parameters. Moreover, using a reference dataset, SERGIO allows fine tuning the added technical noise in an iterative procedure until the generated and reference datasets are matched[4]. While these steps can help improve the resemblance to real datasets, they are often suboptimal and pose an undue burden on the user: for example, in SERGIO, the user must evaluate the similarity based on five different statistics and change three parameters iteratively until desired resemblance is achieved (which itself can be subjective). In addition, since the GRN is imposed on the "clean" dataset, but technical noise is added afterwards, this step may change the encoded causal relationships in a non-trivial way (which may explain why the performance of GRN inference methods dropped close to random when applied to the noisy dataset[4]). Furthermore, SERGIO and BoolODE do not readily pre-

serve gene identities and make simplifying assumptions regarding cooperative regulation (see Discussion). These shortcomings demonstrate the need for simulators capable of generating realistic scRNA-seq data that retain the regulatory dynamics specified by a user-defined GRN. Importantly, with the growing interest in causal inference, simulators that impose causal GRNs are of great need.

GRouNdGAN (GRN-guided in silico simulation of single-cell RNA-seq data using Causal generative adversarial networks) is a causal implicit generative model for reference-based GRN-guided simulation of scRNA-seq data inspired by CausalGAN[31]. Given an input GRN and a reference scRNA-seq dataset, it can be trained to generate simulated data that is both indistinguishable from the reference data and faithful to the causal regulatory interactions of the input GRN. Unlike model-based simulators that rely on simplifying assumptions regarding co-regulatory patterns, in GRouNdGAN these patterns are *learned* through complex functions. This allows it not to compromise on the underlying complexity of the system and to model elaborate regulatory dynamics. GRouNdGAN provides state-of-the-art performance in realistic scRNA-seq data generation, while preserving gene identities, causal gene regulatory interactions, and cellular dynamics (e.g., lineage trajectory and pseudo-time ordering). This is achieved through implicit parameterization without the need for manual fine-tuning. Using GRouNdGAN, we benchmark eight GRN inference methods and find the results to be aligned with BEELINE's experimental results[19]. Furthermore, the causal structure of GRouNdGAN enables it to be used for sampling from both interventional and observational data distributions, enabling in silico knockout experiments.

Results

GRouNdGAN generates scRNA-seq data using causal generative adversarial networks

GRouNdGAN is a deep learning model that generates scRNA-seq data while imposing a user-defined causal GRN to describe the regulatory relationships of the genes and TFs. Its architecture builds on the causal generative adversarial network[31] and includes a causal controller, target generators, a critic, a labeler and an anti-labeler (Fig. 1). Training includes two steps. First, the causal controller that generates TF expression values is pre-trained (Fig. 1B) as the generator of a Wasserstein GAN (WGAN)[32] with gradient penalty (WGAN-GP) (Methods). In the second step (Fig. 1C), TF expression values (generated by the pre-trained causal controller) and randomly generated noise are provided as input to the target generators that produce target genes' expressions, while incorporating the TF-gene relationships of the input GRN. To achieve

this, as input, each target generator only accepts a noise value and the generated expression values of TFs that causally regulate it (Fig. 1C, Methods). The generated expression of genes and TFs are then fed to a library-size normalization (LSN) layer[24].

The two steps of training above are performed based on the same reference experimental scRNA-seq training dataset. The goal of the first step (pre-training, Fig. 1B) is to train the causal controller to learn scRNA-seq data distribution and generate realistic data, irrespective of any GRN. The GRN edges between a set of regulators (e.g., TFs) and their targets are imposed in the second step of training (Fig. 1C). To achieve this, in the second step of training, only the expression of regulators (TFs) outputted by the causal controller are used and its output of target genes' expressions are discarded and instead are re-generated using the target generator neural networks in Fig. 1C to enable imposition of causal relationships.

The critic's role (in both steps of training) is to quantify the Wasserstein distance (Supplementary Notes) between the reference and simulated data. The target generators are trained in an adversarial manner to generate realistic datapoints indistinguishable from the reference datapoints by the critic. The labeler/anti-labeler estimate the TF expression values *only* from the target genes' expressions to ensure that the generated causal TF-gene dependencies are encoded. The combination of the model architecture (in that each target generator only receives a noise value and the simulated expression of its regulating TFs in the GRN) and the labeler/anti-labeler ensure that the GRN edges are causally imposed in the simulated data. Details are provided in Methods, the architectural choices in Supplementary Tables 1 and 2, and an ablation study in Supplementary Notes and Supplementary Table 3.

GRouNdGAN generates realistic scRNA-seq data

We first trained GRouNdGAN on three datasets (results for three additional datasets will be discussed later). The first dataset contained scRNA-seq profiles of 68,579 human peripheral blood mononuclear cells (PBMCs) from 10x Genomics[33] corresponding to eleven cell types ("PBMC-All"). We formed a dataset of the most common cell type in the PBMC-All dataset containing 20773 CD8+ Cytotoxic T-cells ("PBMC-CTL"). Additionally, we obtained the scRNA-seq (MARS-seq) profile of 2730 cells corresponding to differentiation of hematopoietic stem cells to different lineages from mouse bone marrow[34] ("BoneMarrow"). We used GRNBoost[28] to identify fifteen TFs for each gene to form the input GRN. Note that these identified interactions may contain both spurious correlations and causal relationships in the reference dataset. However, when they are used as input to GRouNdGAN, they will correspond to causal interactions in the *simulated* dataset (see Kocaoglu et al. for theoretical details of causality in this architecture[31]). For each dataset, we trained GRouNdGAN on a randomly selected training set and evaluated on a held-out test set (Methods, Supplementary Data 1 –

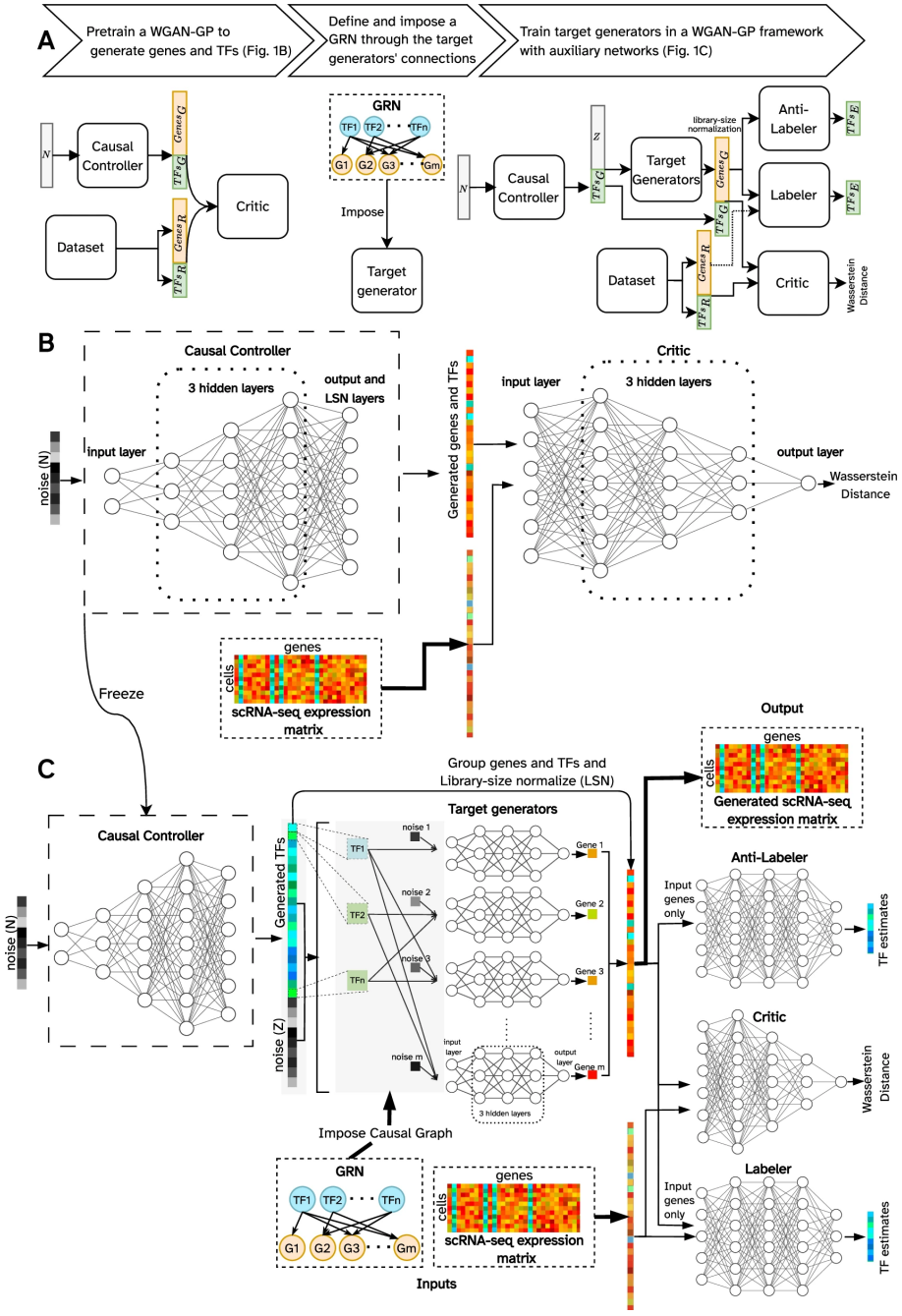


Figure 1: Architecture and training procedure of GRouNdGAN. **A** The flowchart representing the steps of the training procedure and the overall architecture of the model in each step. Subscripts G, R, and E represent generated, real, and estimated, respectively. **B** A WGAN-GP is pre-trained to generate realistic simulated cells using the reference (real) training set. **C** The LSN layer of the generator of the trained WGAN-GP (panel **B**) is removed, its weights are frozen, and is used as the causal controller to generate unnormalized TF expression values (expression of target genes generated by the causal controller are discarded). These TF expression values along with a noise vector are provided as input to the target generators, following the provided causal GRN. The generated gene and TF expression values are reorganized and passed through the LSN layer. The normalized simulated expression vectors and experimental reference data (the same training set as **B** are then passed to the critic to estimate Wasserstein distance between the reference and the generated data distributions. The anti-labeler estimates TF values based on generated target gene expressions. The labeler performs a similar task, but in addition to receiving generated values, it also utilizes target gene expression values from the reference data. Labeler and anti-labeler ensure that the causal GRN is incorporated by the target generators. Details of the model are provided in Methods.

Table 1: Performance of different simulators in generating realistic scRNA-seq data using the PBMC-CTL dataset

Simulator	Cosine distance	Euclidean distance	MMD	RF AUROC	miLISI
GRouNdGAN	**0.00057**	**182**	**0.026**	**0.54**	**1.891**
scGAN[24]	0.00095	222	0.031	0.59	1.888
scDESIGN2[25]	0.00100	229	0.065	0.76	1.736
SPARsim[26]	0.00104	235	0.309	0.95	1.625
Control	0.00019	99	0.012	0.50	1.909

The metrics are calculated between a simulated dataset of 1000 cells and the held-out test set of 1000 real cells (see Supplementary Data 1 – Sheet 2 for training set performance). Each gene in the imposed GRN of GRouNdGAN is regulated by 15 TFs (constructed using GRNBoost2 from the experimental training set). For the first three metrics, a value closer to zero is preferred, for RF AUROC a value closer to 0.5 is preferred, and for miLISI a value closer to 2 is preferred. For the first two metrics, the values correspond to the distance of the mean centroids of the real and simulated cells. The RF AUROC of control corresponds to perfect performance (of a random classifier). The other control metrics are calculated using the two halves of the real test dataset. Best performance values (excluding control) are in bold-face.

Sheets 2 and 3). Figure 2A, B and Supplementary Figs. 1–3 show the t-SNE plots of reference and simulated cells, qualitatively revealing their similarity.

We quantitatively assessed the resemblance using Euclidean distance, Cosine distance, maximum mean discrepancy (MMD)[35], mean integration local inverse Simpson's index (miLISI)[36], and the area under the receiver operating characteristic curve (AUROC) of a random forests (RF) classifier distinguishing simulated and experimental cells (Methods). As a "control" (and to enable calibration of these scores), we also calculated these metrics using two halves of the reference test set (corresponding to real cells). Table 1 shows the performance of GRouNdGAN, control, and three state-of-the-art simulators for the PBMC-CTL dataset (see Supplementary Data 1 – Sheet 2 for training set performance). We did not include BoolODE in these tables since it is a reference-free simulator and is not designed to match a particular reference dataset. We tried generating samples using SERGIO to match the PBMC-CTL dataset, but the simulated dataset did not show high resemblance to the PBMC-CTL dataset (see Supplementary Notes, Supplementary Tables 4 and 5 and Supplementary Fig. 4). However, we decided not to include these results in Table 1, since SERGIO relies on the user's judgement for distribution matching and the quality of the final simulated dataset is user dependent. Instead, we only included reference-based simulators that automatically generate scRNA-seq data resembling experimental data, in order to assess data quality systematically and fairly.

One of the challenges faced by GANs is the possibility of mode collapse: when applied to heterogenous datasets, they may learn to generate only a limited set of examples, instead of generating a variety of examples from the entire distribution of the training data. We trained GRouNdGAN using PBMC-All, which contains multiple cell types, to determine whether it

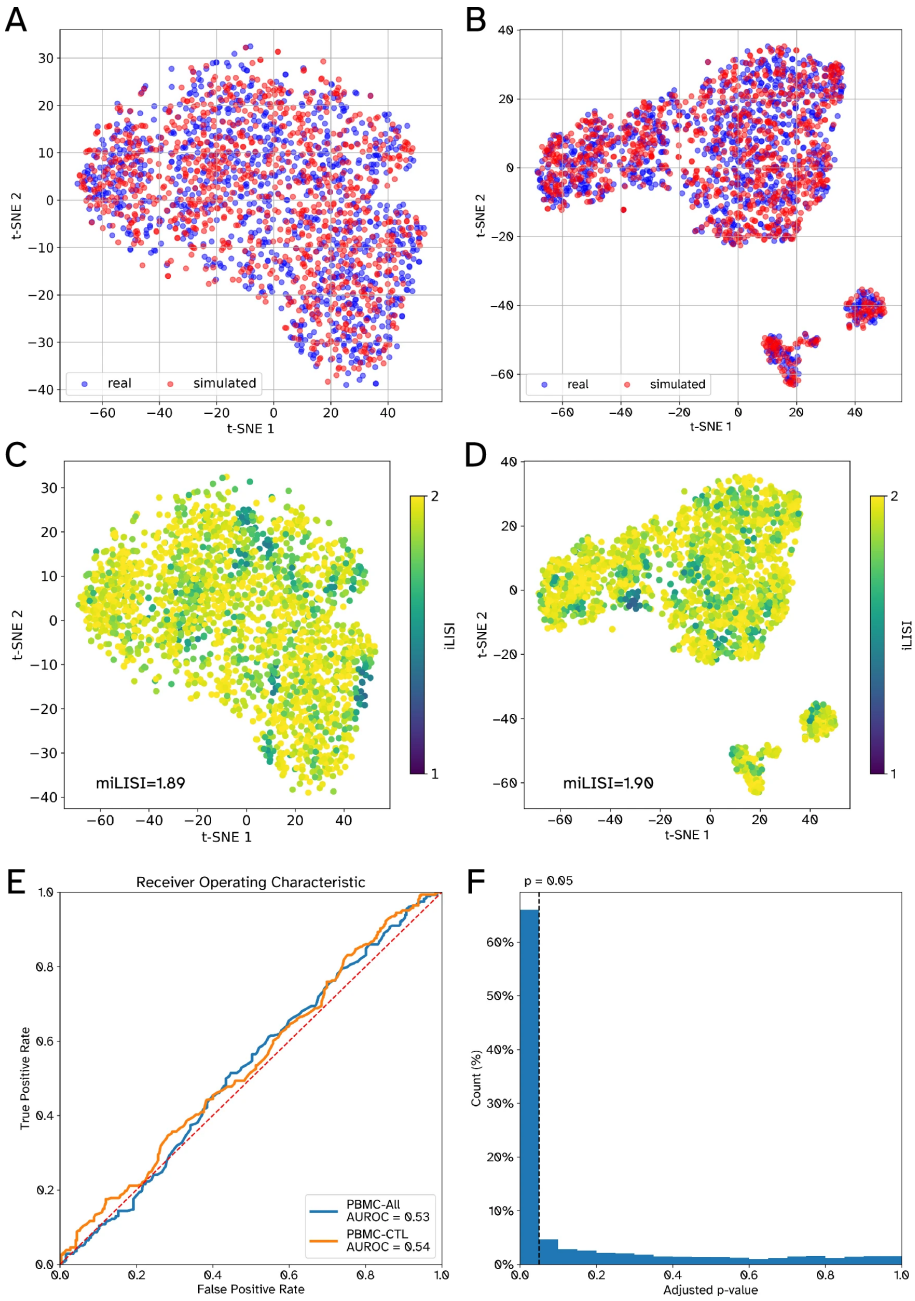


Figure 2: Performance of GRouNdGAN in generating realistic scRNA-seq data. All plots correspond to 1000 real (experimental) held-out test cells and 1000 simulated cells. Each gene in the imposed GRN of GRouNdGAN is regulated by 15 TFs (identified using GRNBoost2 from the experimental training set). Panels **A** and **B** depict the t-SNE plots of real (blue) and simulated (red) cells for PBMC-CTL and PBMC-All, respectively, while panels **C** and **D** also show their iLISI distributions. **E** The ROC curve of a Random Forest classifier in distinguishing between simulated and real cells. **F** The distribution of adjusted p-values corresponding to the TF perturbation study on the PBMC-CTL dataset. The p-values are obtained using two-sided Wilcoxon signed ranked tests and are adjusted for multiple hypotheses following the Benjamini–Hochberg procedure. Source data are provided as a Source Data file.

can generate samples from different cell types present in this dataset, or it will suffer from mode collapse and only generates samples corresponding to the most represented cell type. This is particularly important, since many scRNA-seq datasets are heterogenous and contain different cell types or cells in different states. If cell type information of the reference dataset is available a priori, one can easily generate realistic samples of each cell type separately (similar to the PBMC-CTL analysis) and use either cell type-specific GRNs, a shared GRN, or a combination of both. However, we were particularly interested to know whether our model could generate realistic samples *without* this knowledge, a more challenging task and useful for when such information is unavailable. Also, not requiring cell type information reduces the amount of user involvement (to annotate cell types) and improves model's usability. We repeated the analysis using the PBMC-All dataset, containing eleven cell types. Although GRouNdGAN did not receive cell type/cluster information, we did not observe mode collapse (Supplementary Fig. 2), and it was able to generate cells from distinct clusters (Fig. 2B, test set miLISI = 1.90). GRouNdGAN outperformed all simulators that did not use cell cluster information (Table 2) and compared to those that utilized this information, it was still the top performing based on MMD and RF AUROC and the second best according to the Euclidean distance (6% higher than the best) and miLISI (0.5% lower than the best) (Table 2). Moreover, we observed a high degree of concordance between the cell type marker expressions in the experimental data and GRouNdGAN-simulated data (Supplementary Notes and Supplementary Fig. 5). Finally, GRouNdGAN outperformed all other simulators based on all metrics when we repeated the analysis using the BoneMarrow dataset, containing continuous cell states and much fewer cells (Supplementary Data 1 – Sheet 3). These results show that GRouNdGAN is a stable simulator of scRNA-seq data and even without extra cell type information in heterogenous datasets, it automatically generates realistic samples reflecting different cell types (while imposing a GRN). See Supplementary Notes and Supplementary Data 2 for stability analysis and the effect of different GRN properties (including number of TFs) on the performance.

GRouNdGAN imposes a causal GRN in the simulated data

To assess the ability of GRouNdGAN in imposing the input causal GRN, we performed in silico TF knockout experiments (one of GRouNdGAN's capabilities) on the simulated cells ($n = 1000$) from the PBMC-CTL dataset. We performed a forward pass after setting the expression of each TF to zero (one at a time) in the causal controller output, while keeping all other parameters and TF expressions unchanged, ensuring that the perturbations were performed on the same batch of cells forming matched case/control experiments. There was no change in the expression of genes that were not regulated by the knocked-out TF (as expected), while the expression of the regulated genes changed. Figure 2F shows the distribution of the adjusted p-values (Benjamini–Hochberg) for each TF-gene edge in the GRN, comparing the expression of a gene across all cells before

Table 2: Performance of different simulators in generating realistic scRNA-seq data using the PBMC-All dataset

Simulator	Cell cluster labels/ratios provided as input	Cosine distance	Euclidean distance	MMD	RF AUROC	miLISI
GRouNdGAN	No	0.00028	**168**	**0.017**	**0.53**	1.90
cscGAN[24]	Yes	0.00025	242	0.030	0.56	**1.91**
cWGAN	Yes	0.00053	239	0.027	0.57	1.89
scGAN[24]	No	0.00081	300	0.035	0.58	1.90
scDESIGN2[25]	Yes	0.00032	181	0.046	0.86	1.81
scDESIGN2[25]	No	0.00057	256	0.124	0.91	1.53
SPARsim[26]	Yes	**0.00019**	**158**	0.287	0.96	1.66
SPARsim[26]	No	0.00035	245	0.307	0.96	1.39
Positive control	NA	0.00022	116	0.012	0.50	1.91

Baseline models that enable utilization of cell cluster labels or ratios are run with and without this information. The metrics are calculated between a simulated dataset of 1000 cells and the held-out test set of 1000 real cells. Each gene in the imposed GRN of GRouNdGAN is regulated by 15 TFs (constructed using GRNBoost2 from the experimental training set). For the first three metrics, a value closer to zero is preferred, for RF AUROC a value closer to 0.5 is preferred, and for miLISI a value closer to 2 is preferred. For the first two metrics, the values correspond to the distance of the mean centroids of the real and simulated cells. The RF AUROC of control corresponds to perfect performance (of a random classifier). The other control metrics are calculated using the two halves of the real test dataset. Best performance values (excluding control) are in bold-face.

and after the knockout of one of its regulating TFs (two-sided Wilcoxon signed-rank tests). In the majority (66.5%) of cases, the knockout of a gene's regulating TF significantly (adjusted p-value < 0.05) altered its expression, showcasing that the GRN is indeed imposed in the expression profiles of simulated cells. Note that since each gene is regulated by multiple TFs, we cannot expect that knockout of a single TF result in a significant change in the expression of its target genes in all cases. The analysis above shows that GRouNdGAN does not simply ignore the input TF values to instead rely solely on the noise input to generate realistic cells, and the effect of TF-gene edges are indeed imposed. Moreover, since each gene is regulated by multiple (15) TFs, but the knockout experiment is one TF at a time, this further shows that individual edges are imposed by the model.

GRouNdGAN does not require the input GRN to be the true causal GRN of the training data (we will discuss this further later). To further clarify this, we replaced GRNBoost2 with PPCOR[37] to form the GRN, which is a method based on (partial) correlation for inferring GRNs (see Supplementary Data 1 – Sheet 2 for resemblance metrics). Repeating the knockout experiments above ($n = 1000$ cells) resulted in similar results showing that in the majority (75.2%) of cases, the knockout of a gene's regulating TF significantly altered its expression (Supplementary Fig. 6). Taken together, these results show that GRouNdGAN can successfully impose different GRNs in the simulated data, while generating realistic cells.

The imposed GRN can be reconstructed from the simulated data

To test whether the imposed GRN can be reconstructed from the generated data, we first applied GRNBoost[28] to the reference PBMC-CTL dataset and recorded the top ten TFs for each gene. For each gene, we created two sets of five TFs based on their even/odd parity in the ranked list: TFs ranked 1st, 3rd, 5th, 7th, and 9th were connected to the gene and were imposed ("positive control GRN"), while the remaining five TFs were not ("negative control GRN"), resulting in two GRNs with identical densities and the same number of TFs and target genes. When considering the positive and negative control GRNs as ground truths (separately), the GRN reconstructed using GRNBoost2 from the reference (training) dataset ($n = 19{,}773$) resulted in comparable AUPRC (0.28 and 0.24, respectively, Supplementary Data 3 – Sheet 2), reflecting that they contain edges of comparable importance.

We simulated data ($n = 19{,}773$ cells) using GRouNdGAN by imposing only the positive control GRN, and used GRNBoost2, PIDC[10] GENIE3[9], and PPCOR[37] to reconstruct the underlying GRN (Fig. 3, Supplementary Fig. 7, Supplementary Data 3 – Sheet 2). All methods reconstructed the imposed edges with a much higher AUPRC compared to the reference dataset, while they did not assign high scores to the unimposed edges, resulting in close to random AUPRC. Figure 3 shows the performance of GRNBoost2 applied to the reference, GRouNdGAN-simulated, and scGAN-simulated data ($n = 19{,}773$ cells). The performance for scGAN (both GRNs) was lower than the reference dataset, showcasing that it had not preserved the TF-gene relationships of the experimental dataset well. The performance was relatively consistent between positive (Fig. 3A, C) and negative control (Fig. 3B, D) GRNs using the reference and scGAN-simulated data; however, it was much higher for the positive control GRN (compared to negative control GRN) using GRouNdGAN-simulated data. Similar patterns were observed when we repeated these analyses using the PBMC-All ($n = 67{,}579$ cells) and BoneMarrow ($n = 2230$ cells) datasets (Supplementary Data 3 – Sheets 3 and 4, Supplementary Figs. 8 and 9). These results show that the imposed edges were accentuated by GRouNdGAN, while the unimposed edges were disrupted and could not be found by GRN inference methods.

Next, we asked whether GRNBoost2 can reconstruct the imposed GRN, if GRNs of different densities are used to simulate data. Using data for which each gene was regulated by 15, 10, 5, and 3 TFs, we observed that in all cases the GRN imposed by GRouNdGAN could be inferred from the simulated data (Supplementary Data 3 – Sheets 5 and 6). Finally, we repeated the controlled analysis above, swapping the role of positive and negative control GRNs. Similar to the results of Fig. 3, the imposed edges were inferred by GRNBoost2, while the unimposed edges were not accurately discovered (Supplementary Data 3 – Sheet 7). These results show that 1) GRouNdGAN imposes the causal GRN, 2) the GRN inference methods can identify the imposed edges, and 3) the TF-gene relationships present in the reference dataset but unimposed

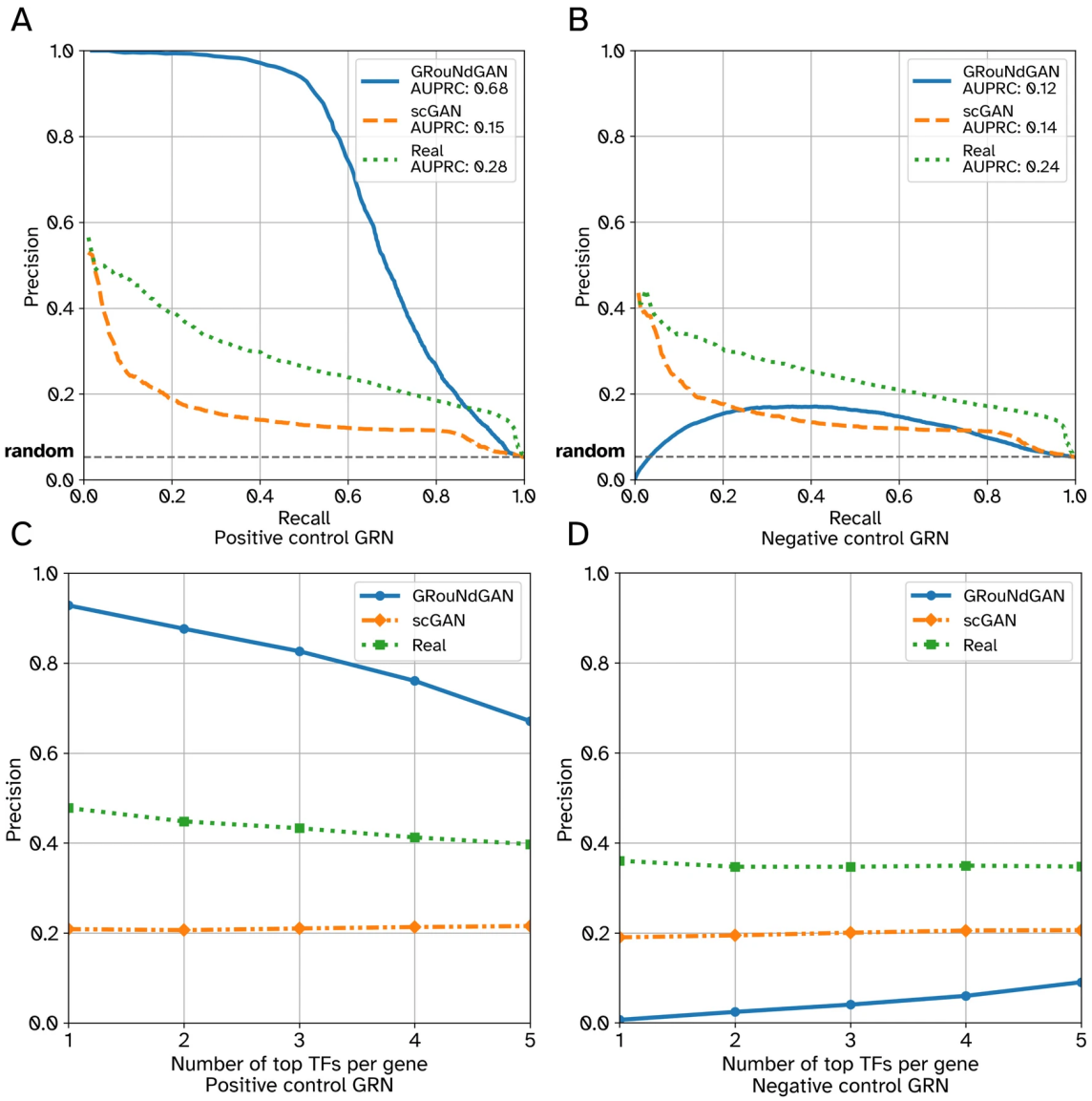


Figure 3: Performance of GRNBoost2 in recovering the imposed edges versus unimposed edges (PBMC-CTL). Left column shows the AUPRC (**A**) and Precision at k (per gene) (**C**) when the imposed edges (positive control GRN) were considered the ground truth. Right column shows the AUPRC (**B**) and Precision at k (per gene) (**D**) when the unimposed edges (negative control GRN) were considered the ground truth. Precision at k (per gene) refers to the precision when top k TFs for each gene is used to form the reconstructed GRN. All three datasets had the same number of cells ($n = 19{,}773$). Source data are provided as a Source Data file.

by GRouNdGAN are disrupted during simulation. The last property is particularly desirable for benchmarking GRN inference methods to ensure that the regulatory relationships present in the reference dataset (but unimposed) do not bias the simulated data, which could result in an inflated false positive rate.

Next, we repeated the analyses above on the PBMC-CTL dataset, but this time with input GRNs learned by PPCOR (a GRN inference based on partial correlation) from the training set. Similar to the results above, GRNBoost2 could reconstruct the imposed edges, and the importance

of the imposed edges were accentuated by GRouNdGAN (compared to the original training dataset) (Supplementary Data 3 – Sheet 8). We also imposed two randomly generated GRNs (with similar density to the GRNs above in which each gene was regulated by 5 TFs). Once again, the imposed edges were accentuated by GRouNdGAN, however imposing a random GRN (whose induced patterns may be inconsistent with real reference data) came at the cost of lower resemblance between simulated and reference data. (See Supplementary Notes and the Discussion for more details on why imposing a GRN that is inconsistent with the reference data (e.g., a random GRN) is a contradictory requirement to generating simulated data resembling the reference data).

GRouNdGAN-simulated data preserves trajectories and can be used for pseudo-time inference

In addition to deciphering discrete states, scRNA-seq data is often used to determine continuous cell transitions during biological processes such as differentiation, using trajectory and pseudo-time inference[38,39,40,41,42,43,44]. We set to determine whether GRouNdGAN-simulated data conforms to the transitional states and pseudo-time of the reference data. This is particularly important for benchmarking GRN inference methods that utilize pseudo-time information[12,13,14]. For this purpose, we used the BoneMarrow dataset corresponding to differentiation of hematopoietic stem cells to different lineages and simulated $n = 2230$ cells using GRouNdGAN (same as the number of real cells in the reference training dataset, to remain consistent).

We used Partition-based graph abstraction (PAGA)[38] for trajectory inference of the reference and GRouNdGAN-simulated data. PAGA computes graph-like maps of data manifolds faithful to the topology of data, retaining its continuous and discrete structures. Supplementary Fig. 10 shows the PAGA graphs, where nodes capture discrete states and the edges signify transitions between them. To compare the identified trajectories, we used markers of the cell types present in this dataset (Supplementary Table 6)[38]. Figure 4 shows the expression patterns of two marker genes for erythroid cells (*Gata1*, *Klf1*), neutrophils (*Mpo*, *Ctsg*), and basophils (*Mcpt8*, *Prss34*) for PAGA trajectory graphs of both datasets (Supplementary Figs. 11–17 show marker genes of all cell types). The comparison showed similar activation patterns between corresponding regions of PAGA graphs of the experimental and GRouNdGAN data, revealing several trajectories (e.g., erythroid and neutrophil branches). Strikingly, even the number of nodes activated in each figure scaled similarly between the two datasets: e.g., the markers of basophils were consistently expressed in only one node in each dataset (Fig. 4E, F).

We annotated different cell types present in both datasets using these marker genes (following PAGA's official tutorial) (Fig. 5A, B). The GRouNdGAN's annotated graph showed similar

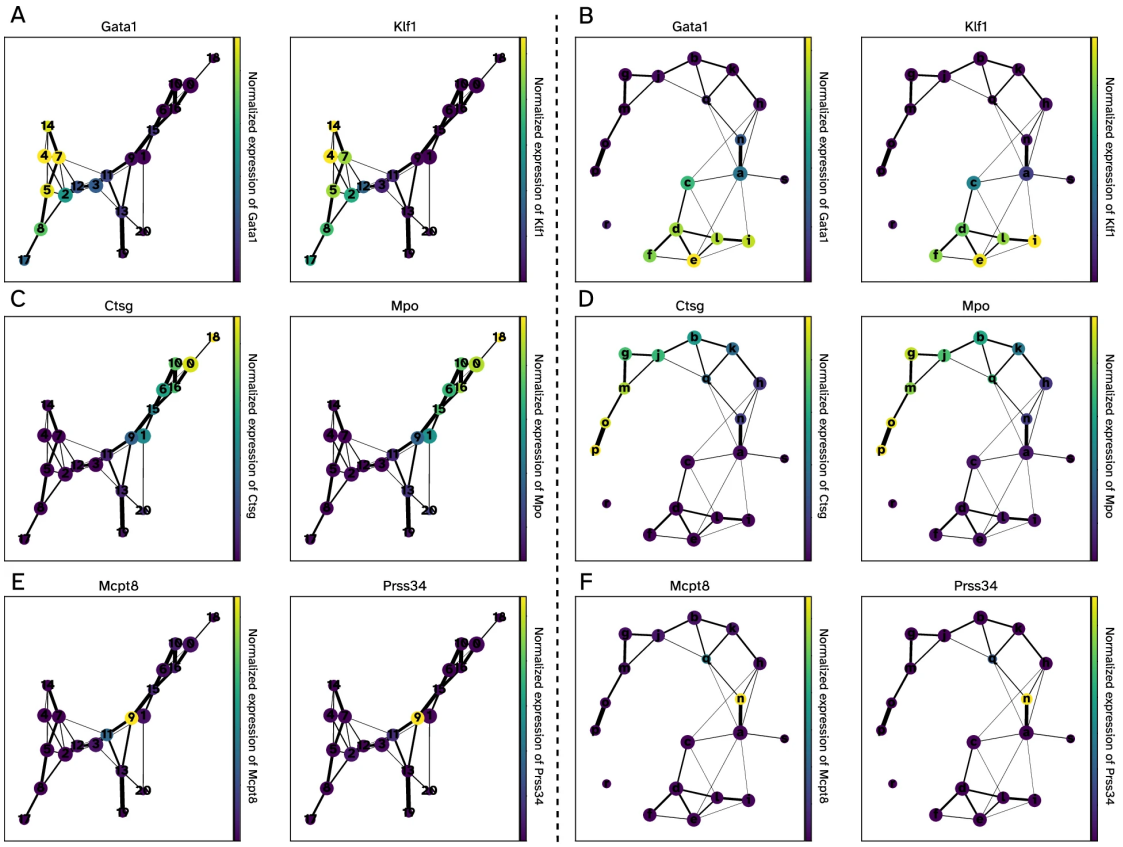


Figure 4: PAGA graphs and cell type marker gene activation patterns in experimental and simulated BoneMarrow dataset. In each figure, nodes represent Louvain clusters (capturing discrete states), edges represent transitions between states, and edge weights indicate confidence in the presence of connections. Low-connectivity edges below a threshold of 0.01 were discarded. Left panels (**A**, **C**, **E**) show the PAGA graph computed on GRouNdGAN-generated cells, and right panels (**B**, **D**, **F**) show the PAGA graph for the original BoneMarrow dataset. Node colors represent normalized expression of marker genes for erythroid cells (**A**, **B**), neutrophils (**C**, **D**), and basophils (**E**, **F**). Source data are provided as a Source Data file.

topological features to that of the reference data and captured known biological properties of hematopoiesis, which were also noted by Wolf et al.[38] using the reference dataset (e.g., proximity between monocytes and neutrophils or association between megakaryocyte and erythroid progenitors). These observations show that the key characteristics of the reference data are retained in GRouNdGAN-simulated data.

We used the diffusion pseudotime algorithm[39] (in scanpy[45]) to infer the progression of cells through geodesic distance (Fig. 5C, D). We noticed strong concordance when examining pseudotime values assigned to cell subgroups in the reference and GRouNdGAN-generated data. To track these patterns, we focused on marker expression changes along erythroid, neutrophil, and monocyte trajectories (Supplementary Figs. 11–13), similar to the analysis performed by Wolf et al.[38]. In both datasets, the activation of neutrophils' marker, *Elane*, and monocytes' marker, *Irf8*, were predominantly observed towards the later stages of their trajectories (Fig.

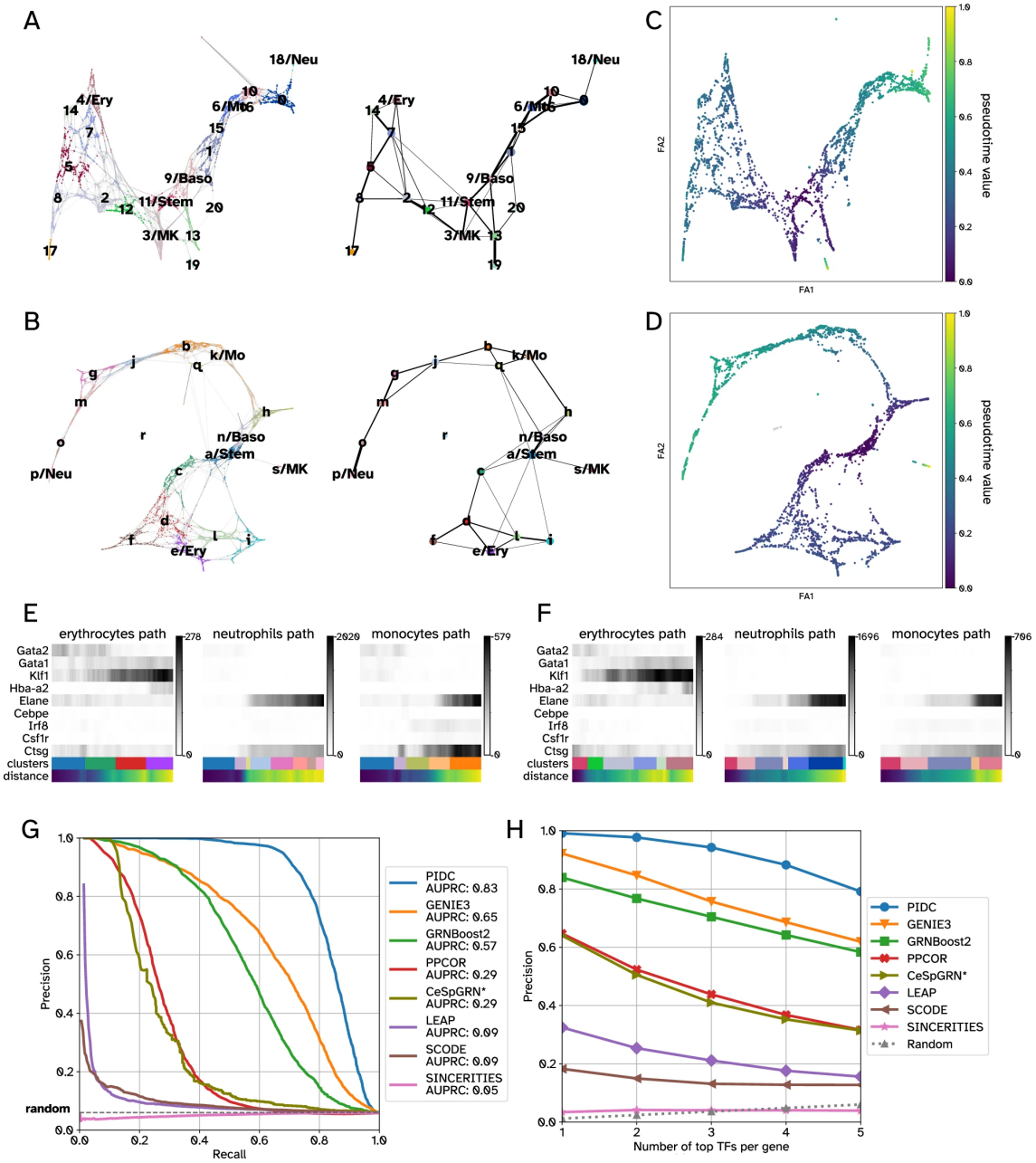


Figure 5: Consistency in trajectories and pseudo-temporal orderings between data generated by GRouNdGAN and the experimental BoneMarrow hematopoietic dataset. Panels **A** and **B** show the cell type annotations of simulated and experimental data, respectively. The figures on the left show the force-directed graphs (obtained using ForceAtlas2[66]) and the figures on the right show the PAGA graphs. Panels **C** and **D** show the PAGA-initialized force-directed graphs with cells colored by their inferred pseudo-time value. The following abbreviations are used: Stem for stem cells, Ery for erythroid cells, Neu for neutrophils, Mo for monocytes, MK for Megakaryocytes, and Baso for Basophils. Panels **E** and **F** display expression changes in the marker genes of erythrocyte (*Gata2*, *Gata1*, *Klf1*, and *Hba-a2*), neutrophil (*Elane* and *Cebpe*), and monocyte (*Irf8*, *Csf1r*, and *Ctsg*) branches for experimental (**E**) and simulated data (**F**). The distance refers to the geodesic distance from a root cell (here a stem cell). A darker shade of gray shows higher expression. Panels **G** and **H** show the performance of different GRN inference methods on data generated by GRouNdGAN using the BoneMarrow dataset. The benchmarking results are based on a dataset with $n = 2230$ simulated cells (except for CeSpGRN). Source data are provided as a Source Data file. * Due to the high memory requirement of CeSpGRN, we were only able to run it on a subset of data containing $n = 1000$ cells and $m = 100$ genes (the set of regulating TFs and the edges connecting them to these genes remained unchanged).

Table 3: Performance of GRouNdGAN in generating realistic scRNA-seq data using the Dahlin, Tumor-All, and Tumor-malignant datasets

	Dahlin		Tumor-All		Tumor-malignant	
	Training	Testing	Training	Testing	Training	Testing
Cosine distance	0.00011	0.00012	0.00141	0.00061	0.00059	0.00098
Euclidean distance	51	53	227	145	134	174
MMD	0.014	0.026	0.020	0.021	0.014	0.015
RF AUROC	0.52	0.54	0.53	0.52	0.53	0.57
miLISI	1.91	1.91	1.90	1.89	1.90	1.89

Each gene in the imposed GRN of GRouNdGAN is regulated by 15 TFs (constructed using GRNBoost2 from the experimental training set). For the first three metrics, a value closer to zero is preferred, for RF AUROC a value closer to 0.5 is preferred, and for miLISI a value closer to 2 is preferred. The metrics are calculated between a simulated dataset of 1000 cells and a set of 1000 real cells (all cells in the test set and 1000 randomly selected cells from the training set). For the first two metrics, the values correspond to the distance of the mean centroids of the real and simulated cells.

5E, F). Additionally, we observed the activation of erythroid maturity marker genes *Gata2*, *Gata1*, *Klf1*, and *Hba-a2* roughly in sequential order along the erythroid trajectory in both datasets (Fig. 5E, F), concordant with previous findings[38]. The striking similarity between the activation patterns of these markers in both datasets and the preservation of their ordering highlights GRouNdGAN's ability to capture dynamic transcriptional properties of scRNA-seq data, leading to correct trajectory inference and pseudo-time ordering.

GRouNdGAN achieves high performance on other datasets

In addition to the datasets discussed earlier, we applied GRouNdGAN to three other datasets. One corresponded to cells undergoing hematopoiesis (similar to the BoneMarrow dataset), but with much larger number of cells ($n = 44{,}802$)[46] (which we called Dahlin dataset). The second dataset corresponded to malignant cells and cells present in the tumor microenvironment of 20 fresh biopsies from follicular lymphoma tumors ($n = 136{,}147$), which we called the Tumor-All dataset[47]. We also formed a dataset containing only the malignant cells present in this dataset ($n = 89{,}203$ cells), which we called the Tumor-malignant dataset. Table 3 and Supplementary Data 1 - Sheet 2 show the performance of GRouNdGAN in generating realistic scRNA-seq data, while Supplementary Figs. 18–20 show the tSNE plots of simulated and real cells. For all three datasets, different metrics show that the generated data has a high degree of similarity to experimental data and there is a small difference between the training and test set performances.

We next repeated the trajectory inference, pseudo-time ordering, and analysis of cell type marker expression along the pseudo-time axis on the Dahlin dataset using both GRouNdGAN-simulated data ($n = 43{,}802$ cells) and real cells ($n = 43{,}802$), as we had done using the BoneMarrow dataset

(Supplementary Fig. 21). Similar to the results from the BoneMarrow dataset, we observed a high degree of concordance between the activation patterns of cell type markers in the simulated and real datasets and the preservation of their orderings. Taken together, these results suggest that GRouNdGAN can be used to generate realistic data from different biological contexts, while imposing a causal GRN and preserving various biological properties.

Benchmarking GRN inference methods using GRouNdGAN confirms prior insights from curated and experimental datasets

The BEELINE study[19] found conflicting results when benchmarking GRN inference methods on curated benchmarks and synthetic ones: top two performing models on synthetic data where among the bottom three on curated benchmarks. We re-investigated this using GRouNdGAN-simulated data with the BoneMarrow dataset and the positive and negative control GRNs (described earlier). We used seven GRN inference methods used by BEELINE, GRNBoost2[8], GENIE3[9], PIDC[10], PPCOR[37], LEAP[12], SCODE[13], and SINCERITIES[14], capturing a wide range of methods with a broad spread of performances reported on synthetic and curated benchmarks in the BEELINE study. We also included CeSpGRN[48], which is a cell-specific GRN inference method. We observed a striking resemblance between the performance pattern (order) of the methods on GRouNdGAN-simulated data and curated benchmarks from BEELINE (Fig. 5G, H and Supplementary Data 3 – Sheet 3 versus Fig. 4 of BEELINE[19]): the only difference was the swapping of LEAP and SCODE's order. We repeated this analysis using the PBMC-CTL dataset (Supplementary Data 3 – Sheet 2), obtaining similar patterns. We also tested these methods using the PBMC-All dataset (Supplementary Data 3 – Sheet 4). While the results were generally consistent, the order of GENIE3 and GRNBoost2 were swapped, which may be due to their ability in working with datasets containing multiple distinct cell types (also see Discussion).

Overall, PIDC[10] outperformed all methods using both datasets, followed by GENIE[39] and GRNBoost[28]. LEAP[12], SCODE[13] and SINCERITIES[14] (which use pseudo-time and could only be applied to the BoneMarrow dataset) performed worse than others, matching their behavior in the BEELINE study on curated benchmarks (but not on synthetic benchmarks). All methods performed close to random on unimposed negative control GRN edges. These results not only show that GRouNdGAN can be used for benchmarking GRN inference methods, but also shows that the insights obtained from it matches those obtained from curated and experimental benchmarks (such as those used in Figs. 4–6 of the BEELINE study), the formation of which requires extensive amount of resources and effort.

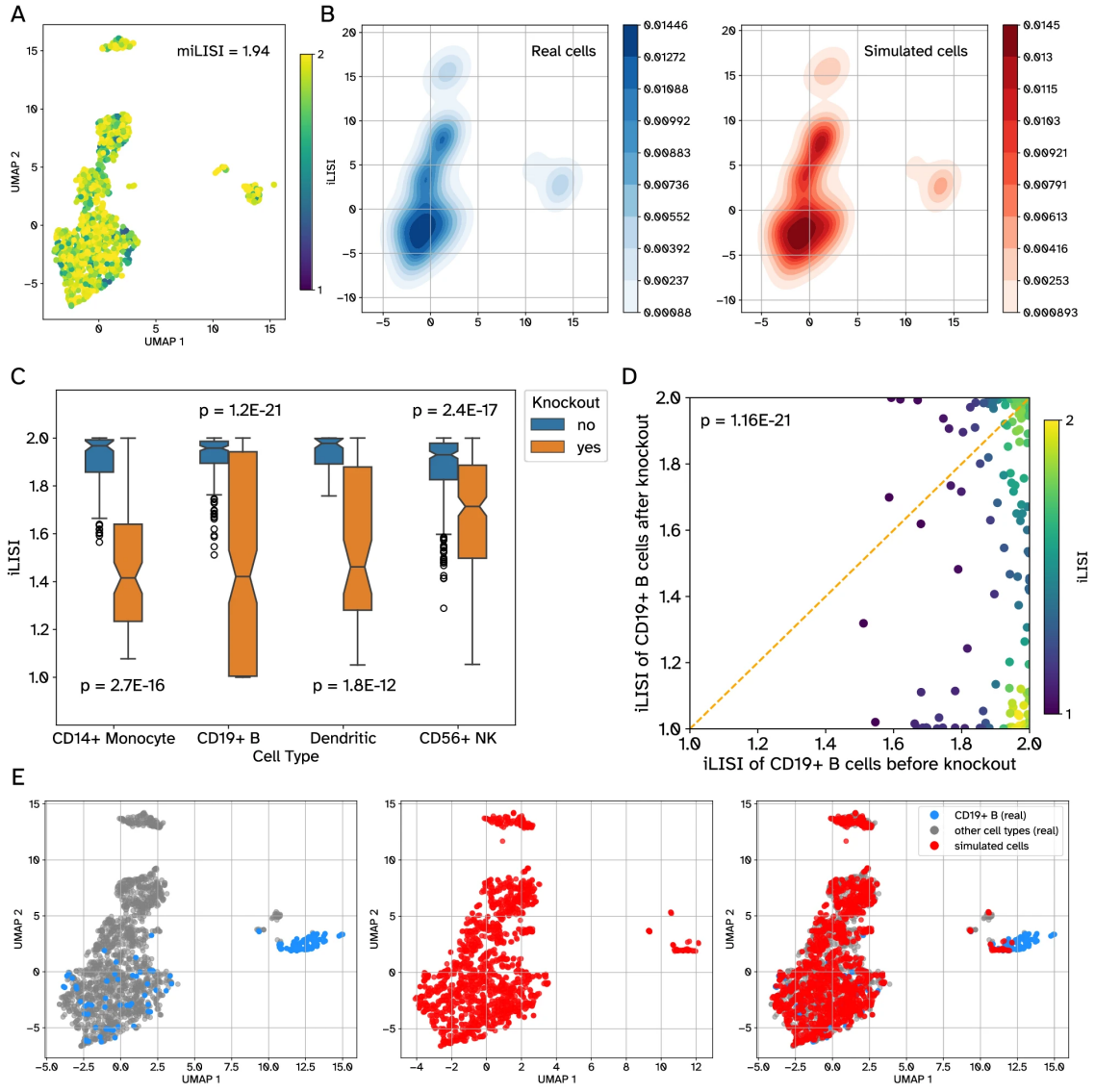


Figure 6: TF knockout experiments and their effect on cell type generation. **A** The UMAP shows the distribution of $n = 2000$ randomly selected cells from the experimental PBMC-All dataset. The color shows the iLISI value of each cell (miLISI = 1.94) calculated from a UMAP embedding, jointly obtained from the experimental cells and the same number of GRouNdGAN-generated cells. TFs were omitted as features when generating UMAP plots. In the simulated data, each gene was regulated by 15 TFs, identified using GRNBoost2. **B** Each plot shows the density of cells in the embedding space: the left plot (blue) corresponds to the real cells and the right plot (red) shows the simulated cells. **C** The boxplots show the distribution of iLISI values of $n = 2000$ real cells, calculated along with $n = 2000$ unperturbed simulated cells (blue) and with $n = 2000$ perturbed simulated cells (orange). For each cell type, top three most differentially expressed TFs were knocked out. The p-values reported in this figure were calculated using one-sided Wilcoxon signed rank tests. The center horizontal line corresponds to the median. Each box spans the interquartile range, and its limits show the upper (Q3) and lower quartiles (Q1). Whiskers extending from the boxes reach 1.5 times above and below the interquartile range. Outliers are shown individually if they fall outside the whiskers. **D** The scatter plot shows the iLISI values of CD19+B cells calculated along with unperturbed simulated cells (x-axis) and along with perturbed simulated cells (y-axis). The circles correspond to real cells and their colors reflect the density of datapoints in that region. The p-value reported in this figure was calculated using a one-sided Wilcoxon signed rank test. **E** The UMAP embedding of real cells (left), simulated cells after knockout of top TFs of CD19+B cells (middle), and all together (right). Source data are provided as a Source Data file.

In silico perturbation experiments using GRouNdGAN

In many studies, one is interested to characterize the relationship between TFs' expression and phenotypic labels (e.g., cell types). Differential expression (DE) analysis is a common approach to identify TFs (or other genes) most associated with a specific cell type. However, to directly test whether perturbing the expression of such candidate TFs change the expression of genes representative of a cell type, perturbation experiments (e.g., TF knockout) are desired. As a causal implicit generative model, GRouNdGAN provides the capability to sample from interventional distributions, thereby enabling in silico perturbation experiments. This functionality is provided at inference time by enabling manipulation of TFs' expressions as the outputs of the causal controller. This change is propagated along the GRN (through the model architecture), impacting the expression of genes regulated by the perturbed TF(s). By enabling a deterministic mode of operation, GRouNdGAN allows for exact comparison of gene expressions before and after perturbation while maintaining the invariance of other parameters (Methods).

To test this functionality, we asked whether the knockout of top 3 TFs most differentially expressed (Mann Whitney U test) between each cell type and the rest (identified from the reference dataset) would result in less cells being generated in the vicinity of the cells of that type. If positive, this implies that the perturbation in the TFs' expression has modified the gene expression profile of generated cells so that they no longer resemble that particular cell type (confirming our expectation). Figure 6A shows the UMAP embedding of the reference dataset (miLISI = 1.94 when compared with simulated data) and Fig. 6B shows the density plots of the experimental and simulated data, revealing a high degree of resemblance. We focused on the changes in the iLISI values of experimentally profiled cells of a specific type (annotated in the original study), when jointly mapped to the same embedding space with simulated data, before and after knockout. For the changes in the iLISI values to be observable, it was necessary to focus on cell types that occupy relatively distinct regions of the embedding space: CD14+ monocytes, CD19+ B cells, Dendritic cells, CD56+ natural killer cells (Supplementary Fig. 22); otherwise, the presence of other cell types in that region would confound the results.

Figure 6C shows the iLISI distributions for each cell type (only for real cells), calculated along with unperturbed simulated cells (blue) and perturbed simulated cells (orange). Figure 6D, E show the iLISI values and the distribution of generated cells after the knockout experiment for CD19+ B cells as an example (see Supplementary Figs. 23–26 for other cell types). In all cases, the iLISI values significantly reduced after the knockout experiment (Fig. 6C, Supplementary Table 7), showcasing that after the knockout of top TFs associated with a cell type, GRouNdGAN generated much fewer cells resembling that cell type (even though it did not have the knowledge of the cell type annotations). Instead, the simulated cells were dispersed into other cell types, yet they retained meaningful positions within the overall dataset embedding space. Also, the

miLISI of the other cell types remained relatively unchanged (Supplementary Table 7).

Next, we repeated the analyses above using Tumor-All dataset that contained different cell types including malignant cells, but also other cell types present in the tumor microenvironment. We considered cell types that made up of at least 0.25% of the dataset (Supplementary Table 8). When cells of two or more types populated the same regions of the embedding space, we kept the cell type containing more cells. Malignant cells, T cells, B cells, and plasma cells satisfied the conditions above, which we included in our analyses. Similar to the previous analyses, we observed that the KO of top 3 TFs of each cell type results in a significant reduction in the iLISI values (Supplementary Fig. 27, Supplementary Table 8).

These analyses suggest that in silico TF perturbation experiments using GRouNdGAN produces results concordant with results directly obtained from real biological data, a direction that we will explore further in the future studies.

Discussion

GRouNdGAN is a causal implicit generative model designed to simulate realistic scRNA-seq datasets based on a user-defined GRN and a reference dataset. By incorporating GRN connections in its architecture and including auxiliary tasks, it imposes causal TF-gene relationships in the gene expression profile of simulated cells. These causal relationships are verifiable by in silico knockout experiments and identifiable by GRN inference methods. (In addition to the experimental analyses performed in this study, an interested reader should refer to Kocaoglu et al.[31] for theoretical evidence of causality of data generated by this architecture.)

We demonstrated that GRouNdGAN achieves state-of-the-art performance on various tasks including realistic scRNA-seq data generation, GRN inference benchmarking, and in silico knockout experiments. Even when applied to heterogenous datasets, it generated realistic datasets without utilizing information regarding cell clusters or cell types, achieving best or second-best performance compared to models that used this information. We should note that it is trivially possible to use cell type/cluster information with GRouNdGAN: one can simply provide subsets of the experimental dataset corresponding to a cell type/cluster as a reference, one at a time, with a cell-type specific (or shared) GRN (similar to our analyses using PBMC-CTL and PBMC-NaiveT in Supplementary Data 2 – Sheet 2). However, the fact that this model does not require this information broadens its applicability. We showed that GRouNdGAN-generated data preserves complex cellular dynamics and patterns of the reference dataset such as lineage trajectories, pseudo-time orderings, and patterns of cell type markers' activation. One consideration in generating realistic and generalizable cells is the number of training examples. In the BoneMarrow dataset (which had only 2.7k cells), the performance of all simulators

dropped on the test set compared to the training set. When we generated cells using another similar hematopoiesis dataset (Dahlin), but with much larger number of cells (45k), the testing and training performance difference significantly reduced (Table 3). Similarly, the training and testing performance on all the other datasets that had more than ten thousand cells remained on par with each other, suggesting that a large number of cells improves the realism of the simulated data (which is expected). In spite of this, the cells generated by GRouNdGAN could capture trajectories and pseudo-time orderings not only in the larger Dahlin dataset, but also in the smaller BoneMarrow dataset.

GRouNdGAN imposes user-defined causal regulatory interactions, while excluding those present in the reference dataset, but not in the input GRN. This makes GRouNdGAN-simulated datasets ideal for benchmarking GRN inference algorithms with the input GRN as the ground truth. We benchmarked eight GRN inference algorithms on the simulated BoneMarrow dataset and showed that our results coincide with results reported by BEELINE[19] on curated and experimental benchmarks. This highlights GRouNdGAN's ability to bridge the gap between experimental and simulated datasets for GRN benchmarking without requiring excessive user manipulation. We assessed five of these methods (that did not require pseudo-time) on the PBMC-CTL dataset and observed that their ordering was consistent with the simulated BoneMarrow data. We then applied them to the PBMC-All dataset, containing multiple distinct cell types. While the general trend was consistent with the other benchmarks, the order of GENIE3 and GRNBoost2 switched. This could be related to the ability of these methods on handling distinct cell types. One should also note that the results of benchmarking on PBMC-All is not conclusive, and follow-up studies are required to fully characterize the effect of distinct cell types on the performance of GRN inference methods. In particular, GRNs are cell type- and context-specific, and such a study should impose different GRNs on different subpopulations of the dataset. Due to the flexibility provided by GRouNdGAN, one can evaluate a variety of GRN inference methods including supervised methods[49,50] or cell-specific GRN inference methods[48,51]. We included CeSpGRN[48] in our benchmarks, but a comprehensive study that evaluates this and other methods of this class from different perspectives is of great interest.

Since GRouNdGAN generates data based on a structural causal model, it can be used to sample synthetic cells from both observational and interventional distributions. Using this property, we conducted in silico perturbation experiments by knocking out key regulating TFs in various cell types and observed that the targeted cell-states were suppressed in the generated dataset. These results, while preliminary, show potential for developing a method based on GRouNdGAN to predict the results of TF knockout experiments. However, such method would inevitably need access to a high quality GRN underlying the real data, requiring additional data modalities such as (sc)ATAC-seq or TF ChIP-seq.

GRouNdGAN maintains key advantages over existing simulators that incorporate GRNs such

as SERGIO and BoolODE. BoolODE, unlike GRouNdGAN, is a reference-free (de novo) and rule-based simulator that relies on user-defined parameters instead of a reference dataset to artificially introduce properties of single-cell data (e.g., dropout, batch effect, existence of distinct populations, and variation among experimental samples and conditions). As a rule-based simulator, it allows for flexible selection of simulation parameters such as choice of noise strength, dropout rate, or kinetic parameters (for mRNA transcription, protein translation and mRNA and protein degradation rates). However, extensive user knowledge is required to leverage this flexibility to avoid generating unrealistic datasets and recent studies have argued that one should use discretion when interpreting results obtained from such de novo simulations[52]. This may explain why some of GRN inference methods that performed best on BoolODE's synthetic benchmarks, were among the worst on the experimental benchmarks of BEELINE. Moreover, as a reference-free simulator, BoolODE is not designed to match a specific reference dataset. As a result, it does not lend itself to benchmarking GRN inference methods in distinct contexts characterized by experimental reference datasets.

On the other hand, SERGIO is a hybrid method that contains elements of reference-free simulators, but also enables matching the simulated cells to a specific reference dataset by adding technical noise. SERGIO simulates data by first generating a "clean" dataset based on SDEs. Then, the user would iteratively fine-tune three noise parameters until the simulated and reference datasets are matched. However, this data-matching procedure puts extra burden on the user and might lead the user to overly rely on fine-tuning technical noise parameters, instead of SDE parameters. Our experiments with SERGIO (described in Supplementary Notes) showed that in spite of our best efforts for distribution matching, there were clear differences between the SERGIO-simulated data and the reference dataset. Another consideration is that over relying on added technical noise to achieve data-matching can potentially corrupt the data, rendering the underlying regulatory patterns unobservable in the final simulated dataset. This might explain the near-random GRN inference performance of various methods reported by SERGIO on their noisy datasets[4].

In contrast to BoolODE and SERIGO, during training, GRouNdGAN's parameters are optimized to minimize any mismatch between real and simulated data distributions automatically. This enables it to replicate scRNA-seq data characteristics (e.g., technical and biological noise) without any user intervention or post-processing. Moreover, GRouNdGAN is trained such that each of its generated genes or TFs mimics the expression patterns of a corresponding gene or TF in the reference dataset, preserving gene/TF identities. In contrast, BoolODE and SERGIO simulate pseudo-genes whose identities cannot be readily mapped to their counterparts in the reference dataset. Gene identity preservation can be attained in these simulators by fine-tuning the SDE parameters until the expression patterns of a given generated gene aligns with a corresponding gene in the reference dataset. This, however, places further burden on the user. Finally,

GRouNdGAN learns complex non-linear co-regulatory patterns from the data, instead of relying on simplifying assumptions or user input. In contrast, SERGIO assumes that the combined effect of multiple regulatory TFs is simply the sum of their individual effects. Similarly, BoolODE requires the user to provide a truth table specifying combinatory TF co-regulation rules for each target gene. In contrast, GRouNdGAN's neural network-based generator enables it to implicitly learn complex non-linear co-regulatory patterns that may represent co-expression, co-inhibition, or co-activation of individual genes. This is achieved during the training of GRouNdGAN without user intervention.

Several key points need to be considered when using GRouNdGAN. First, our analyses showed that the architectural and hyperparameter choices of the presented model result in good performance across different datasets. We performed hyperparameter tuning on one dataset and used the same set of values for all datasets and for different choices of GRNs, obtaining state-of-the-art performance. However, when trained on a completely new dataset, or datasets containing significantly different number of genes and TFs, it is advisable to try different choices of the hyperparameters using a validation set. Second, the architecture of the model includes LSN layers and, as such, the model generates expression values library-size normalized across the considered genes and TFs. Our ablation study (Supplementary Notes) and our experiments showed that the LSN layer improves the GRN imposition, the realism of the generated samples, and stabilizes the training (an observation also made by the scGAN study). In the future iterations of the model, we are planning to remove this restriction to enable generating unnormalized count data. Third, an important consideration for any causal inference problem is that given an observational dataset, its underlying causal graph is not necessarily unique. When benchmarking GRN inference methods, this problem exists regardless of how the ground truth GRN is obtained (even using knockout/knockdown experiments), and a GRN inference algorithm may find a plausible causal GRN, yet not the one that generated the data. Despite this, our analysis using the positive/negative control GRNs showed that GRN inference methods could detect the imposed edges while ignoring others that were unimposed. Based on this observation, we posit that the imposed GRN by GRouNdGAN, while not the only causal GRN capable of generating the simulated dataset, is the most probable one. However, verifying this requires theoretical analyses that are well beyond the scope of this study. Fourth, GRouNdGAN is a generative model and is not a predictive model. As such, its goal is to simulate cells similar to a reference training set (while imposing a user-defined GRN), and not to predict the scRNA-seq profile of cells in a new context or an independent dataset. Given a new dataset, the model needs to be retrained to generate realistic samples.

Finally, although GRouNdGAN can generate data with any GRN, the resemblance of the data to the reference dataset deteriorates if the imposed TF-gene relationships are significantly different from those consistent with the patterns in the reference dataset. The reason is that in such a

scenario, generating simulated datapoints resembling the reference training dataset and imposing the GRN act as contradictory requirements. This is a fundamental problem that is not unique to GRouNdGAN and is applicable to any simulator with these two requirements. However, simulators that do not try to impose a causal graph (e.g., scGAN) can focus on generating realistic scRNA-seq samples, even at the expense of disrupting some of the interactions present in the reference dataset (as we observed in Fig. 3 with scGAN). As a result, for GRN inference method benchmarking, we recommend first finding a set of regulatory edges from the reference dataset (e.g., using GRNBoost2 or even other data modalities) and imposing a selected subset of those in the simulated data. This ensures that the imposed edges are encoded in the simulated data, the unimposed (or undetected) edges are disrupted (as shown in our analyses), and the simulated dataset resembles the reference dataset. It is important to note that the regulatory edges inferred from the reference dataset do not need to be causal (and most likely, some are not); some of them could represent spurious correlations that are observable in the data. In fact, most GRN inference methods do not identify causal edges, yet can be used to form the input GRN. When a subset of the (potentially) non-causal edges of the reference dataset are imposed by GRouNdGAN, they are imposed in a *causal* manner, representing the causal graph underlying the *simulated* data (and not the reference data).

The current version of GRouNdGAN only allows imposing a bipartite GRN and does not support a multi-layer GRN, an assumption that we plan to relax with the future versions of the model. Moreover, we intend to augment the model's architecture in the future to enable imposing a causal graph with three layers of nodes capturing TF-gene-phenotype relationships. This is particularly useful for development and evaluation of phenotype-relevant GRNs, a concept that we have introduced in the past[11]. Additionally, a conditional version of GRouNdGAN can be developed, allowing the user to generate cells of specific cell type or on a specific stage along a lineage trajectory. Finally, by incorporating DAG structural learning[53], we intend to enable GRouNdGAN to infer a GRN while simultaneously learning to mimic the reference dataset.

Methods

Datasets and preprocessing

We downloaded the human peripheral blood mononuclear cell (PBMC Donor A) dataset containing the single-cell gene expression profiles of 68579 PBMCs represented by UMI counts from the 10x Genomics (`https://support.10xgenomics.com/single-cell-gene-expression/datasets/1.1.0/fresh_68k_pbmc_donor_a`). This dataset contains a large number of well-annotated cells and has been used by other models to gener-

ate synthetic scRNA-seq samples[24]. We also downloaded the scRNA-seq profile of 2730 cells corresponding to differentiation of hematopoietic stem cells to different lineages from mouse bone marrow[34] ("BoneMarrow" dataset) from Gene Expression Omnibus (GEO) (accession number: GSE72857). We also obtained another haematopoietic dataset corresponding to the scRNA-seq (10x Genomics) profiles of 44,802 mouse bone marrow hematopoietic stem and progenitor cells (HSPCs) differentiating towards different lineages from GEO (accession number: GSE107727)[46] (called Dahlin dataset here). Finally, we obtained the batch-corrected scRNA-seq (10x Genomics) profile of 136,147 cells corresponding to malignant cells as well as cells in the tumor microenvironment (called Tumor-All dataset here) from 20 fresh core needle biopsies of follicular lymphoma patients from (https://cellxgene.cziscience.com/collections/968834a0-1895-40df-8720-666029b3bbac)[47].

We followed pre-processing steps similar to those of scGAN and cscGAN[24] using scanpy version 1.8.2[45]. In each dataset, cells with nonzero counts in less than ten genes were removed. Similarly, genes that only had nonzero counts in less than three cells were discarded. Top 1000 highly variable genes were selected using the dispersion-based method described by Satija et al.[54]. Finally, library-size normalization was performed on the counts per cell with a library size equal to 20,000 in order to be consistent with previous studies[24]. See Supplementary Data 1 - Sheet 3 for the number of cells and highly variable genes and TFs present in each final dataset.

GRouNdGAN's model architecture

GRouNdGAN's architecture consists of 5 components, each implemented using separately parameterized neural networks: a causal controller, target generators, a critic, a labeler, and an anti-labeler (Fig. 1C).

Causal controller

The role of causal controller is to generate the expression of TFs that causally control the expression of their target genes based on a user-defined gene regulatory network (Fig. 1C). To achieve this, it is first pre-trained (Fig. 1B) as the generator of a Wasserstein GAN with gradient penalty (WGAN-GP) (see Supplementary Notes for the formulation of the Wasserstein distance). GANs are a class of deep learning models that can learn to simulate non-parametric distributions[55]. They typically involve simultaneously training a generative model (called the "generator") that produces new samples from noise, and its adversary, a discriminative model that tries to distinguish between real and generated samples (called the "discriminator"). The generator's goal is to generate samples so realistic that the discriminator cannot determine whether it is real or simulated (an accuracy value close to 0.5). Through adversarial training, the generator and

discriminator receive feedback, allowing them to co-evolve in a symbiotic manner.

The main difference between a WGAN and a traditional GAN is that in the former, a Wasserstein distance is used to quantify the similarity between the probability distribution of the real data and the generator's produced data (instead of Kullback–Leibler or Jensen–Shannon divergences). Wasserstein distance has been shown to stabilize the training of WGAN without causing mode collapse[32]. The detailed formulation of the Wasserstein distance used as the loss function in this study is provided in Supplementary Notes. In addition, instead of a discriminator, WGAN uses a "critic" that estimates the Wasserstein distance between real and generated data distributions. In our model, we added a gradient penalty term for the critic (proposed by Gulrajani et al.[56] as an alternative to weight clipping used in the original WGAN) in order to overcome vanishing/exploding gradients and capacity underuse issues.

In the pretraining step, we trained a WGAN-GP with a generator (containing an input layer, three hidden layers, and an output layer), a library-size normalization (LSN) layer[24], and a critic (containing an input layer, three hidden layers, and an output node). A noise vector of length 128, with independent and identically distributed elements following a standard Gaussian distribution, was used as the input to the generator. The output of the generator was then fed into the LSN layer to generate the gene and TF expression values. The details of hyperparameters and architectural choices of this WGAN-GP are provided in Supplementary Table 1. Although we were only interested in generating expression of TFs using the generator of this WGAN-GP (in the second step of pipeline), the model was trained using all genes and TFs to properly enforce the library-size normalization. Once trained, we discarded the critic and the LSN layer, froze the weights of the generator and used it as the "causal controller"[31] to generate expression of TFs (Fig. 1C).

Target generators

The role of target generators is to generate the expression of genes causally regulated by TFs based on the topology of a GRN. Consider a target gene G_j regulated by a set of $TFs = \{TF_1, TF_2, ..., TF_n\}$. Under the causal sufficiency assumption and as a result of the manner by which TFs' expressions are generated from independent noise variables, we can write $E_{G_j} = f_{G_j}(E_{TF_1}, E_{TF_2}, ..., E_{TF_n}, N_{G_j})$ and $E_{TF_i} = f_{TF_i}(N_{TF_i})$ for $i = 1, 2, ..., n$, where E represents expression, N represents a noise variable, and f represents a function (to be approximated using neural networks). All noise variables are jointly independent. Following the theoretical and empirical results of CausalGAN[31], we can use feedforward neural networks to represent functions f by making the generator inherit its neural network connections from the causal GRN. To achieve this, we generate each gene in the GRN by a separate generator such that target gene generators do not share any neural connections (Fig. 1C).

As input, the generator of each gene accepts a vector formed by concatenating a noise variable and a vector of non-library size normalized TF expressions from the causal controller, corresponding to the TFs that regulate the gene in the imposed GRN (i.e., its parents in the graph). The expression values of TFs and the generated values of target genes are arranged into a vector, which is passed to an LSN layer for normalization. We used target generators with three hidden layers of equal width. The width of the hidden layers of a generator is dynamically set as twice the number of its regulators (the noise variable and the set of regulating TFs). If the imposed GRN is relatively dense and contains more than 5000 edges, we set the depth to 2 and the width multiplier to 1 to be able to train on a single GPU. The details of hyperparameters and architectural choices are provided in Supplementary Table 2.

In practice, generating each target gene's expression using separate neural networks introduces excess overhead. This is because in every forward pass, all target genes' expressions must be first generated before collectively being sent into the LSN layer. As a result, instead of parallelizing the generation of each target gene's expression (which due to the bottleneck above does not provide a significant computational benefit), we implemented target generators using a single large sparse network. This allows us to reduce the overhead and the training time to train the model on a single GPU, and to benefit from GPU's large matrix multiplication. We mask weights and gradients to follow the causal graph, while keeping the generation of genes independent from each other. From a logical standpoint, our implementation has the same architecture described earlier, but is significantly more computationally efficient. See Supplementary Notes for details.

Critic

Similar to a traditional WGAN, the objective of GRouNdGAN's critic (Fig. 1C) is to estimate the Wasserstein distance between real and generated data distributions. We used the same critic architecture as the WGAN-GP trained in the first stage.

Labeler and anti-labeler

Although the main role of the target generators is to produce realistic cells to confuse the critic, it is crucial that they rely on the TFs' expression (in addition to noise) in doing so. One potential risk is that the target generators disregard the expression of TFs and solely rely on the noise variables. This is particularly probable when the imposed GRN does not conform to the underlying gene expression programs of real cells in the reference dataset; in such a scenario, and to make realistically looking simulated cells, it is more convenient for a WGAN to simply ignore the strict constraints of the GRN and solely rely on noise.

To overcome this issue, we used auxiliary tasks and neural networks known as "labeler" and "anti-

labeler"[31]. The task of these two networks is to estimate the causal controller's TF expressions (here called labels) from the target genes' expressions alone, by minimizing the squared L2 norm between each element's TF estimates and their true value. More specifically, the corresponding loss for a batch of size N_{batch} is of the form $\frac{1}{N_{batch}} \sum_{i=1}^{N_{batch}} ||\hat{y}_i - y_i||_2^2$, where $\hat{y}_i$ is the estimated vector of TF expression values generated by the labeler or anti-labeler. For anti-labeler, y_i corresponds to the TF expression values outputted by the causal controller; for labeler, this vector can also correspond to TF expression values from the real training data. This resembles the idea behind an autoencoder and ensures that the model will not disregard the expression of TFs in generating the expression of their target genes. The anti-labeler is trained solely based on the outputs of the target generators, while the labeler utilizes both the outputs of target generators and the expression of real cells (Fig. 1C). They are both implemented as fully connected networks with a width of 2000 and a depth of 3 and optimized using the AMSGrad[57] algorithm. Each layer, except the last one, utilizes a ReLU activation function and batch normalization. In addition to WGAN-GP losses, we add labeler and anti-labeler losses to the generator to minimize both. This is different from the approach used in CausalGAN, where the anti-labeler's loss is maximized in the early training stages to ensure that the generator doesn't fall into label-conditioned mode collapse. In GRouNdGAN, the causal controller is pretrained and generates continuous labels (TFs expression) and does not face a similar issue. As a result, we instead minimized the loss of the anti-labeler from the beginning and as such the labeler and anti-labeler both act as auxiliary tasks to ensure the generated gene expression values take advantage of TFs expression.

Training procedure and hyperparameter tuning

We follow a two-step training procedure comprising of training two separate WGAN-GPs (Fig. 1) to train GRouNdGAN. The generators and critics in both GANs are implemented as fully connected neural networks with rectified linear unit (ReLU) activation functions in each layer, except for the last layer of the critic. The weights were initialized using He initialization[58] for layers containing ReLU activation and Xavier initialization[59] for other layers (containing linear activations). We used Batch Normalization[60] to normalize layer inputs for each training minibatch, except for the critic. This is since using it in the critic invalidates the gradient penalty's objective, as it penalizes the norm of the critic's gradient with respect to the entire batch rather than to inputs independently. An LSN layer[24] was used in both WGAN-GPs (Fig. 1B, C) to scale counts in each simulated cell to make it consistent with the library size of input reference dataset. This normalization results in a dramatic decrease in convergence time and smooths training by mitigating the inherent heterogeneity of scRNA-seq data.

We would like to point out that both steps of training (i.e., the pretraining of Fig. 1B and training of Fig. 1C) are performed on the exact same training set, and when data corresponding to a new

dataset is to be generated, these steps need to be repeated. The goal of the first step is to train the causal controller to learn the distribution of training set and generate realistic TF expression values (without imposing TF-gene relationships). The goal of the second step is to use the TF expression values generated by the trained causal controller to generate expression of target genes while imposing the TF-gene causal relationships.

GRouNdGAN solves a min-max game between the generator (f_g) and the critic (f_c), with the following objective function:

$$\min_{f_g} \max_{\|f_c\|_L \leq 1} \mathbb{E}_{x \sim \mathbb{P}_r} f_c(x) - \mathbb{E}_{x \sim f_g(\mathbb{P}_{\text{noise}})} f_c(x) \tag{1}$$

The training objective of the critic involves maximizing the difference between the average score assigned to real ($\mathbb{E}_{x \sim \mathbb{P}_r} f_c(x)$) and generated samples ($\mathbb{E}_{x \sim f_g(\mathbb{P}_{noise})} f_c(x)$) with respect to its parameters following Eq. (1). On the contrary, the generator attempts to minimize the average score that the critic assigns to real and generated samples. Through this adversarial game, both the critic and generator co-evolve and the generator learns a mapping f_g from a simple standard Gaussian noise distribution ($\mathbb{P}_{noise}$) to a distribution $f_g(\mathbb{P}_{noise})$ that approximates the real data distribution $\mathbb{P}_r$. An important point to consider is that the critic does not directly compute the Wasserstein distance (mathematically defined in Supplementary Notes - Details regarding the Wasserstein distance). Instead, it is encouraged to provide a meaningful estimate of the Wasserstein distance between the distribution of real data ($\mathbb{P}_r$) and the distribution of generated data ($f_g(\mathbb{P}_{noise})$) through its training process.

We alternated between minimizing the generator loss for one iteration and maximizing the critic loss for five iterations. We employed the AMSGrad[57] optimizer with the weight decay parameters $\beta_1 = 0.5$, $\beta_2 = 0.9$ and employed an exponentially decaying learning rate for the optimizer of both the critic and generator.

The hyperparameters were tuned using a validation set consisting of 1000 cells from the PBMC-CTL dataset (Supplementary Tables 1 and 2) based on the Euclidean distance and the RF AUROC score, which were consistently in accord. The same hyperparameters were used for all other analyses and datasets.

Causal GRN preparation

This section describes the creation of the causal graph inputted to GRouNdGAN used to impose a causal structure on the model. GRouNdGAN accepts a GRN in the form of a bipartite directed acyclic graph (DAG) as input, representing the relationship between TFs and their target genes. In this study, we created the causal graph using the 1000 most highly variable genes

and TFs identified in the preprocessing step (Supplementary Data 1 - Sheet3). First, the set of TFs among the highly variable genes were identified based on the AnimalTFDB3.0 database[61] and a GRN was inferred using GRNBoost2[8] (with the list of TFs provided) from the training reference dataset. It is important to note that the regulatory edges identified from the reference dataset using GRNBoost2 are not necessarily causal edges (and they do not need to be for the purpose of forming the input GRN), but they are consistent with the patterns of the data. However, when this (potentially non-causal) GRN is imposed by GRouNdGAN, it is imposed in a causal manner and represents the causal data generating graph of the simulated data (and not the reference data).

Evaluation of the resemblance of real and simulated cells

We evaluated all models using held-out test sets containing randomly selected cells from each reference dataset (500 cells from BoneMarrow and 1000 cells for all other datasets) (see Supplementary Data 1 - Sheet 3 for other statistics about the datasets). To quantify the similarity between real and generated cells, we employed various metrics. For each cell represented as a datapoint in a low dimensional embedding (e.g., t-SNE or UMAP), the integration local inverse Simpson's Index (iLISI)[36] captures the effective number of datatypes (real or simulated) to which datapoints of its local neighborhood belong based on weights from a Gaussian kernel-based distributions of neighborhoods. The miLISI is the mean of all these scores and in our study ranges between 1 (poor mixing of real and simulated cells) and 2 (perfect mixing of real and simulated cells). Additionally, we calculated the cosine and Euclidean distances of the centroids of real cells and simulated cells, where the centroid was obtained by calculating the mean along the gene axis (across all simulated or real cells).

To estimate the proximity of high-dimensional distributions of real and simulated cells without creating centroids, we used the maximum mean discrepancy (MMD)[35]. Given two probability distributions p and q and a set of independently and identically distributed (i.i.d.) samples from them, denoted by X and Y, MMD with respect to a function class $\mathcal{F}$ is defined as:

$$\mathrm{MMD}\left[\mathcal{F}, X, Y\right] := \sup_{f \in \mathcal{F}} \left(\mathbb{E}_x \left[f(x) \right] - \mathbb{E}_y \left[f(y) \right] \right), \tag{2}$$

where *sup* refers to supremum and $\mathbb{E}$ denotes expectation. When the MMD function class $\mathcal{F}$ is a unit ball in a reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ with kernel k, the population MMD takes a zero value if and only if $p = q$ and a positive unique value if $p \neq q$. The squared MMD can be written as the distance of mean embeddings μ_p, μ_q of distributions p and q, which

can be expressed in terms of kernel functions:

$$\begin{aligned} \mathrm{MMD}^2[F, p, q] &= \|\mu_p - \mu_q\|_H^2 \\ &= \mathbb{E}_{x,x'}\big[k(x, x')\big] - 2\mathbb{E}_{x,y}\big[k(x, y)\big] + \mathbb{E}_{y,y'}\big[k(y, y')\big] \end{aligned} \tag{3}$$

Following existing implementations of MMD in the single-cell domain[24,62], we chose a kernel that is the sum of three Gaussian kernels to increase sensitivity of the kernel to a wider range:

$$k\left(x, y\right) = \sum\nolimits_i \left(\frac{||x - y||^2}{\sigma_i^2}\right), \; i = 1, 2, 3 \tag{4}$$

where $_i$ denote standard deviations and were chosen to be the median of the average distance between a point to its 25 nearest neighbors divided by factors of 0.5, 1, and 2 in the three kernels, respectively.

We also used a random forests (RF) classifier and used its area under the receiver operating characteristic (AUROC) curve to determine whether the real and simulated cells can be distinguished from each other. Consistent with previous studies[24,63], we first performed a dimensionality reduction using principal component analysis (PCA) and used the top 50 PCs of each cell as the input features to the RF model, which improves the computational efficiency of this analysis. The RF model was composed of 1000 trees and the Gini impurity was used to measure the quality of a split.

Baseline simulator models

We compared the performance of GRouNdGAN to scDESIGN2[25], SPARsim[26], and three GAN-based methods: scGAN[24], cscGAN with projection-based conditioning[24], and a conditional WGAN (cWGAN). The cWGAN method conditions by concatenation following the cGAN framework[64]. More specifically, it concatenates a one-hot encoded vector (representing the cluster number or cell type) to the noise vector input to the generator and cells forwarded to the discriminator. We did not train the cWGAN or cscGAN on the PBMC-CTL dataset, since it contains only one cell type. For the PBMC-All and the BoneMarrow dataset, we trained all models above. Additionally, we simulated data using scDESIGN2 and SPARsim with and without cell cluster information, as they allow providing such side information in their training.

To train models that utilized cell cluster information, we performed Louvain clustering and provided the cluster information and ratio of cells per cluster during training. Clustering was done by following the cell ranger pipeline[33], based on the raw unprocessed dataset (and independent of the pre-processing steps described earlier for training simulators). First, genes with no UMI count in any of the cells were removed. Then the gene expression profile of each

cell was normalized by the total UMI of all (remaining) genes, and highly variable genes were identified. The gene expression profile of each cell was then re-normalized by the total UMI of retained highly variable genes, and each gene vector (representing its expression across different cells) was z-score normalized. Given the normalized gene expression matrix above, we found top 50 principal components (PCs) using PCA analysis. These PCs were then used to compute a neighborhood graph with a local neighborhood size of 15, which was used in Louvain clustering. We ran the Louvain algorithm with a resolution of 0.15.

For SPARSim, we set all sample library sizes to 20000 and estimated gene expression level intensities and simulation parameters by providing it with both raw and normalized count matrices. When cell cluster information was provided, distinct SPARSim simulation parameters were estimated per cell for each cluster. scDESIGN2 accepts input matrices where entries are integer count values; we thus performed rounding on the expression matrix before fitting scDESIGN2. With cluster information provided, a scDESIGN2 model was fit separately for cells of each cluster, and similar to conditional GANs, the ratio of cells per cluster was provided to the method.

In silico perturbation experiments using GRoundGAN

To perform perturbation experiments using GRoundGAN, we put the trained model in a deterministic mode of operation. This is necessary to ensure that the perturbation experiments are performed on the same batch (i.e., replicate) of generated cells to form matched case/control experiments. To do this, we performed a forward pass through the generator and then saved the input noise to the causal controller, the input noise to the target generators, the TF expression values generated by the causal controller, and the per-cell scaling factor of the LSN layer. Subsequent passes through the generators used the saved parameters so that ensuing runs always output the same batch of cells (instead of generating new unmatched cells).

Trajectory inference and pseudo-time analysis

Following official PAGA tutorial for the BoneMarrow dataset (`https://github.com/scverse/scanpy-tutorials/blob/master/paga-paul15.ipynb`), we used (Partition-based graph abstraction) PAGA[38] for trajectory inference and analysis. We built force-directed graphs[65] (with ForceAtlas2[66]) using the top 20 principal components of the data (using principal component analysis or PCA) and a neighborhood graph of observations computed using UMAP (to estimate connectivities). We next denoised the graph by representing it in the diffusion map space and computed distances and neighbors as before using this new representation. After denoising, we then ran the Louvain clustering algorithm with a resolution of 0.6. Finally, we

ran the PAGA algorithm on the identified clusters and used the obtained graph to initialize and rebuild the force-directed graph.

GRN inference methods

In our GRN benchmarking analysis, we focused on eight GRN inference algorithms: GENIE3[9], GRNBoost2[8], PPCOR[37], PIDC[10], LEAP[12], SCODE[13] and SINCERITIES[14], which were used in the BEELINE study[19], as well as CeSpGRN[48]. Of these methods, LEAP requires pseudo-time ordering of cells, while SCODE and SINCERITIES require both pseudo-time ordering and pseudo-time values. Since not all algorithms inferred the edge directionality or its sign (activatory or inhibitory nature), we did not consider these factors in our analysis to be consistent among different models.

For the methods available in the BEELINE study, we ran them as docker containers using the docker images provided by BEELINE's GitHub (`https://github.com/Murali-group/Beeline`)[19] with the default parameters used in BEELINE. These methods were applied to nine datasets simulated by GRouNdGAN and scGAN, and the original training real dataset corresponding to PBMC-CTL, PBMC-All, and BoneMarrow. To ensure consistency, the same number of cells as the real training set were simulated using GRouNdGAN and scGAN for each dataset: $n = 19773$ for PBMC-CTL, $n = 67579$ for PBMC-All, and $n = 2230$ for BoneMarrow. Number of genes and TFs present in the GRN for each dataset is provided in Supplementary Data 1 - Sheet 3. To benchmark algorithms requiring pseudo-time ordering of cells, we computed the pseudo-times of GRouNdGAN-simulated data (based on the BoneMarrow dataset) using PAGA[38] and diffusion pseudotime[39], following the methodology described earlier. In the GRN inference benchmark analysis, we did not provide the list of TFs to GRNBoost2 to make it consistent with other GRN inference methods.

We also included CeSpGRN, which is a cell-specific GRN inference method. Since this method first generates one GRN for each cell, it requires a high amount of memory to run. As a result, we were only able to benchmark it using a subset of data consisting of only $n = 1000$ cells (for any of the three datasets) and 100 genes (we did not change the number of TFs, or the GRN edges connecting them to the considered genes). Following the method described in the original study, we then averaged the total absolute edge weights across all cells to form a consensus GRN using CeSpGRN.

Statistics and reproducibility

The sample sizes were selected by original studies producing datasets used for training GRouNdGAN. Preprocessing steps are described in the datasets and preprocessing section. Other than standard cell-level and gene-level filtering, there were no data exclusions. Statistical tests used for each analysis are described in the corresponding sections and include Wilcoxon signed rank and Mann Whitney U tests.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The PMBC reference dataset is available from the 10x Genomics repository (corresponding to healthy donor A) from the following link: `https://support.10xgenomics.com/single-cell-gene-expression/datasets/1.1.0/fresh_68k_pbmc_donor_a?` (related to PBMC-All, PBMC-CTL, and PBMC-NaiveT). The BoneMarrow reference dataset is available in the Gene Expression Omnibus (GEO) repository under accession number GSE72857. The Dahlin reference dataset is available in GEO (accession number: GSE107727). The Tumor-All (and Tumor-malignant) dataset is available from cellxgene (`https://cellxgene.cziscience.com/collections/968834a0-1895-40df-8720-666029b3bbac`). Simulated data corresponding to PBMC-All, PBMC-CTL, BoneMarrow, Dahlin, Tumor-All, and Tumor-malignant and their corresponding imposed GRNs are provided on GRouNdGAN's website (`https://emad-combine-lab.github.io/GRouNdGAN/benchmarking`) and can be used for benchmarking different GRN Inference methods. Source data are provided with this paper.

Code availability

Our implementation and evaluation of GRouNdGAN in Python 3.9.6 using the PyTorch framework[67] along with a tutorial is freely available under the GNU Affero General Public License v3.0 on GitHub (`https://github.com/Emad-COMBINE-lab/GRouNdGAN`) which is archived in Zenodo under record number 11068246[68] (`https://doi.org/10.5281/zenodo.11068246`). The repository also contains a Docker image for reproducibility

and our PyTorch implementation of scGAN, projection-based conditioning cscGAN[24] and a variation of cscGAN that uses conditioning by concatenation (cWGAN). More information about the installation, a detailed tutorial, and simulated datasets that can be readily used are provided in (`https://emad-combine-lab.github.io/GRouNdGAN`).

References

1. Lee, T. I. et al. Transcriptional regulatory networks in Saccharomyces cerevisiae. *Science* **298**, 799–804 (2002).

2. Che, D. et al. Dynamic and modular gene regulatory networks drive the development of gametogenesis. *Brief. Bioinform.* **18**, 712–721 (2017).

3. Olson, E. N. Gene regulatory networks in the evolution and development of the heart. *Science* **313**, 1922–1927 (2006).

4. Dibaeinia, P. & Sinha, S. SERGIO: a single-cell expression simulator guided by gene regulatory networks. *Cell Syst.* **11**, 252–271. e211 (2020).

5. Yang, Y. et al. Gene knockout inference with variational graph autoencoder learning single-cell gene regulatory networks. *Nucleic Acids Res.* **51**, 6578–6592 (2023).

6. Madhamshettiwar, P. B., Maetschke, S. R., Davis, M. J., Reverter, A. & Ragan, M. A. Gene regulatory network inference: evaluation and application to ovarian cancer allows the prioritization of drug targets. *Genome Med.* **4**, 1–16 (2012).

7. Manuel, A. M., Dai, Y., Jia, P., Freeman, L. A. & Zhao, Z. A gene regulatory network approach harmonizes genetic and epigenetic signals and reveals repurposable drug candidates for multiple sclerosis. *Hum. Mol. Genet.* **32**, 998–1009 (2023).

8. Moerman, T. et al. GRNBoost2 and Arboreto: efficient and scalable inference of gene regulatory networks. *Bioinformatics* **35**, 2159–2161 (2019).

9. Huynh-Thu, V. A., Irrthum, A., Wehenkel, L. & Geurts, P. Inferring regulatory networks from expression data using tree-based methods. *PloS one* **5**, e12776 (2010).

10. Chan, T. E., Stumpf, M. P. & Babtie, A. C. Gene regulatory network inference from single-cell data using multivariate information measures. *Cell Syst.* **5**, 251–267. e253 (2017).

11. Emad, A. & Sinha, S. Inference of phenotype-relevant transcriptional regulatory networks elucidates cancer type-specific regulatory mechanisms in a pan-cancer study. *NPJ Syst. Biol. Appl.* **7**, 9 (2021).

12. Specht, A. T. & Li, J. LEAP: constructing gene co-expression networks for single-cell RNA-sequencing data using pseudotime ordering. *Bioinformatics* **33**, 764–766 (2017).

13. Matsumoto, H. et al. SCODE: an efficient regulatory network inference algorithm from single-cell RNA-Seq during differentiation. *Bioinformatics* **33**, 2314–2321 (2017).

14. Papili Gao, N., Ud-Dean, S. M., Gandrillon, O. & Gunawan, R. SINCERITIES: inferring gene regulatory networks from time-stamped single cell transcriptional expression profiles. *Bioinformatics* **34**, 258–266 (2018).

15. Aibar, S. et al. SCENIC: single-cell regulatory network inference and clustering. *Nat. Methods* **14**, 1083–1086 (2017).

16. Bravo González-Blas, C. et al. SCENIC+: single-cell multiomic inference of enhancers and gene regulatory networks. *Nat. Methods* **20**, 1355–1367 (2023).

17. Kartha, V. K. et al. Functional inference of gene regulation using single-cell multi-omics. *Cell Genomics* **2**, 100166 (2022).

18. Badia-i-Mompel, P. et al. Gene regulatory network inference in the era of single-cell multi-omics. *Nat. Rev. Genet.* **24**, 739–754 (2023).

19. Pratapa, A., Jalihal, A. P., Law, J. N., Bharadwaj, A. & Murali, T. Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nat. Methods* **17**, 147–154 (2020).

20. Emmert-Streib, F., Dehmer, M. & Haibe-Kains, B. Gene regulatory networks and their applications: understanding biological and medical problems in terms of networks. *Front. Cell Dev. Biol.* **2**, 38 (2014).

21. Kolmykov, S. et al. GTRD: an integrated view of transcription regulation. *Nucleic Acids Res.* **49**, D104–D111 (2021).

22. Xu, H. et al. ESCAPE: database for integrating high-content published data collected from human and mouse embryonic stem cells. *Database* **2013**, bat045 (2013).

23. Kanehisa, M. & Goto, S. KEGG: kyoto encyclopedia of genes and genomes. *Nucleic Acids Res.* **28**, 27–30 (2000).

24. Marouf, M. et al. Realistic in silico generation and augmentation of single-cell RNA-seq data using generative adversarial networks. *Nat. Commun.* **11**, 1–12 (2020).

25. Sun, T., Song, D., Li, W. V. & Li, J. J. scDesign2: a transparent simulator that generates high-fidelity single-cell gene expression count data with gene correlations captured. *Genome Biol.* **22**, 1–37 (2021).

26. Baruzzo, G., Patuzzi, I. & Di Camillo, B. SPARSim single cell: a count data simulator for scRNA-seq data. *Bioinformatics* **36**, 1468–1475 (2020).

27. Schaffter, T., Marbach, D. & Floreano, D. GeneNetWeaver: in silico benchmark generation and performance profiling of network inference methods. *Bioinformatics* **27**, 2263–2270 (2011).

28. Marbach, D. et al. Revealing strengths and weaknesses of methods for gene network inference. *Proc. Natl Acad. Sci.* **107**, 6286–6291 (2010).

29. Marbach, D. et al. Wisdom of crowds for robust gene network inference. *Nat. Methods* **9**, 796–804 (2012).

30. Chen, S. & Mar, J. C. Evaluating methods of inferring gene regulatory networks highlights their lack of performance for single cell gene expression data. *BMC Bioinforma.* **19**, 1–21 (2018).

31. Kocaoglu, M., Snyder, C., Dimakis, A. G. & Vishwanath, S. CausalGAN: Learning causal implicit generative models with adversarial training. Preprint at https://arxiv.org/abs/1709.02023 (2017).

32. Arjovsky, M., Chintala, S. & Bottou, L. Wasserstein generative adversarial networks. In *International conference on machine learning*. 214–223 (PMLR, 2017).

33. Zheng, G. X. et al. Massively parallel digital transcriptional profiling of single cells. *Nat. Commun.* **8**, 1–12 (2017).

34. Paul, F. et al. Transcriptional heterogeneity and lineage commitment in myeloid progenitors. *Cell* **163**, 1663–1677 (2015).

35. Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B. & Smola, A. A kernel two-sample test. *J. Mach. Learn. Res.* **13**, 723–773 (2012).

36. Korsunsky, I. et al. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat. Methods* **16**, 1289–1296 (2019).

37. Kim, S. ppcor: an R package for a fast calculation to semi-partial correlation coefficients. *Commun. Stat. Appl. Methods* **22**, 665 (2015).

38. Wolf, F. A. et al. PAGA: graph abstraction reconciles clustering with trajectory inference through a topology preserving map of single cells. *Genome Biol.* **20**, 1–9 (2019).

39. Haghverdi, L., Büttner, M., Wolf, F. A., Buettner, F. & Theis, F. J. Diffusion pseudotime robustly reconstructs lineage branching. *Nat. Methods* **13**, 845–848 (2016).

40. Trapnell, C. et al. Pseudo-temporal ordering of individual cells reveals dynamics and regulators of cell fate decisions. *Nat. Biotechnol.* **32**, 381 (2014).

41. Kumar, P., Tan, Y. & Cahan, P. Understanding development and stem cells using single cell-based analyses of gene expression. *Development* **144**, 17–32 (2017).

42. Saelens, W., Cannoodt, R., Todorov, H. & Saeys, Y. A comparison of single-cell trajectory inference methods. *Nat. Biotechnol.* **37**, 547–554 (2019).

43. Street, K. et al. Slingshot: cell lineage and pseudotime inference for single-cell transcriptomics. *BMC Genom.* **19**, 1–16 (2018).

44. Hwang, B., Lee, J. H. & Bang, D. Single-cell RNA sequencing technologies and bioinformatics pipelines. *Exp. Mol. Med.* **50**, 1–14 (2018).

45. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* **19**, 1–5 (2018).

46. Dahlin, J. S. et al. A single-cell hematopoietic landscape resolves 8 lineage trajectories and defects in Kit mutant mice. *Blood J. Am. Soc. Hematol.* **131**, e1–e11 (2018).

47. Han, G. et al. Follicular lymphoma microenvironment characteristics associated with tumor cell mutations and MHC class II expression. *Blood Cancer Discov.* **3**, 428–443 (2022).

48. Zhang, Z., Han, J., Song, L. & Zhang, X. Inferring cell-specific gene regulatory networks from single cell gene expression data. Preprint at https://www.biorxiv.org/content/10.1101/2022.03.03.482887v3.full (2022).

49. Yuan, Y. & Bar-Joseph, Z. Deep learning for inferring gene relationships from single-cell expression data. *Proc. Natl Acad. Sci.* **116**, 27151–27158 (2019).

50. Chen, J. et al. DeepDRIM: a deep neural network to reconstruct cell-type-specific gene regulatory network using single-cell RNA-seq data. *Brief. Bioinforma.* **22**, bbab325 (2021).

51. Zhang, S. Y. & Stumpf, M. P. Learning cell-specific networks from dynamical single cell data. Preprint at https://www.biorxiv.org/content/10.1101/2023.01.08.523176v2 (2023).

52. Crowell, H. L., Morillo Leonardo, S. X., Soneson, C. & Robinson, M. D. The shaky foundations of simulating single-cell RNA sequencing data. *Genome Biol.* **24**, 1–19 (2023).

53. Gao, Y., Shen, L. & Xia, S.-T. DAG-GAN: Causal structure learning with generative adversarial nets. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 3320-3324 (IEEE, 2021).

54. Satija, R., Farrell, J. A., Gennert, D., Schier, A. F. & Regev, A. Spatial reconstruction of single-cell gene expression data. *Nat. Biotechnol.* **33**, 495–502 (2015).

55. Goodfellow, I. et al. Generative adversarial networks. *Commun. ACM* **63**, 139–144 (2020).

56. Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V. & Courville, A. C. Improved training of wasserstein gans. In *Advances in neural information processing systems*, 30 (NeurIPS, 2017).

57. Reddi, S. J., Kale, S. & Kumar, S. On the convergence of adam and beyond. Preprint at https://arxiv.org/abs/1904.09237 (2019).

58. He, K., Zhang, X., Ren, S. & Sun, J. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE international conference on computer vision*. 1026–1034 (IEEE, 2015).

59. Glorot, X. & Bengio, Y. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*. 249-256 (JMLR Workshop and Conference Proceedings, 2010).

60. Ioffe, S. & Szegedy, C. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *International conference on machine learning*. 448-456 (PMLR, 2015).

61. Hu, H. et al. AnimalTFDB 3.0: a comprehensive resource for annotation and prediction of animal transcription factors. *Nucleic Acids Res.* **47**, D33–D38 (2019).

62. Shaham, U. et al. Removal of batch effects using distribution-matching residual networks. *Bioinformatics* **33**, 2539–2546 (2017).

63. Shu, H. et al. Modeling gene regulatory networks using neural network architectures. *Nat. Comput. Sci.* **1**, 491–501 (2021).

64. Mirza, M. & Osindero, S. Conditional generative adversarial nets. Preprint at https://arxiv.org/abs/1411.1784 (2014).

65. Islam, S. et al. Characterization of the single-cell transcriptional landscape by highly multiplex RNA-seq. *Genome Res.* **21**, 1160–1167 (2011).

66. Jacomy, M., Venturini, T., Heymann, S. & Bastian, M. ForceAtlas2, a continuous graph layout algorithm for handy network visualization designed for the Gephi software. *PloS One* **9**, e98679 (2014).

67. Paszke, A. et al. Pytorch: An imperative style, high-performance deep learning library. In *Advances in neural information processing systems* 32 (NeurIPS, 2019).

68. Zinati, Y., Takiddeen, A. & Emad, A. GRouNdGAN. Zenodo https://doi.org/10.5281/zenodo.11068246 (2024). article-v1.0.0.

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Author contributions

All authors contributed to the design of the study and the algorithm, the analyses, and the writing of the manuscript. A.E. supervised the study. Y.Z. implemented the model and performed the statistical analyses. Y.Z., A.T., and A.E. read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Background and Context for "*Causal single-cell RNA-seq simulation, in silico perturbation, and GRN inference benchmarking using GRouNdGAN-Toolkit*"

While RAPPPID represented a meaningful advance in our ability to make accurate predictions on datasets which controlled for data leakage, generalization remained (and continues to remain) a challenge for PPI inference models. Following the publication of RAPPPID, we began to look to specific areas in which PPI inference models fail to generalize. Predicting interactions between proteins of different organism, particularly those of very evolutionary distance species, had been previously identified as a challenge for PPI inference models [81, 155]. What's more, these "cross-species" interactions are of great value to researchers. The vast majority of PPIs studied belong to a small number of model organisms representing a small fraction of all species, representing less than a third of taxa at the class rank as measured by the STRING core dataset [9]. We've named this discrepancy in elucidated PPI's the "species gap". It follows, then, that it would be greatly advantageous to construct computational methods for expanding our knowledge of PPIs in species other than a select few model organisms.

The cross-species PPI inference models of which I was aware at the time, however, conducted their evaluations on datasets which suffered from data leakage [81, 155]. Further, when evaluated on a strict, C3 dataset which controlled for data leakage, performance severely deteriorated [9].

This is the context from which our method INTREPPPID arose [9]. INTREPPPID builds upon the lessons and architecture of RAPPPID, leaving unchanged its AWD-LSTM encoder, expanding the model such that it now also modelled the orthology between proteins. Orthology here can be thought of similarities in protein sequence across species due them sharing a common ancestor [165]. Modelling orthology proved to have many synergistic effects with the cross-species task, lending itself particularly well to both theoretical and practical concerns. Theoretically, homology is a natural place to begin when comparing the interactions of two species, and forms the basis of many early analyses such as Date and Marcotte's work as well as that of Sharan *et al.* [74, 166].

Practically, the orthology modelling task can be argued to provide a regularizing effect when viewed through the lens of contrastive self-supervised learning [9, 167, 168]. Due to our novel "quintuplet" architecture, whereby an encoders weights are shared across a twin network task as well as as a triplet contrastive network, the number of parameters used in the INTREPPPID network were ostensibly identical to that of RAPPPID.

INTREPPPID successfully improved on RAPPPID's predictive performance on testing pairs between proteins of the same and differing organisms [9]. Experiments also demonstrated that the improvements in predictive performance that INTREPPPID enjoyed were due to having successfully modelled orthology, rather than any spurious or unintended cause resulting from the novel "quintuplet" architecture. INTREPPPID's efficient architecture and resulting low computational requirements has allowed us to create an online service, PPI.bio, for using INTREPPPID and RAPPPID to infer PPI interactions. Together, the work presented in the INTREPPPID manuscript represent a concrete step in the direction of narrowing the species gap.

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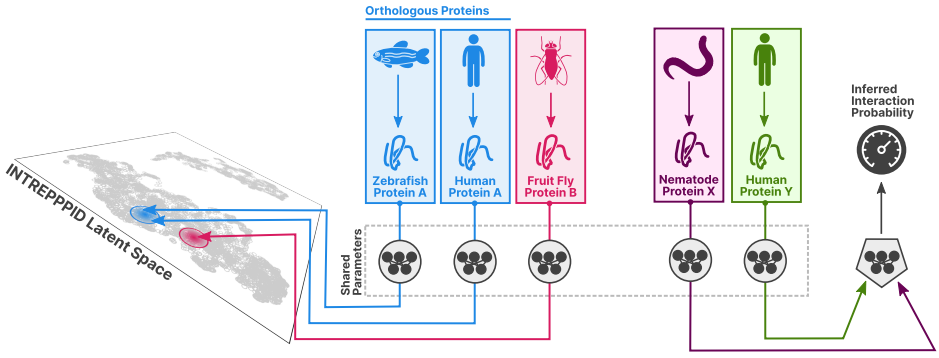
Chapter 4

Causal single-cell RNA-seq simulation, in silico perturbation, and GRN inference benchmarking using GRouNdGAN-Toolkit

Abstract

An overwhelming majority of protein–protein interaction (PPI) studies are conducted in a select few model organisms largely due to constraints in time and cost of the associated 'wet lab' experiments. In silico PPI inference methods are ideal tools to overcome these limitations, but often struggle with cross-species predictions. We present INTREPPPID, a method that incorporates orthology data using a new 'quintuplet' neural network, which is constructed with five parallel encoders with shared parameters. INTREPPPID incorporates both a PPI classification task and an orthologous locality task. The latter learns embeddings of orthologues that have small Euclidean distances between them and large distances between embeddings of all other proteins. INTREPPPID outperforms all other leading PPI inference methods tested on both the intraspecies and cross-species tasks using strict evaluation datasets. We show that INTREPPPID's orthologous locality loss increases performance because of the biological relevance of the orthologue data and not due to some other specious aspect of the architecture. Finally, we introduce PPI.bio and PPI Origami, a web server interface for INTREPPPID and a software tool for creating strict evaluation datasets, respectively. Together, these two initiatives aim to make both the use and development of PPI inference tools more accessible to the community.

Graphical Abstract



Keywords: protein-protein interactions; orthology; cross-species; deep learning

Introduction

Studying how proteins work in concert has yielded deeply valuable biological insights. Not least of these insights is the implication of protein interactions in the appearance of biological phenotypes and relatedly, how disrupting these interactions can lead to disease [1, 2]. Knowledge of these interactions can, in turn, be used to design drugs and therapies to treat the diseases for which they are implicated [3, 4].

As the relevant technologies have improved, studies which experimentally verify protein–protein interactions (PPIs) have increased in size. In fact, multiple recent studies have published so-called 'all-by-all' studies whereby the interaction of entire proteomes are experimentally evaluated. In September 2014, researchers assembled a reference network of ~14 000 interactions from a search space of 13 000 human genes through a combination of yeast-two-hybrid (Y2H) assays and other validation experiments [5]. Five years later, the same network ('HuRI') expanded to ~53 000 interactions from a search space of 17 500 proteins [6].

While these studies provide an invaluable wealth of data, they represent a tremendous investment of time, labour, and money. A major consequence of the high cost of 'all-by-all' studies is that it is prohibitively expensive to exhaustively perform them for all known species. In fact, it is currently only practical to limit these studies to a small number of well-studied model organisms. This is a problem common to many aspects of the life sciences, which we've taken to calling 'the species gap'.

As of February, 2024 there are currently 793 217 taxa spanning 108 340 genera enumerated by the NCBI Taxon Database [7]. Few, if any, experimental studies are capable of accounting for most, let alone all, of these taxa. For this reason, scientists rely on select few 'model' organisms

that exhibit desirable properties that lend themselves to a field of study or methodology. These selection criteria further bias our already small sampling of all possible living subjects of study. To demonstrate the extent of the species gap, we quantified the number of distinct organisms that are absent from the STRING Core PPI database [8], by calculating the intersection of taxa in this database and taxa in the NCBI Taxon Database. We found that only 30% of all taxa at the class rank and 16% at the order rank were present in the STRING Core database (Fig. 1), underscoring the lack PPI information from the vast majority of species.

Recent advances in co-fractioning and cross-linking mass-pectrometry (CF-MS and XL-MS, respectively) methods have greatly increased the throughput of PPI experiments. Both methods have many advantageous properties for PPI prediction, including their speed, not requiring over-expression, and the ability to identify protein complexes [10].

In silico models of protein interaction have many favourable characteristics that may help narrow the species gap. Relative to 'wet lab' experiments such as Y2H or CF/XL-MS, which may take on the order of days or months to complete and analyse, *in silico* methods can take as little as seconds or minutes. Furthermore, most *in silico* PPI prediction methods operate entirely unattended, resulting in much lower labour costs. Finally, inferring PPIs is often computationally inexpensive and can leverage commodity hardware. In summary, *in silico* methods and their low cost lend themselves to the task of exhaustively searching protein interaction spaces.

Many *in silico* methods, however, face challenges that make them unsuitable for less-studied organisms. Namely, many methods, especially those dependent on machine learning, require a training dataset comprised of PPIs. These datasets must be sufficiently large to provide a representative sample of interactions. They must also scale with the number of parameters being learned lest the model overfit. Other methods that do not rely on machine learning have similar requirements.

Regrettably, a major barrier to bridging the species gap is that there are often too few interaction

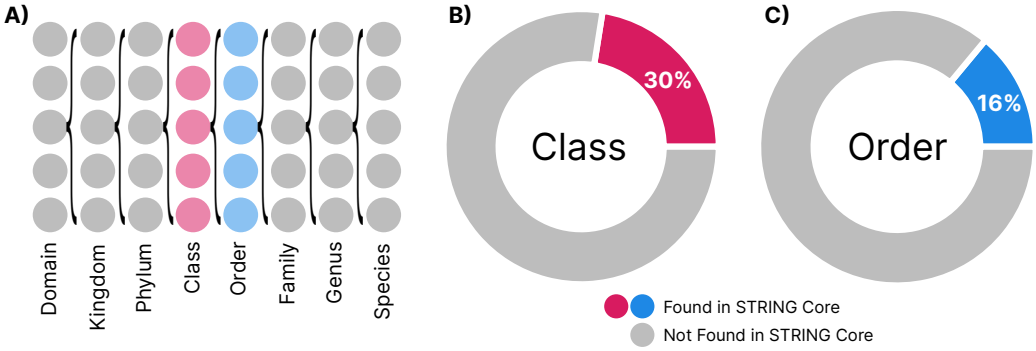


Figure 1: The species gap as illustrated by the STRING PPI database. (A) the hierarchy of biological taxa [9]. **(B, C)** Doughnut charts indicating the proportion of all taxa represented in STRING Core at the class rank **(B)** and order rank **(C)**.

data to train PPI inference models for lesser-studied organisms. In order for computational models to address the species gap, they must make accurate PPI inferences on less-studied organisms, while using existing PPI data from well-studied organisms during training. Such 'cross-species' inference can be used to infer the intraspecies PPI network of the less-studied organism, all while only using the model organism as training data, addressing the species gap. Cross-species inference requires that methods perform accurate out-of-distribution (OOD) inference. Historically, PPI inference methods have failed to make accurate predictions on interactions between OOD proteins [11–13].

In 2012, a letter published by Park and Marcotte outlined the severity of pervasive data leakage that was obscuring the inability of methods of the time to generalize to interactions between OOD proteins [11]. The cause of the data leakage Park and Marcotte observed was the common 'random splitting' approach to preparing cross-validation folds. Random splitting of interactions (between pairs of proteins) into folds allows for the same proteins appearing in both the training and testing sets, resulting in data leakage. Park and Marcotte found that when methods of the time were trained and tested on 'strict' datasets where proteins present in the training set never appear in any of the other folds, the accuracy of most methods dropped to random levels. This precipitous drop in performance indicates that all the models tested had problems dealing with OOD proteins.

While Park and Marcotte found that OOD performance was poor across the landscape of methods available in 2012, studies by Dunham and Ganapathiraju in 2021 and Bernett *et al.* in 2023 corroborate the 12-year-old findings [12, 13]. In both studies, contemporary sequence-based PPI inference methods were tested on datasets that satisfy Park and Marcotte's strictest validation schemes. Predictive power dramatically degraded in all cases, just as Park and Marcotte observed in 2012. While models for PPI inference have changed drastically since 2012 with the notable introduction of deep learning methods, reported performance of methods continue to be inf lated as a result of the random splitting of interactions (as opposed to stricter splitting based on proteins).

Furthermore, work by Hamp and Rost as well as Bernett *et al.* underscore an additional source of data leakage: distinct proteins with sequences which are nearly identical [13, 14]. Amino acid sequences can be overwhelmingly redundant between distinct proteins. An important consequence of this is that two proteins with all but identical sequences may appear in both sets, thereby leaking data in the same way randomly-splitting PPI pairs does. The challenges presented by sequence similarity can be overcome by ensuring that proteins in the training set do not have a sequence identity above some threshold with proteins in the testing set.

One of the earliest approaches for sequence-based PPI inference is Marcotte *et al.*'s 'Rosetta Stone' method [15]. The authors outline a theory whereby interacting proteins of one species

sometimes appear as fused homologous proteins in other species. Using this technique, Marcotte *et al.* inferred interactions between proteins from *Escherichia coli* using information of proteins of other genomes that had similar sequences to the putative pair. The concept of 'interologue' was soon after introduced to describe proteins whose interactions are conserved among their respective orthologues [16]. Interologues allow us to extend knowledge of interactions of one species to other species by identifying both sequences and interactions that are conserved between them [17, 18]. These homology-based methods, while effective at identifying conserved interactions, have limitations related to unknown interactions, as acknowledged in the work that first introduced the term interologue: 'Although potentially powerful, applying the interologue concept alone in estimating the significance of potential interactions precludes the finding of novel connections previously unidentified in other model organisms' [16].

Some of the first machine-learning methods for predicting PPIs were kernel-based methods operating on sequence features. The support vector machine (SVM) is an algorithm for finding the maximum-margin hyperplane for a classification or regression problem, relying on kernel transformations of the input data to solve nonlinear problems [19]. Effectively applying SVMs to the problem of PPI classification depends greatly on the design of the kernels used to represent protein pairs. These kernels, designed by expert knowledge, may compare individual proteins to each other ('genomic kernels') or pairs of proteins to other pairs ('pairwise kernels'). Furthermore, these kernels may compare a variety of features of the gene products being studied, including motifs, gene ontology (GO) annotations, sequence similarity, or some combination thereof [20].

As advances in hardware and algorithms made research into deep learning more feasible, they've proven to be effective in a number of problem domains, not least of which being PPI prediction. Richoux *et al.* demonstrated that twin neural-networks are far more efficient at PPI inference than other network architectures [21]. Twin neural networks are models that consider pairs of inputs using the same trainable parameters rather than specifying parameters for each input and are commonly used to make comparisons between two inputs [22, 23]. Twin networks can reduce the risk of overfitting and aiding few-shot performance [24]. Many twin PPI prediction methods rely on convolutional neural networks (CNNs) or recurrent neural networks (RNNs) to encode feature sequences. The PIPR method by Chen *et al.*, however, eschews these architectures for a hybrid: the recurrent CNN (RCNN) [25]. The RCNN comprises of an initial CNN layer whose output is fed into a bidirectional gated recurrent unit (GRU) RNN [26].

We've previously described RAPPPID, a method for PPI prediction that outperforms leading methods on strict validation sets that conform to the guidelines set out by Park and Marcotte [11, 27]. RAPPPID is a twin network that learns latent representations of sequences and inputs them into a classifier network to infer whether two proteins interact. RAPPPID achieves its high performance largely due to its adoption of averaged weight-decayed LSTM (AWD-LSTM) and the Unigram tokenization algorithm [28, 29]. The former is a combination of regularization

techniques that are effective for long short-term memory (LSTM) RNNs. The input is tokenized using subword regularization, which regularizes the network by sampling from a distribution of different segmentations of the input [30].

In spite of the aforementioned progress in the field, most computational models are not designed to address the species gap and exhibit reduced performance when tested on the cross-species task. This makes them unsuitable to address the species gap problem and as such, the knowledge of the protein interactions that govern fundamental biological processes of ill-studied non-model organisms remains beyond our grasp. To address this issue, we introduce an extension of RAPPPID named INTREPPPID (Incorporating Triplet Error for Predicting PPIs using Deep Learning). INTREPPPID is specifically designed to improve cross-species PPI prediction performance with the aim of reducing the species gap. In this study, 'cross-species' PPI prediction refers to predicting the intraspecies interactions of proteins of one species using a computational model that is trained on intraspecies interactions of a different species. This term does not refer to predicting the interaction between one protein from one species and another protein from a different species.

We designed INTREPPPID to learn latent representations of amino acid sequences that are useful for predicting PPIs as well as identifying orthologues. Alongside RAPPPID's original twin network for inference, INTREPPPID adds a triplet network that minimizes the Euclidean distance between latent representations of orthologous sequences while maximizing it for nonorthologous sequences. Parameters are shared between both the twin and triplet networks, resulting in five networks with a single set of shared parameters, which we call a 'quintuplet' network. INTREPPPID's architecture learns a latent space that encodes both PPI and orthology information. This allows us to enforce an inductive bias loosely based on the concept of interologues.

The contributions of this study are as follows. First, we introduce INTREPPPID, a deep learning method based on a novel quintuplet architecture and a carefully designed loss function that makes out-of-distribution predictions of PPIs to address the species gap. Our analysis shows that INTREPPPID outperforms all leading methods tested on both intraspecies and cross-species PPI predictions. Second, we demonstrate that INTREPPPID effectively learns the orthology of proteins and its performance improvement is due to learning biologically relevant information. This suggests the use of orthology information as a powerful approach for making out-of-distribution predictions in this domain. Third, we have released to the community PPI.bio, an open-source interface for INTREPPPID and RAPPPID hosted on the internet. This allows researchers to infer PPIs without installing any specialized software or using the command line. Finally, we have released PPI Origami, a computer programme for creating reproducible strict PPI databases in an automated fashion. By automating the dataset generation process, we hope to reduce common sources of data leakage in datasets used for PPI inference.

Materials and methods

The INTREPPPID architecture

INTREPPPID integrates RAPPPID's twin network for the PPI Classification task with a triplet neural network used for a novel Orthologous Locality task. Learning the two networks jointly results in five encoder layers with shared parameters in what we've come to call a quintuplet network architecture.

INTREPPPID uses the same AWD-LSTM encoder as the one used in RAPPPID to learn vector representations of protein sequences. This encoder is a regularized RNN that has been shown to improve performance on tasks that require inferring interactions between proteins absent from the training set [27]. Before encoding, sequences are first tokenized using the Unigram algorithm as implemented by the SentencePiece library [29, 30]. INTREPPPID uses the vector representations generated by the encoder for two downstream tasks: the 'orthologous locality' task and the 'PPI classification' task. The encoder as well as the two tasks are trained in an end-to-end fashion.

INTREPPPID's quintuplet architecture means that training the network relies on additional orthology data. At inference time, however, only the protein classification twin network is used, so no information other than the amino acid sequences of the proteins in question are required as input.

The protein–protein interaction classification task

INTREPPPID infers PPIs by learning latent representations of the candidate protein sequences which can be classified by a fully connected neural network outputting the interaction probability. The two amino acid sequences (s_β and s_γ) are first tokenized into vectors x_β and x_γ using the Unigram algorithm. The Unigram algorithm segments the amino acid of the protein sequence into groups using a vocabulary of 250 tokens. The Unigram algorithm also employs subword regularization. Unigram draws tokens for a given sequence from a probabilistic distribution of all possible combination of tokens in the vocabulary, meaning that the same sequence can be tokenized differently between successive calls to the tokenizer.

Once the tokenized vectors x_β and x_γ are obtained, they are inputted into an encoder ($e(\cdot)$) that outputs a latent vector for each protein (z_β and z_γ , respectively):

$$z_\beta = e(x_\beta)$$
$$z_\gamma = e(x_\gamma).$$

Finally, the latent vectors z_β and z_γ are averaged and inputted into a classifier network $g(\cdot)$ that outputs an interaction probability $\hat{y}$:

$$y = g\left(\left(z_\beta + z_\gamma\right)/2\right).$$

The classifier $g(\cdot)$ is a two-layer, fully connected neural network whose hidden layer is activated by the Mish function [31]. The fully connected neural network outputs a single real number which, once activated by the sigmoid function, represents the probability that the proteins represented by the amino acid sequences s_β and s_γ interact. This interaction probability as well as the true label are inputted into the binary cross-entropy loss to obtain a loss for the PPI classification network:

$$\ell_{\text{class}} = (\hat{y}, y) = -y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y}),$$

where y is a binary label representing whether the two proteins truly interact ($y = 1$) or not ($y = 0$).

The orthologous locality task

INTREPPPID introduces a novel task to help inform the PPI classification network. The orthologous locality task is to learn a latent space whereby the Euclidean distance between latent representations of proteins that are orthologous to one another are within some margin m. Conversely, nonorthologous proteins are to have a Euclidean distance greater than m.

To achieve this, we adopt a triplet network and a triplet margin loss ℓ_{ortho} following the description of [32]:

$$\ell_{\text{ortho}}(z_a, z_p, z_n) = \max(d(z_a, z_p) - d(z_a, z_n) + m, 0),$$

where z_a is the latent vector of some 'anchor' protein for which there is a known orthologous protein with a latent vector z_p, z_n is the latent vector of a randomly chosen protein that is not an

orthologue of z_a, m is a margin factor and d is the Euclidean distance:

$$d(u, v) = ||u - v||^2.$$

For the experiments conducted herein, the margin m was fixed at a value of 1.

The principle of the triplet network is to, at every training step, maximize the distance between two nonorthologous proteins (the 'anchor' and 'negative' proteins), as well as minimize the distance between two orthologous proteins (the 'anchor' and 'positive' proteins).

The anchor protein, with amino acid sequence s_a, can be any protein in the dataset for which orthology data are known. All orthology data for the experiments conducted herein are from the Orthologous Matrix (OMA) database of orthologues [33]. In the event that orthologues are known for the protein that s_β encodes, s_β is chosen to be s_a.

The positive and negative proteins (with amino acid sequences s_p and s_n, respectively) are chosen relative to s_a. s_p is chosen to be the sequence of a randomly chosen protein that, according to the OMA database, belongs to the set of sequences of proteins orthologous to the anchor protein. At every training step, s_p is drawn randomly from this set of proteins orthologous to s_a, meaning that the positive protein for a given anchor protein may not be the same between training steps or epochs. The negative protein is sampled in a similar manner to the positive protein, but rather than being sampled from the set of proteins orthologous to s_a, is sampled from the set of proteins not orthologous to s_a. The amino acid sequence of the anchor, positive, and negative proteins (s_a, s_p, and s_n) are tokenized and encoded in the same manner as s_β and s_γ , resulting in their latent representations z_a, z_p, and z_n.

Optimizing the triplet margin loss in this manner maximizes the Euclidean distance between latent vectors of proteins that are not orthologues, while minimizing the distance between vectors of orthologues. Our adoption of the triplet margin loss follows evidence that such a loss is an effective way of creating feature descriptors that are 'spread out' over the latent space, which in turn increases performance on downstream classification tasks [32, 34]. Furthermore, orthologues typically share the same protein functions and therefore offer a good interspecies basis for the downstream PPI classification task [35, 36].

INTREPPPID's loss function

The final INTREPPPID loss function that is optimized at training time is the sum of ℓ_{ortho} and ℓ_{class},

$$\hat{L}(\hat{y}, y, z_a, z_p, z_n) = \ell_{\text{class}}(\hat{y}, y) + \ell_{\text{ortho}}(z_a, z_p, z_n).$$

In this way, INTREPPPID optimizes both the orthologous locality and PPI classification losses simultaneously, in an end-toend fashion, ensuring that latent vectors simultaneously encode orthology and PPI information.

Preparing strict protein–protein interaction datasets

Experiments were run using datasets with strict data splitting folds, described as 'C3' by Park and Marcotte [11]. Namely, we ensured that there was no overlap of proteins between any of the training, validation, or testing datasets. Additionally, we controlled for sequence similarity of proteins between the training, validation, and testing sets that is a known vector for information leakage [13, 14]. We specifically control to ensure that proteins with > 90% sequence identity are not present between splits to ensure that proteins that differ only by name are not accidentally present in two different splits. Another reason for choosing a threshold of 90% sequence identity is that it allows us to test the ability of INTREPPPID to learn motifs and domains relevant to interactions. Individuals who wish to create testing datasets that limit the exposure of similar motifs and domains can create those with more strict sequence identity requirement (*e.g.* a 50% threshold) using PPI Origami (see Generating Strict Protein–Protein Interaction Datasets with Protein–Protein Interaction Origami). Protein interaction data are obtained from the STRING database, version 12 [8]. Interactions with a confidence score < 90% are filtered. Supplementary Methods provide details of how we created such strict datasets, and Supplementary Table S1 outlines summary statistics for the datasets used in this study.

Generating strict protein–protein interaction datasets with protein–protein interaction origami

We have written a command-line interface (CLI) tool named 'PPI Origami' with the aim of making the task of preparing strict PPI datasets in a replicable and automated fashion. PPI Origami has three primary functions: 'downloading' primary databases, 'processing' primary databases into strict datasets, and an 'analysis' function.

PPI Origami allows for the downloading of primary databases as an integrated part of its pipeline. By directly handling downloading third-party databases, PPI Origami ensures that different users of the software are downloading the same files in the same way. PPI Origami also stores downloaded files in a distinct location from the processed files, isolating primary files from

accidental changes.

PPI Origami's main function is to process and transform primary databases into strict PPI datasets. It can create datasets that satisfy both criteria of strict PPI datasets, filtering low-confidence positive examples, and drawing negative examples as previously described.

PPI Origami allows for simple reporting of basic statistics of PPI datasets it generates. These reports allow for convenient verification of whether datasets meet both criteria of strict datasets, as well as other statistics of interest such as the ratio of interactions between dataset cross-validation subsets.

We hope that PPI Origami encourages other researchers to incorporate strict PPI datasets in the future development and analyses of supervised PPI inference methods.

Training procedure

INTREPPPID was trained using the Ranger21 optimizer, which uses AdamW with a look-ahead mechanism [37, 38]. An 'explore-and-exploit' learning rate scheduler was used, with a maximum learning rate of 0.01 [39]. INTREPPPID was trained for 100 epochs. The weights of the epoch with the lowest validation loss were used in the analyses herein (and for evaluation on the test set). Hyperparameters used to train INTREPPPID were informed by those selected during the development of RAPPPID and are listed in Supplementary Table S2.

Results

INTREPPPID outperforms leading models for prediction of human protein–protein interaction

First, we set out to evaluate the performance of INTREPPPID (Fig. 2A) in an intraspecies PPI prediction task: trained and tested on *Homo sapiens* interactions in the STRING database (see Materials and Methods). Table 1 shows the performance of INTREPPPID against five leading PPI prediction methods. These methods include RAPPPID, our previous method upon which INTREPPPID expands, D-SCRIPT, a method designed for the cross-species task, PIPR, a twin network with an RCNN encoder, SPRINT, a sequence-similarity-based method, and the twin networks proposed by Richoux *et al.* [21, 25, 27, 40, 41]. INTREPPPID performed slightly better than RAPPPID across all metrics and outperformed all other methods (Table 1, Fig. 2B, Supplementary Table S3). Richoux *et al.*, D-SCRIPT, and SPRINT perform similarly on average, with D-SCRIPT exhibiting a particularly high degree of variability across the different datasets tested.

We observed severe over-fitting while training and testing PIPR on all datasets. Precision–recall and receiver-operator Curves are reported in Supplementary Figs S1 and S2, respectively.

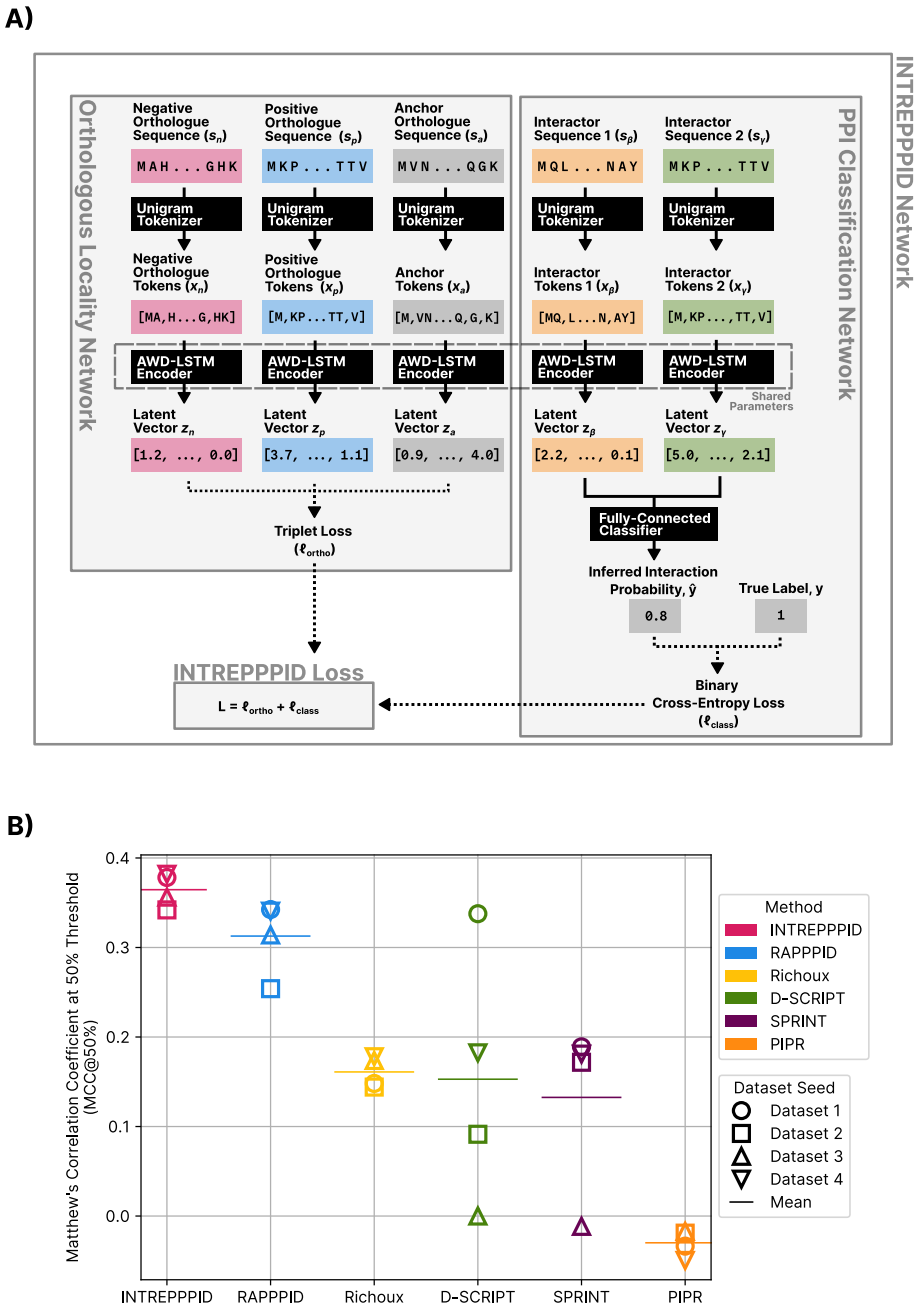


Figure 2: INTREPPPID's architecture and performance. (A) A schematic overview of INTREPPPID's quintuplet architecture. The leftmost panel depict s the triplet network responsible for the orthologous locality task, while the rightmost panel depicts the triplet network responsible for the PPI classification task. During training, INTREPPPID learns both tasks jointly through an overall loss function, which is the sum of the losses of the two tasks. **(B)** Test (C3) Matthew's correlation coefficient (MCC) of leading PPI inference methods trained and tested on *H. sapiens* datasets.

Unlike our benchmarks, it is estimated that there are far more pairs of proteins that do not interact than those that do [6, 42]. Accordingly, we tested INTREPPPID's performance at increasing negative-to-positive interaction ratios and found that no appreciable change in performance was

Table 1: Average performance of five leading PPI inference methods trained and tested on four *H. sapiens* PPI datasets. The performance metrics are calculated on strict C3 test sets (see Materials and Methods). The upward arrow (▲) indicates higher values are better for a metric, whereas a downward arrow (▼) indicates lower value are better. Boldface indicates the best value. The following abbreviations are used: AUROC: area under the receiver operating characteristic; AP: average precision; MCC: Matthew's correlation coefficient, SD: standard deviation.

Method	INTREPPID	RAPPPID	D-SCRIPT	Richoux *et al.*	SPRINT	PIPR
▲ AUROC	**0.752**	0.732	0.634	0.606	0.597	0.482
Mean (±SD)	(± 0.017)	(± 0.024)	(± 0.125)	(± 0.009)	(± 0.061)	(± 0.006)
▲ AP	**0.730**	0.717	0.627	0.614	0.641	0.484
Mean (±SD)	(± 0.022)	(± 0.015)	(± 0.086)	(± 0.015)	(± 0.064)	(± 0.003)
▲ MCC@50%	**0.365**	0.313	0.153	0.161	0.133	-0.030
Mean (±SD)	(± 0.067)	(± 0.041)	(± 0.144)	(± 0.017)	(± 0.096)	(± 0.015)
▲ F_1@50%	**0.697**	0.630	0.506	0.370	0.566	0.318
Mean (±SD)	(± 0.019)	(± 0.025)	(± 0.282)	(± 0.030)	(± 0.048)	(± 0.010)
▼ Brier Score	**0.200**	0.215	0.402	0.274	0.493	0.487
Mean (±SD)	(± 0.004)	(± 0.006)	(± 0.116)	(± 0.011)	(± 0.003)	(± 0.005)

found (Supplementary Table S4).

INTREPPPID maintains high cross-species performance on multiple organisms

INTREPPPID was designed to improve on RAPPPID's ability to make predictions across species. We therefore sought to compare INTREPPPID's cross-species performance with that of RAPPPID's as well as other leading PPI prediction methods.

When trained on *H. sapiens* and tested on various species (C3), ranging from *Mus musculus* (house mouse) to *Arabidopsis thaliana* (thale cress), INTREPPPID predominantly outperformed other methods (Fig. 3A, Supplementary Table S5), with only RAPPPID rarely surpassing its performance in some metrics. Figure 3A shows the mean AUROC (across three datasets) of different methods, with INTREPPPID significantly outperforming all other methods (one-sided Wilcoxon signed rank test versus RAPPPID = 0.03). On the cross-species task, the performance of PIPR and SPRINT were close to a random classifier. Both D-SCRIPT and the Richoux *et al.* method performed similarly to one another, but consistently scored below RAPPPID and INTREPPPID.

In many applications, one is particularly interested in the precision of a model among the highest-ranking predicted edges (*e.g.* to validate them experimentally). To evaluate INTREPPPID's suitability for such tasks, we measured the precision of predictions whose scores are above various percentiles (Fig. 3B). Considering the average across all species tested, INTREPPPID

achieved a precision > 90% among predictions above the 85th percentile, and even when we considered the 75th percentile, its mean precision remained > 85%. These observations confirm that INTREPPPID is suitable for applications where very high precision is paramount.

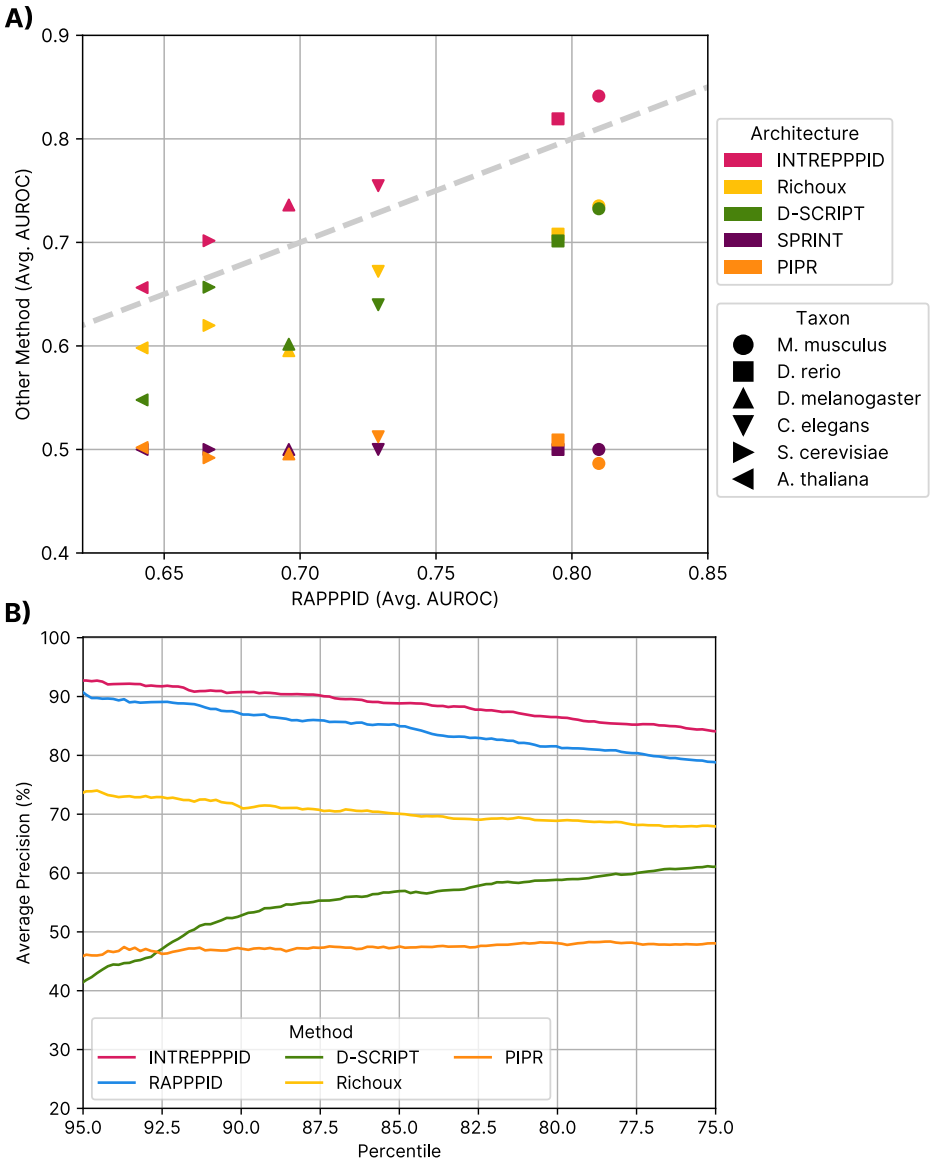


Figure 3: **(A)** Average cross-species AUROC of leading PPI inference methods. The y-axis shows the AUROC of each method (shown as different colours) against the AUROC of RAPPPID (x-axis). Each model was trained on H. sapiens and tested on various species (indicated by the different markers). Markers above the dashed grey line indicate instances where a method outperformed RAPPPID, while markers below the line indicate under-performance. **(B)** The precision of inferred interactions among the kth percentile of the scores inferred by the method indicated by the line colour. The x-axis shows k and the y-axis shows precision along the y-axis. Interactions were inferred for datasets from seven different species. The mean across these datasets is reported by the solid line.

We sought to further confirm INTREPPPID's ability to generalize by measuring the number of interactions in additional PPI databases that INTREPPPID was capable of recovering. When considering interactions with confidence scores of 95% or higher, INTREPPPID was capable of

correctly identifying 72% of interaction in the HIPPIE database and 69% of interaction in the IntAct database. These results are only slightly lower than the 74% true-positive rate INTREPPPID achieves on our benchmark dataset.

INTREPPPID learns protein representations that exhibit strong orthologous locality

Through its orthologous locality task, INTREPPPID learns latent representations of orthologues that are near to one another in the Euclidean space, while ensuring latent representations of nonorthologous sequences remain distant. To qualitatively assess the locality of orthologues in latent space, we calculated the embeddings of the sequences of 1000 randomly sampled proteins present in the OMA database of orthologues and projected them onto a 2D space using UMAP and PCA (Supplementary Fig. S3) [43]. The projections show that embeddings of orthologues cluster tightly in INTREPPPID's latent space, while embeddings from RAPPPID (which has a similar architecture, but without the triplet network) do not.

A more quantitative analysis of embedding distances further confirmed that INTREPPPID's latent space exhibits strong orthologous locality. Namely, the Euclidean distance between latent representations of orthologues was on average 0.822, while the distance between the representations of nonorthologous proteins is on average 5.31 (Supplementary Fig. S3C). Nearly 80% of INTREPPPID's latent representations of orthologues fell under the triplet loss margin of 1, while virtually all nonorthologues had a distance greater than the margin between them.

Together, these analyses confirm that INTREPPPID successfully accomplishes the orthologous locality task.

INTREPPPID's performance gain is due to the biologically relevant information encoded by orthology

To further assess the role of orthology information on the performance of INTREPPPID, we conducted an experiment in which we trained INTREPPPID on two separate datasets: one constructed using orthologue data from the OMA database (called 'not scrambled') and one where 'random' pairs of proteins from the OMA database were labelled orthologues ('scrambled'). After training and testing on both variants of the *H. sapiens* dataset, we found that INTREPPPID's performance dropped to near random in the 'scrambled' case (Supplementary Table S6, AUROC = 0.524). Moreover, on this dataset, neither the classification loss nor the triplet loss significantly changed over 30 000 training steps (Supplementary Fig. S3D).

INTREPPPID's low testing performance on 'scrambled' data suggests that the mere exposure of sequences from various species is not sufficient to increase PPI classification performance. The impact of the 'scramble' dataset on both the classification and triplet losses suggests that the orthologous locality task is complimentary to the PPI classification task. Ablating the biological meaning of the orthology data conversely results in the orthologous locality task negatively impacting PPI classification performance. These results show that the encoding of orthology information increases INTREPPPID's performance due to its biological relevance.

Having established that the central contribution of incorporating orthologues in the INTREPPPID model is the biological information they encode, we continued to test the extent to which they are relevant to INTREPPPID's performance. To do so, we constructed INTREPPPID datasets of *H. sapiens* protein interactions much as we have for previous cross-species analyses herein. For this analysis, however, orthology data related to proteins belonging to any of the testing species (*H. sapiens*, *M. musculus*, *Danio rerio*, *Drosophila melanogaster*, *Caenorhabditis elegans*, *Saccharomyces cerevisiae*, *A. thaliana*) were omitted. Relative to INTREPPPID trained on dataset that included all available orthology data, the altered dataset resulted in worse testing performance metrics across every species tested (Supplementary Table S7). The decrease in performance was least severe when testing on *H. sapiens* (*i.e.* intraspecies inference), having measured a −9.00% change in MCC and −1.36% change in AP scores. Conversely, the performance drop was much greater on testing sets of species different from the training species (*i.e.* cross-species inference), with observed changes as low as −38.85% for MCC and −17.58% for AP scores in some instances. This further confirms that orthology information is used by the model to make predictions.

INTREPPPID learns interactions of evolutionarily conserved functions that RAPPPID does not

To better understand the biological characteristics of the interactions that account for INTREPPPID's performance gains and the role that its orthology loss plays, we analysed the pairs of interacting proteins from our testing datasets in which INTREPPPID predicted a higher probability of interaction than RAPPPID by the widest margin. We found that, on average, INTREPPPID assigns higher scores to interacting pairs in 73% (±11%) of all cases. Of those inferred interactions, the top interactions were disproportionately between proteins that are evolutionarily conserved between species.

Our analysis began with pairs of proteins in the testing dataset that interact and were correctly identified by INTREPPPID (*i.e.* a score ≥ 0.5), and incorrectly identified by RAPPPID (*i.e.* a score < 0.5). To examine cases in which most typified INTREPPPID are performing well while RAPPPID are falling short, we further filtered the aforementioned interactions such that only protein pairs

with the largest differences between the INTREPPPID and RAPPPID scores remained (*i.e.* in the 90th percentile or higher). These scored protein pairs will be referred to as 'top INTREPPPID unique positive hits' hereafter.

A statistical analysis of over-representation of Gene Ontology (GO) Molecular Function annotations was performed on the top INTREPPPID-unique positive hits using the PANTHER Classification System [44–46]. This analysis provided us with an understanding of which GO Molecular Function annotations were over-represented among proteins that comprised the top INTREPPPID-unique positive hits, relative to those in the testing set as a whole.

We found that, across the species tested, the GO Molecular Function annotations that were over-expressed were primarily regarding ubiquinone activity, ribosome structure, and RNA binding (Supplementary Table S8). These functions all belong to pathways that are consistently found to be highly conserved across a broad spectrum of species [18, 47, 48].

An example of one of these interactions is one between the large ribosomal subunit proteins uL24A and uL10 found in *S. cerevisiae* encoded by the genes *RPL26A* and *RPP0*, respectively. INTREPPPID correctly predicts that these two proteins interact with a probability of 94.6% while RAPPPID assigns a probability of 4.52%, both models having been trained exclusively on interactions between human proteins. The two proteins in question are highly conserved, with 131 and 798 orthologues recorded in the OMA database for each protein, respectively. In both cases, a human orthologue is identified by OMA (*RPL26L1* and *RPLP0* genes, respectively).

These observations suggests that INTREPPPID correctly identifies interactions that have functions crucial to pathways such as protein degradation and protein translation that are highly conserved between species, that RAPPPID does not. In addition, these observations provide further support that the discrepancy between INTREPPPID and RAPPPID's performance among interactions composed of highly conserved proteins is driven by the similarity of embeddings among orthologous proteins (*i.e.* the orthologous locality property), which is the primary distinction between INTREPPPID and RAPPPID.

A web server and interface for RAPPPID & INTREPPPID

While we have made best efforts to make RAPPPID and INTREPPPID as simple as possible to run on a local computer with minimal computational requirements, it currently requires knowledge of the CLI and the Python computer language. These prerequisites, while accessible to researchers with relevant experience, can be too great a hurdle for others.

To address this, we have created a website which acts as a graphical interface for a server that is capable of performing proteome-scale inference in under 15 min (https://ppi.bio). The source

code that runs PPI.bio is available publicly on GitHub and can be found by visiting the internet address: https:// github.com/Emad-COMBINE-lab/ppi.bio. Many online academic servers are no longer available online and do not provide their source openly, resulting in the total loss of these servers (see Supplementary Table S9 for a review of 22 such resources, 50% of which are no longer alive). We hope to enable others to run their own server in the event that https://ppi.bio goes off line by releasing its code and documentation.

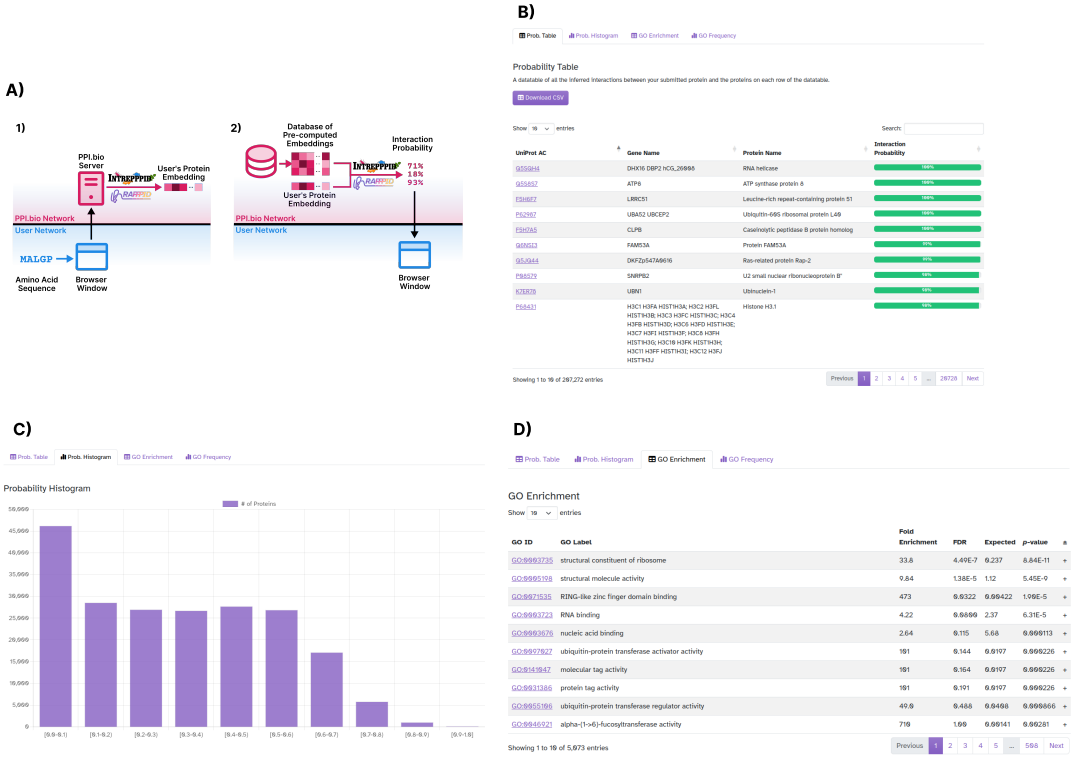


Figure 4: The webserver of PPI.Bio supporting RAPPPID and INTREPPPID at whole proteome scale. **(A)** Overview of the operation of the PPI.Bio PPI inference server. In the first stage (leftmost panel), amino acid sequences are encoded into protein embeddings using either RAPPPID or INTREPPPID, depending on the user's preference. In the second stage (rightmost panel), precomputed protein embeddings of a specified species proteome are loaded from a database. The classification head of either RAPPPID or INTREPPPID is then used to compute the probability of interaction of the user-specified protein and the precomputed embeddings. Screenshot of the **(B)** data table of interaction probabilities, **(C)** probability histograms, and **(D)** GO enrichment analysis powered by PANTHERDB, as seen on PPI.Bio after inferring proteome-wide interactions.

The process for making proteome-wide inferences on user-submitted amino acid sequence is as follows. Users input amino acid sequences that are then submitted to the PPI.bio server (Fig. 4). The PPI.bio server in turn encodes the sequence using the user's choice of either RAPPPID or INTREPPPID's encoder to obtain a protein embedding. The embedding is then paired with precomputed embeddings of all proteins belonging to the proteome of the user's choosing (currently, proteins for six organisms have been computed). The probability of these pairs of proteins interacting are then inferred using RAPPPID or INTREPPPID's classification head (based on the user's choice), and returned to the user's browser window.

PPI.bio also enables users to conduct GO enrichment analysis on the set of proteins that have an interaction probability greater than some user-specified threshold. PPI.bio uses the PANTHER online API [44] to calculate GO enrichment statistics. This functionality allows users to find instances of GO annotations being statistically over-represented among proteins that were inferred to interact with the protein supplied by the user.

Discussion

This study introduces three tools for researchers in their pursuit of better understanding PPIs: a cross-species method for PPI inference (INTREPPPID), a software package for creating strict PPI datasets (PPI Origami), and a web server for inferring PPIs using INTREPPPID and RAPPPID (PPI.bio). These tools and their code are all freely available under open licenses with complete documentation.

We view INTREPPPID's novel contributions as being two-fold. First, INTREPPPID is among the few sequence-based PPI inference methods that are specifically designed with the purpose of addressing the challenge of accurately inferring protein interactions between proteins of species other than that it was trained on (*i.e.* cross-species inference). The only other such methods specifically designed for this purpose known to us are D-SCRIPT and PIPE4 [40, 49]. Of these two methods, only D-SCRIPT's software is made publicly available, which has made its inclusion in the analyses herein possible. We believe that INTREPPPID joining D-SCRIPT in being the only publicly available sequence-based cross-species PPI inference methods represents a significant contribution to the community.

Secondly, INTREPPPID introduces a novel orthologous locality loss and quintuplet architecture which, as we demonstrated, is responsible for a significant increase in both intra- and crosss-pecies PPI inference performance under strict evaluation conditions. While INTREPPPID has a quintuplet network, there is no appreciable increase in trainable parameters compared to RAPPPID's twin network construction due to the parameter sharing between encoders (2.20E5 and 1.88E5 parameters, respectively). By sharing parameters, INTREPPPID reduces both its GPU memory footprint as well as the risk of over-fitting via over-parameterization.

The orthologous locality task can be viewed through the lens of contrastive self-supervised learning (SSL) techniques like 'SimCLR' or Time-Contrastive Networks [50, 51]. Specifically, orthologous sequences can be thought of as multiple views of the same protein augmented by an evolutionary process in a way that is analogous to how SimCLR compares pairs of images that have been augmented by, *e.g.* cropping or blurring. In both instances, the model is learning invariance between entities that are similar in their desired feature space, but are significantly different in the input space. By creating a feature space that is invariant to the effects of speciation

on amino acid sequences, we position our model to capture more interologues. While SimCLR only uses pairs of contrasting examples, SSL methods like time-contrastive networks have adopted a triplet loss to more efficiently learn embeddings that better fit downstream tasks.

Sequence-based methods rely on a large number of PPI examples to learn to recognize features in pairs of amino acid sequences that indicate two pairs of proteins are likely to interact. The more examples provided, the better models become at identifying features that we know from experimental knowledge are important for mediating interactions.

As is the case with all sequence-based methods for PPI inference, INTREPPPID risks becoming brittle and only memorizing sequences found in the training set or learning trivial solutions such as reporting the sequence similarity of two proteins as their likelihood of interaction. In the absence of more experimental data to train on, ensuring the data are of high-quality, validating strictly and judicious application of regularization are the primary ways to address these generalization concerns. Our previous work on RAPPPID underscores how these methods can be used to create generalized, performant models that do not rely on trivial solutions like comparing sequence similarity nor memorizing sequences.

While INTREPPPID is shown here to outperform leading methods in the intra- and cross-species tasks, it is important to note that deep learning, sequence-based methods like ours can sometimes struggle with the changes observed by certain protein variants and isoforms. Future work might help address this by constructing architectures that provide inductive biases relevant to learning binding domains and residues.

Additionally, all of the sequence-based methods tested infer interactions that occur under unspecified intrinsic (*e.g.* protein conformation, post-translational modifications) and extrinsic (*e.g.* subcellular localization, pH, temperature) conditions (a property inherited from the training STRING dataset). While databases such as Reactome richly annotate interactions, including information about, *e.g.* PTMs and subcellular localizations, the number of interactions and organisms in the database is limited due to the high-labour costs associated with manual curation [52]. Future work towards reconciling and augmenting existing PPI datasets and methods with such annotations could possibly help to provide valuable context to the interactions provided.

The second tool presented in this work is PPI Origami, a framework for creating new strict PPI datasets. While many reference or 'gold-standard' PPI datasets exist, they each bring with them different underlying assumptions and limitations [11–13]. Researchers might wish to take advantage of large, comprehensive PPI databases like the STRING database used to create the aforementioned strict PPI datasets or alternatively take a specialist approach and interrogate smaller curated networks that describe certain organisms, diseases, or cell types [53–55]. They may also wish to derive their training set from one dataset and their evaluation datasets from another, much as we did over the course of our experiments. PPI Origami provides tools for

creating PPI datasets from sources the user specifies, while automating the process of ensuring that the data conform to the criteria outlined herein. By providing tools and frameworks rather than one set of datasets, we hope to impower researchers to adopt PPI datasets that (i) do not leak data and (ii) conform to the specific needs of the researcher.

The final major contribution herein is PPI.bio, a web server for inferring PPIs using RAPPPID and INTREPPPID. PPI.bio infers interactions entirely on the Central Processing Unit (CPU), greatly reducing server costs. Both RAPPPID and INTREPPPID facilitate this by allowing known protein sequences to be precomputed, only requiring a forward pass of the shallow fully connected classification network when the embeddings for both proteins has already been calculated. This allows for significant speed-ups when doing proteome-wide inferences.

As time passes, online services published in academic articles tend to go off line or otherwise become inaccessible. After conducting a search for the terms 'protein-protein interaction portal' and 'protein-protein interaction website' on the online academic search engine Semantic-Scholar.org and randomly sampling 22 articles that introduce online tools, 45% of the URIs either did not resolve or no longer offered the service introduced (Supplementary Table S9). To help ensure that PPI.bio remains accessible for years to come, we've made the code that runs PPI. bio available under a free and open license. To help facilitate the deployment of PPI.bio, the server is designed to be easily deployed to the Platform as a Service (PaaS) service Heroku as well as the self-hosted alternative Dokku.

Key Points

- Most PPI studies are conducted in a select few organisms, leading to a 'species gap'.
- Performant cross-species PPI methods are required to bridge this species gap.
- We introduce INTREPPPID, which improves cross-species performance using a novel quintuplet architecture to incorporate orthology data.
- INTREPPPID outperforms leading PPI inference methods on both intra and cross-species tasks across multiple metrics.
- We also have released an online interface for INTREPPPID (PPI.bio) and a tool for making PPI datasets (PPI Origami).

Supplementary data

Supplementary data are available at *Briefings in Bioinformatics* online.

Author contributions

A.E. and J.S. contributed to project conceptualization, methodology as well as writing, editing, and reviewing. A.E. contributed to supervision and funding acquisition. J.S. contributed to software, data curation, investigation and visualization. All authors read and approved the final article.

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Software & data availability

INTREPPPID and the PPI.bio server are freely available under the Affero General Public License version 3.0. PPI Origami is licensed under the GNU General Public License version 3.0. They can be downloaded from https://github.com/Emad-COMBINE-lab/intrepppid, https://github.com/Emad-COMBINE-lab/ppi.bio, and https://github.com/Emad-COMBINE-lab/ppi_origami, respectively. All code and datasets have been deposited in the Zenodo repository under record numbers 10652231, 10652229, 10652233, and 10594149.

References

1. Cheng F, Zhao J, Wang Y, *et al.* Comprehensive characterization of protein–protein interactions perturbed by disease mutations. *Nat Genet* 2021;**53**:342–53. https://doi.org/10.1038/s41588-020-00774-y.

2. Richards AL, Eckhardt M, Krogan NJ. Mass spectrometry-based protein–protein interaction networks for the study of human diseases. *Mol Syst Biol* 2021;**17**:e8792. https://doi.org/10.15252/msb.20188792.

3. Gordon DE, Jang GM, Bouhaddou M, *et al.* A SARS-CoV-2 protein interaction map reveals targets for drug repurposing. *Nature* 2020;583:459–68. https://doi.org/10.1038/s41586-020-2286-9.

4. Mabonga L, Kappo AP. Protein-protein interaction modulators: advances, successes and remaining challenges. *Biophys Rev* 2019;11:559–81. https://doi.org/10.1007/s12551-019-00570-x.

5. Rolland T, Taşan M, Charloteaux B, *et al.* A proteome-scale map of the human Interactome network. *Cell* 2014;159:1212–26. https://doi.org/10.1016/j.cell.2014.10.050.

6. Luck K, Kim D-K, Lambourne L, *et al.* A reference map of the human binary protein interactome. *Nature* 2020;580:402–8. https://doi.org/10.1038/s41586-020-2188-x.

7. Schoch CL, Ciufo S, Domrachev M, *et al.* NCBI taxonomy: a comprehensive update on curation, resources and tools. *Database* 2020;**2020**:baaa062. https://doi.org/10.1093/database/baaa062.

8. Szklarczyk D, Kirsch R, Koutrouli M, *et al.* The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Res* 2023;**51**:D638–46. https://doi.org/10.1093/nar/gkac1000.

9. International commission on zoological nomenclature. *International code of zoological nomenclature* 1999;**1999**:1–344.

10. Low TY, Syafruddin SE, Mohtar MA, *et al.* Recent progress in mass spectrometry-based strategies for elucidating protein-protein interactions. Cell *Mol Life Sci* 2021;**78**:5325–39. https://doi.org/10.1007/s00018-021-03856-0.

11. Park Y, Marcotte EM. Flaws in evaluation schemes for pair-input computational predictions. *Nat Methods* 2012;**9**:1134–6. https://doi.org/10.1038/nmeth.2259.

12. Dunham B, Ganapathiraju MK. Benchmark evaluation of protein–protein interaction prediction algorithms. *Molecules* 2021;**27**:41. https://doi.org/10.3390/molecules27010041.

13. Bernett J, Blumenthal DB, List M. Cracking the black box of deep sequence-based protein-protein interaction prediction. *Briefings in Bioinformatics* 2024;**25**:bbae076. https://doi.org/10.1093/bib/bbae076.

14. Hamp T, Rost B. More challenges for machine-learning protein interactions. *Bioinformatics* 2015;31:1521–5. https://doi.org/10.1093/bioinformatics/btu857.

15. Marcotte EM, Pellegrini M, Ng H-L, *et al.* Detecting protein function and protein-protein interactions from genome sequences. Science 1999;285:751–3. https://doi.org/10.1126/science.285.5428.751.

16. Walhout AJM, Sordella R, Lu X, *et al.* Protein interaction mapping in C. Elegans using proteins involved in Vulval development. *Science* 2000;**287**:116–22. https://doi.org/10.1126/science.287.5450.116.

17. Matthews LR, Vaglio P, Reboul J, *et al.* Identification of potential interaction networks using sequence-based searches for conserved protein-protein interactions or 'interologs'. *Genome Res* 2001;**11**:2120–6. https://doi.org/10.1101/gr.205301.

18. Sharan R, Suthram S, Kelley RM, *et al.* Conserved patterns of protein interaction in multiple species. *Proc Natl Acad Sci* 2005;**102**: 1974–9. https://doi.org/10.1073/pnas.0409522102.

19. Cortes C, Vapnik V. Support-vector networks. *Mach Learn* 1995;20:273–97. https://doi.org/10.1007/BF00994018.

20. Ben-Hur A, Noble WS. Kernel methods for predicting proteinprotein interactions. *Bioinformatics* 2005;**21**:i38–46. https://doi.org/10.1093/bioinformatics/bti1016.

21. Richoux F, Servantie C, Borès C, *et al.* Comparing two deep learning sequence-based models for protein-protein interaction prediction. *arXiv:190106268 [cs, q-bio, stat]* 2019.

22. Bromley J, Bentz JW, Bottou L, *et al.* Signature verification using a "siamese" time delay neural network. *Int J Patt Recogn Artif Intell* 1993;**07**:669–88. https://doi.org/10.1142/S0218001493000339.

23. Baldi P, Chauvin Y. Neural networks for fingerprint recognition. *Neural Comput* 1993;**5**:402–18. https://doi.org/10.1162/ neco.1993.5.3.402.

24. Koch G, Zemel R, Salakhutdinov R. Siamese Neural Networks for One-shot Image Recognition. *Proceedings of the 32nd International Conference on Machine Learning*. Lille, France, 2015;**37**.

25. Chen M, Ju CJ-T, Zhou G, *et al.* Multifaceted protein–protein interaction prediction based on Siamese residual RCNN. *Bioinformatics* 2019;**35**:i305–14. https://doi.org/10.1093/bioinformatics/btz328.

26. Cho K, van Merrienboer B, Gulcehre C. *et al.* Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation. *arXiv:1406.1078 [cs.CL]* 2014.

27. Szymborski J, Emad A. RAPPPID: towards generalizable protein interaction prediction with AWD-LSTM twin networks. *Bioinformatics* 2022;**38**:3958–67. https://doi.org/10.1093/bioinformatics/btac429.

28. Merity S, Keskar NS, Socher R. Regularizing and optimizing LSTM language models. *arXiv:170802182 [cs]* 2017.

29. Kudo T, Richardson J. SentencePiece: A simple and language independent subword tokenizer and detokenizer for Neural Text Processing. In: Blanco E, Lu W (eds), *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations.* Brussels, Belgium: Association for Computational Linguistics, 2018;66–71.

30. Kudo T. Subword regularization: improving neural network translation models with multiple subword candidates. *arXiv:180410959 [cs]* 2018.

31. Misra D. Mish: a self regularized non-monotonic activation function. *arXiv:190808681 [cs, stat]* 2020.

32. Zhang X, Yu FX, Kumar S, *et al.* Learning Spread-Out Local Feature Descriptors. 2017 *IEEE International Conference on Computer Vision (ICCV)*. Venice, Italy: IEEE, **2017**;4605–13.

33. Altenhoff AM, Train C-M, Gilbert KJ, *et al.* OMA orthology in 2021: website overhaul, conserved isoforms, ancestral gene order and more. *Nucleic Acids Res* 2021;**49**:D373–9. https://doi.org/10.1093/nar/gkaa1007.

34. Schroff F, Kalenichenko D, Philbin J. FaceNet: a unified embedding for face recognition and clustering. 2015 *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 2015;815–23.

35. Forslund K. *The relationship between orthology, protein domain architecture and protein function.* Sweden: University of Stockholm in Stockholm, 2011.

36. Fang G, Bhardwaj N, Robilotto R, *et al.* Getting started in gene Orthology and functional analysis. *PLoS Comput Biol* 2010;**6**:1–8. https://doi.org/10.1371/journal.pcbi.1000703.

37. Wright L, Demeure N. Ranger21: a synergistic deep learning optimizer, *arXiv:2106.13731 [cs.LG]* 2021. http://arxiv.org/abs/2106.13731.

38. Loshchilov I, Hutter F. Decoupled Weight Decay Regularization. *7th International Conference on Learning Representations* 2019.

39. Iyer N, Thejas V, Kwatra N. *et al.* Wide-minima Density Hypothesis and the Explore-Exploit Learning Rate Schedule *arXiv:2003.03977 [cs.LG]* 2021.

40. Sledzieski S, Singh R, Cowen L, *et al.* D-SCRIPT translates genome to phenome with sequence-based, structure-aware, genome-scale predictions of protein-protein interactions. cels 2021;**12**:969–982.e6. https://doi.org/10.1016/j.cels.2021.08.010.

41. Li Y, Ilie L. SPRINT: ultrafast protein-protein interaction prediction of the entire human interactome. *BMC Bioinformatics* 2017;**18**:485. https://doi.org/10.1186/s12859-017-1871-x.

42. Venkatesan K, Rual J-F, Vazquez A, *et al.* An empirical framework for binary interactome mapping. Nat Methods 2009;**6**:83–90. https://doi.org/10.1038/nmeth.1280.

43. McInnes L, Healy J, Melville J. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction *arXiv:1802.03426 [stat.ML]* 2020.

44. Thomas PD, Ebert D, Muruganujan A, *et al.* PANTHER: making genome-scale phylogenetics accessible to all. Protein Sci 2022;**31**: 8–22. https://doi.org/10.1002/pro.4218.

45. Mi H, Muruganujan A, Huang X, *et al.* Protocol update for large-scale genome and gene function analysis with PANTHER classification system (v.14.0). *Nat Protoc* 2019;**14**:703–21. https://doi.org/10.1038/s41596-019-0128-8.

46. The Gene Ontology Consortium, Aleksander SA, Balhoff J, *et al.* The gene ontology knowledgebase in 2023. *Genetics* 2023;**224**:iyad031. https://doi.org/10.1093/genetics/iyad031.

47. Kelley BP, Sharan R, Karp RM, *et al.* Conserved pathways within bacteria and yeast as revealed by global protein network alignment. *Proc Natl Acad Sci* 2003;**100**:11394–9. https://doi.org/10.1073/pnas.1534710100.

48. Siepel A, Bejerano G, Pedersen JS, *et al.* Evolutionarily conserved elements in vertebrate, insect, worm, and yeast genomes. *Genome Res* 2005;**15**:1034–50. https://doi.org/10.1101/gr.3715005.

49. Dick K, Samanfar B, Barnes B, *et al.* PIPE4: fast PPI predictor for comprehensive inter- and cross-species Interactomes. *Sci Rep* 2020;**10**:1390. https://doi.org/10.1038/s41598-019-56895-w.

50. Chen T, Kornblith S, Norouzi M. *et al.* A Simple Framework for Contrastive Learning of Visual Representations *arXiv:2002.05709 [cs.LG]* 2020.

51. Sermanet P, Lynch C, Chebotar Y. *et al.* Time-Contrastive Networks: Self-Supervised Learning from Video *arXiv:1704.06888 [cs.CV]* 2018.

52. Jassal B, Matthews L, Viteri G, *et al.* The reactome pathway knowledgebase. *Nucleic Acids Res* 2019;**48**:D498–D503. https://doi.org/10.1093/nar/gkz1031.

53. Alanis-Lobato G, Möllmann JS, Schaefer MH, *et al.* MIPPIE: the mouse integrated protein–protein interaction reference. *Database (Oxford)* 2020;*2020*:baaa035. https://doi.org/10.1093/database/baaa035.

54. Kusari M, Dey L, Mukhopadhyay A. ChikvInt: a chikungunya virus–host protein–protein interaction database. *Lett Appl Microbiol* 2022;**74**:992–1000. https://doi.org/10.1111/lam.13677.

55. Kalman ZE, Dudola D, Mészáros B, *et al.* PSINDB: the postsynaptic protein–protein interaction database. Database 2022;**2022**:baac007. https://doi.org/10.1093/database/baac007.

Chapter 5

Discussion

The studies which comprise this thesis leave its author with a several key insights into PPI inference methods that underscore their current limitations and paths towards addressing them. These insights are outlined here in detail.

Appropriate regularization methods

The application of regularization is a central component to the out-of-distribution performance of both RAPPPID and INTREPPPID, as demonstrated by ablation experiments. To be effective, however, the method of regularization must be chosen to appropriately reflect the architecture of the model and the underlying data. RAPPPID and INTREPPPID offer an interesting look into how progressively overlapping, specifically-targeted regularization can result in gains in model robustness and generalization.

Regularization in RAPPPID

Much of the regularization applied in the training of RAPPPID emerged from work by Stephen Merity *et al.* on adapting existing methods of regularization for use with LSTMs [164]. Among the methods Merity *et al.* describe, the one that resulted in the greatest improvement in their testing is employing weight-dropped LSTMs. Weight-dropping, also known as DropConnect, is a method of regularization similar to Dropout, but operates at the level of model weights rather than network activations [170, 171]. Comparisons to other architectures show that dropping out the hidden activations of LSTMs greatly underperforms weight-dropping [164]. Indeed, applying dropout to LSTM activations has been identified for quite some as a point of frustration when training recurrent models time, resulting in "dramatic" changes in model dynamics [172].

In our own ablation experiments conducted as part of the development of RAPPPID, we found that tokenizing amino acid sequences had the greatest effect on model performance on an out-of-distribution testing set. Tokenization improves RAPPPIDs performance by simultaneously reducing the number of timesteps, thereby mitigating the risk of vanishing gradients. Tokenization increases the pool of possible inputs at every timestep, as they no longer are limited to 22 amino acids, but any number of their permutations. Increasing the vocabulary like so can exacerbate the risk of overfitting. Luckily in the case of RAPPPID, the regularization opportunities afforded by tokenization greatly outweighed the risk of overfitting. The Unigram tokenization algorithm via the SentencePiece library was used to tokenize amino acid sequences. This algorithm allows for subword regularization, a process by which multiple segmentations for a sequence are sampled. For instance, the first 10 residues of the sequence for Human insulin (`MALWMRLLPL`) can be represented by the following sequence of tokens:

- `M, AL, W, M, R, L, L, PL`
- `M, A, L, W, M, R, LL, P, L`
- `M, A, L, W, M, RL, L, P, L`

and *etc*. One of all possible segmentations of the sequence are then chosen at random and used as the input to the model in question. The noise introduced by this process is regularizing and helps avoid the problem of rare tokens, whereby some tokens are under sampled at training time. Additionally, as part of the recommendations of the Merity *et al.* study, "embedding dropout" is employed, which is a process by which the embeddings of certain tokens are randomly zeroed out. This regularizes the network in a similar way to random dropout on the activations or weights of a network, but particularly targets the problem of overfitting to the large vocabulary that was created using the Unigram algorithm.

Regularization in INTREPPPID

The focus on choosing regularization methods for RAPPPID centred on their appropriateness for the overall architecture. INTREPPPID inherited the architecture-focused regularization of RAPPPID, but also explored regularization methods that were informed by biological expert knowledge and inductive biases. Specifically, we sought to leverage our knowledge of orthologous protein sequences to improve the generalization of the model on testing examples from species other than those it was trained.

We took inspiration from a common regularization technique used in the field of computer vision where images are subtly augmented such that many pixels are altered, but their semantic meaning

is unchanged [173]. Orthologous proteins are similarly altered subtly by evolution, but typically retain their function and purpose. Incidentally, the application of augmentation techniques to proteins was simultaneously explored by Sun *et al.*, albeit by directly and randomly altering sequences rather than by using orthologous proteins [174]. This direct mutation approach scales much more easily than our orthologous method, as you are not limited by the number of orthologous sequences found in curated knowledge-bases. Unlike using orthologous protein sequences, randomly altering sequences can however result in proteins which lose their semantic meaning — much as an image of the number 9, when rotated 180°, loses its semantic meaning and becomes a 6.

The final result of adopting the orthologous locality loss introduced in the INTREPPPID study is a regularized latent space where proteins which are similar in function across species share similar vector embeddings.

Questions regarding sequence identity and its role in data leakage

The three studies reproduced in this thesis are greatly indebted to the work of Park and Marcotte, which identified their strictest dataset type (C3) as having "test pairs sharing neither protein with the training set" [13]. Membership of a protein in either the testing, validation, set can be defined a number of ways. For instance, membership can be determined by shared UniprotKB identifiers of proteins. The trouble, however, in using protein names and most identifiers is that many protein names, while unique, have sequences that are identical, or nearly so. For instance, the January 2025 release of Uniprot has two entries for the human protein "Cytochrome c oxidase subunit 3" with identifiers `A0A023Q4K2` and `Q4EZD1` which differ by one additional serine residue in the former.

To help identify proteins that are identical in all but name, clustered identifiers such as those pre-computed in the UnirRf100 database, are commonly used. The two Uniprot identifiers for human Cytochrome c oxidase subunit 3 map to a single UniRef100 identifier (`UniRef100_A0A023Q4K2`). By mapping Uniprot identifiers to UniRef100 identifiers, it is possible to avoid considering proteins with identical sequences as distinct proteins.

A consequence of the high degree of homology between proteins is that many proteins approach, but fall short of 100% sequence identity. These homologous proteins often share the same function and interact with similar proteins, and therefore likely are a source of data leakage and warrant removal. Highly-conserved proteins like insulin (`P01308`) are prolific examples of this phenomena, sharing at least 90% sequence identity with 14 other proteins in Uniprot. Matters

get much worse when one expands the similarity threshold to at least 50%, revealing 53 other homologous proteins in Uniprot from species as diverse as *Rattus norvegicus* (Rat) and *Gorilla gorilla gorilla* (Western lowland gorilla). Once more, the UniRef database allows us to map similar proteins to the same identifiers. For example, proteins with at least 90% sequence identity and 50% sequence identity to insulin can be mapped to the identifiers `UniRef90_P01308` and `UniRef50_P01308`, respectively.

Park and Marcotte used, without a stated justification, a threshold of 40% sequence identity to determine protein similarity [13]. This threshold is used by many others when establishing a similarly strict training/testing split without much or often any discussion as to how it was selected. For instance, Bernett *et al.* used the CD-HIT clustering software to establish validation folds with no more than 40% sequence identity between them [15, 175]. Prior to that study, Hamp and Rost used a sequence-structure alignment database (HSSP) "...correspond[ing] to 40% pairwise sequence identity for 250 aligned residues" [14, 176, 177].

While the rationale for these seemingly arbitrary thresholds is not provided in those studies, one begins to emerge after a thorough review of the relevant literature, though there is no guarantee it is shared with the original authors. It is possible that these thresholds, particularly in the case of Hamp and Rost, were influenced by a 1991 study by Sander and Schneider which found that protein sequences with more than 30% sequence identity often share similar structure [176]. Subsequent work by Burkhard Rost found that proteins identified as having between 20-35% sequence identity, a so-called protein-sequence alignment "twilight zone", are far more likely to be "false negatives" [178]. In this context, false negatives are taken to mean proteins which, despite falling below the sequence identity threshold, are structurally similar and false positive are proteins above the threshold which do not share structural similarity. Rost reports:

> The number of false positives rose by a factor of 5 000 (Figure 2B). Hence, separating true and false positives switched from a trivial task (above 35% [pairwise sequence identity]) to the problem of finding needles in a haystack (20–30%). [178]

In light of these findings, a threshold of 40% sequence identity appears to be a fair compromise between identifying as many structurally similar proteins as possible while greatly reducing the number of proteins false-positives.

While we have identified a possible rationale for why the 40% sequence identity threshold has been identified — it is a proxy for the presence of structural similarity — we have yet to examine any evidence as to whether it is reasonable or useful. I argue here that eliminating nonindependence between the training and testing set must be balanced with creating unduly biased datasets.

Supervised machine learning algorithms rely on example input-output pairs to learn a generalized mapping between the two. These training examples are required to be representative of the

variation observed in the universe of input-output pairs that are expected to be correctly inferred. To understand whether a trained model is performing at the level intended, a set of evaluation input-output pairs must be chosen. If it is to give us an accurate representation of the model's performance, the evaluation set must simultaneously (1) be representative of the problem space and (2) avoid data leakage. Data leakage is a topic well discussed throughout this thesis, however it is useful here to use the taxonomy outlined by Kapoor and Narayanan to identify some of its common subtypes [160]. In the framework of this taxonomy, the Park and Marcotte study identified examples of L3.2 and L1.4 errors — *i.e.*, "nonindependence between training and test samples" and "duplicates in datasets", respectively [13, 160].

The L1.4, or duplication, error describes instances where duplicate data is present in the dataset increasing the risk that "the same data could exist in the training set and the test set" [160]. As explored previously, homologous sequences result in nearly identical sequences, and therefore PPI datasets which do not account for homology will almost certainly observe an L1.4 data leakage. By applying a maximum sequence identity threshold between testing and training datasets of 40%, Park and Marcotte side-step the issue of duplicates. A 100% threshold, however, would also eliminate the presence of duplicate sequences. A further 5-10% allowance for small variations corresponding to threshold between 95-90% would be appropriately sufficient to eliminate near-duplicates.

The L3.2, or nonindependence, error describes instances where testing samples are not drawn independently from the distribution of the problem space, but rather from one present in the training set that is not central to the scientific claim of the model. Examples provided by Kapoor and Narayanan include histology classification models which are evaluated on datasets of patients also present in the training set, or protein function classification models where a given protein family is present in both training and testing sets [179, 180]. In the case of PPI inference models, it is uncontroversial to assert that Park and Marcotte identified the L3.2 error caused by shared proteins in testing and training sets and remediated it by removing the dependence between the training and testing examples. It remains to be explored, however, which precise dependency was eliminated. Was it a dependency between shared *proteins* in the testing and training set, or rather a dependency between *homologous* proteins? Depending on one's interpretation of the role of the chosen sequence identity threshold, perhaps even a dependency on shared protein families is being avoided. If the former, most threshold values would be sufficient to mitigate the error. If the latter, then 40% would appear to be somewhat of an upper-bound among threshold values that would capture structural similarity, as was discussed earlier in reviewing the works of Sander and Schneider and Rost [176, 178].

The murkiness of this question is to be anticipated, according to Kapoor and Narayanan, who

warn:

> Handling nonindependence between the training and test sets in general (*i.e.*, without any assumptions about independence in the data) is a hard problem, because we might not know the underlying dependency structure of the task in many cases.[160]

Indeed, it is not easy to disentangle which aspects of the training and evaluation set distribution is intrinsic to learning a generalized model, and which serve only to artificially inflate the testing performance of the model via data leakage. While it may be tempting at first to resort to lower thresholds in the range of 30-50%, as many studies have done, there is a risk that models are being needlessly deprived of training data causing gaps in the coverage protein families which have representative samples in the training set. An over-correction could result in exacerbating the risk of biasing the training data, reducing its magnitude and variance, and ultimately resulting in overfitting.

Additionally, there is a question of whether stringent sequence identity thresholds are effective methods for excluding proteins on the basis of their protein family membership. Rost reminds us that his findings echo those of the earlier Sander and Schneider study: "...high levels of sequence similarity or identity do *not* ascertain structural similarity..." [178].

Further study into the extent of dependence between the training and testing splits of PPI datasets, at what level it occurs, and possible mitigation strategies, is gravely needed. Until then, interpretation of the findings in the works presented herein offer a suggestion of what such a study might confirm.

In the study in which it was introduced, RAPPPID was trained and tested on a C3-style dataset, however protein homology was not accounted for [10]. This is to say, proteins were excluded solely on the basis of shared STRING protein identifiers across training and testing datasets. The INTREPPPID study, by contrast, saw a RAPPPID model trained on a C3-style dataset with a 90% maximum sequence identity threshold between training and testing splits [9]. Introducing the 90% maximum sequence identity threshold results in PPI inference performance, as measured by AUROC, dropping by $\approx$ 8%. The last study presented in this thesis includes an analysis in the change of performance of PLLM models when trained on PPI datasets with maximum sequence identity threshold between their training and testing sets ranging between 50% to 90% [181]. Over that range, very little variation is observed. While further studies are required to draw any conclusions, we are left with the suggestion that, while it is pertinent to account for protein homology, little if any data leakage is mitigated by varying the maximum sequence identity threshold over the range of 50% to 90%.

Distal impacts of efficient architectures

As a means to improve generalization

RAPPPID's encoder, which make up the majority of the network by neuron count, is constructed using a twin neural network architecture. By sharing weights between the encoders of each protein, RAPPPID effectively halves the number of weights needed to infer PPIs. While the forward pass requires the same number of floating point operations (FLOPS), the number of FLOPS required for the backward-pass is halved. Seeing as the back-propagation is far more computational-intensive than the forward pass, the backward-pass is the more relevant target for optimization. A study by Richoux *et al.* confirms that twin neural network constructions observe far less overfitting than a twin construction [138]. This finding exemplifies the so-called "bias/variance dilemma" that relates the increase in neurons of a neural network to an increase in variance, leading to overfitting [182].

The INTREPPPID expands this concept by adding an additional three passes to compute an orthology-related task, forming what we've come to call a "quintuplet" network. This allows us to leave the number of effective weights in the network unchanged, increase the amount of data the network can learn from, and add an additional source of inductive bias. The result is a network that generalizes better than RAPPPID, despite maintaining the same number of weights, requires the same number of FLOPS for PPI inference, and requires a comparable amount of time to train.

High-quality, unbiased sources of biological data are difficult to come by due to the cost associated with generating such data, and PPI pairs are no exception. After filtering out low confidence interaction pairs and creating a strict validation dataset, only 39 229 training PPI pairs were used to train INTREPPPID. This stands in comparison to the 220 000 parameters that make-up INTREPPPID's encoder, alone. By increasing the amount of data the model is able to learn from (in the form of the orthologous locality task) while leaving the number of parameters unchanged, INTREPPPID reduces the degree of over-parameterization that is common among contemporaneous PPI inference methods.

It's important to note, however, that recent studies have found that over-parameterization in deep neural networks, past some (ill-defined) point, can cease to cause overfitting and result in well-calibrated, generalized models [183]. Further, it has been found that the variance-bias trade-off is not universally applicable [184]. These developments are, in part, leveraged in the study presented in Chapter 5. Briefly, PLLM's offer an interesting solution to the over-parameterization problem that plagues PPI inference models, along with many other protein-related tasks which suffer from constrained training datasets.

Lowering barriers to PPI inference research

> *"Necessity is the mother of invention."*
> — Proverb

Computational resources are not equally distributed amongst researchers, not at the global scale nor most regional scales. An indication of this is that the top 10 super-computers by performance on the LINPACK benchmark all belong to countries in the Global North, according to TOP500 as of June 2025 [185, 186]. While not a perfect measure of the accessibility of computational resources, it follows that access is greater in wealthy countries that can and do allocate more funds to public and private research in the form of grants and tax credits.

Access to computational hardware is an important factor in being able to explore and successfully study research questions in the field of deep learning. In her influential essay "The Hardware Lottery", Sara Hooker argues that "[i]n the field of artificial intelligence research ... it is our tooling which has played a disproportionate role in deciding what ideas succeed (and which fail)" [187]. Over the course of my doctoral studies, I've had the good fortune to have had far greater access to computational resources than the majority of individuals globally. Despite this, I had access to many orders of magnitude fewer GPU hours and video random-access memory (VRAM) than leading research groups, particularly in the private sector. To narrow this gap, I applied a great focus on efficient deep learning architectures to make the most of the resources made available to me. This approach is advocated by Hooker as a way to mitigate the disparity in access to computational resources:

> Software needs to do more work, but it is also well positioned to do so. We have neglected efficient software throughout the era of Moore's law, trusting that predictable gains in compute would compensate for inefficiencies in the software stack. This means there are many low hanging fruit as we begin to optimize for more efficient code... [187]

A out-sized contributor to this hardware inequity on the side of software is the overwhelming popularity of the compute-hungry transformer architecture [123]. An oft-cited concern regarding the computational intensity of transformers is the self-attention mechanism at their heart, whose complexity scales quadratically with the length of its input sequence [188, 189]. As a result, the attention mechanism of the transformer architecture is often the target of improvement and optimization [190, 191, 192, 193].

Despite this, Iandola *et al.* found that the fully-connected layers in the feedforward network is responsible for the overwhelming share of FLOPS (73%) and latency on mobile devices (69.4%) in traditional transformer architectures [194]. They proceed to develop a convolution-based

alternative to the fully-connected layers in the feedforward modules of the transformer architecture [194]. Using this construction, we were capable of training SqueezeProt, a pLLM with less than 7% of the number of parameters of leading pLLM ProtBERT, computes tokens 5× faster, and achieves comparable PPI inference accuracy. VRAM is often the primary hardware bottleneck when training large neural networks and is commonly the main price differentiator between GPUs of the same hardware architecture. While SqueezeProt-U50 was trained on four Nvidia RTX 3090 GPUs, the relatively-low number of parameters allows for SqueezeProt to be trained on one such GPU. The RTX 3090 is a consumer GPU which had an MSRP of $1 499 USD at launch [195]. By comparison, ProtBERT was trained on two supercomputers — one of which is, as its authors describe, "the world's second fastest computer" comprised of 27 708 Nvidia Tesla V100 GPUs [148]. The Nvidia Tesla V100 is a data center GPU with an MSRP of $10 000 USD at launch [196].

Through the early adoption of efficient neural network architectures, we were able to conduct research on topics which are dominated by laboratories with asymmetric access to computational resources. Despite this victory, the inequities of the hardware lottery are unlikely to be optimized away by efficient architectures like the ones presented in this thesis. Hooker concludes:

> ...the hardware lottery has not gone away, and the gap between the winners and losers will grow increasingly larger. In order to avoid future hardware lotteries, we need to make it easier to quantify the opportunity cost of settling for the hardware and software we have. [187]

Environmental impacts

> *"...invention is the mother of necessity."*
> — Thorstein Veblen
> The Instinct of Workmanship and the State of the Industrial Arts

As established in the previous subsection, deep learning is a power-hungry endeavour that relies on specialized hardware. Both the high energy and semiconductor demands of deep learning algorithms have associated environmental footprint. In this subsection, I explore briefly what the relevance of the efficient architectures adopted in this thesis may have on their relative environmental impact.

On the topic of energy consumption, both SqueezeProt and the top-ranking pLLMs analysed herein cause an Nvidia RTX 3090 GPU to consume just under its Thermal Design Power (TDP) of 350 watts. SqueezeProt can compute embeddings at a far faster rate than the top-ranking pLLMs, however, which results in energy requirements for embedding computation that is orders of magnitude lower for the same amino acid sequence. Specifically, SqueezeProt requires

approximately 8.33 millijoules (mJ) per amino acid embedded whereas ProtBERT, ESM2, and ProtT5 require 37.4 mJ, 52.3 mJ, and 67.5 mJ, respectively. As a practical example, it would require the same energy to power the average Canadian home for approximately 4 days to compute embeddings for all 253 061 696 proteins currently in UniProt using SqueezeProt[1]. This number balloons to over two weeks (16 days) in the case of ProtBERT and just under a month (29 days) for ProtT5. By eschewing transformers entirely, power consumption could be greatly reduced. RAPPPID causes an RTX 3090 to consume about half of its TDP according to my measurements, resulting in the use of a mere 0.543 mJ per amino acid embedded. In our UniProt-wide embedding scenario, RAPPPID would require the same energy as the typical Canadian household uses in 5 hours and 36 minutes. Notably, all network training in this study was conducted in the province of Québec where 99% of power delivered to the energy grid is produced by non-emitting sources [199].

Energy consumption is but one of the environmental consequences of deep-learning-based PPI inference methods. Another important ecological concern is the life-cycle impact of the manufacturing of the specialized semiconductors required to train and operate these models. According to Jennifer Gabrys, Chair in Media, Culture and Environment at the University of Cambridge, "[t]o produce a two-gram memory microchip, 1.3 kilograms of fossil fuels and materials are required" [200]. The aforementioned materials are severely injurious to human and animal health, with sweeping implications for individuals born in regions contaminated by electronic waste — including, but not limited to: "...suppressed immune response, hearing loss, altered hepatic and renal function, decline in oral anti-inflammatory ability, and abnormal [fasting blood glucose level]" as well as adverse birth outcomes [201]. By reducing the dependence on such specialized semiconductor hardware, we can hope to lessen the ecological and health impacts caused by their manufacturing and disposal. Gabrys writes:

> While the substances used in the manufacture of microchips contribute to pollution at both bodily and environmental levels, this condition intensifies with the sheer quantity of microchips manufactured. [200]

As discussed in the previous sub-section, VRAM is often the most relevant bottleneck when considering GPUs. Models, particularly LLMs, often require more VRAM than afforded by one GPU. For this reason, multiple GPUs are often required. By adopting an efficient architecture which requires a small fraction of the number of parameters of other leading pLLMs, it is possible to train SqueezeProt on a single consumer-class GPU rather than many.

It may be tempting to see these economies in the amount of energy and number of microchips required to train and operate the models described in this thesis and declare that there has

[1] Assuming an average amino acid length of 400 amino acids and a yearly household energy consumption of 85.4 gigajoules [197, 198].

been an overall improvement in the environmental impact of the model — that the ecological consequences have been partly mitigated. To do so would be premature. Merely observing gains of efficiency is not sufficient to come to such a conclusion. Gains in efficiency of a process can indeed increase the demand and usage of its inputs. William S. Jevons famously observed that the demand, and consequently the usage, of iron and coal increased as machines who required them became more efficient. Jevons' paradox, as it so became known, can be imagined in the context of this work: if pLLM-based PPI inference methods become more accurate and less costly, researchers and industry may adopt them where they had not before. This would result in energy and environment costs that would not have been incurred if pLLM-based PPI inference methods were inefficient, and therefore less desirable. Jevons' paradox, and the so called "rebound" debate which has arisen from it, is far from a settled matter and lacks broad expert consensus [202]. Also to be considered are the environmental impacts of the waste and pollution caused by the experimental techniques used to discover PPIs that inference methods might replace [203, 204, 205, 206].

Whether pLLM-based PPI inference, even efficient ones as I've described, are an improvement over their alternatives requires systematic study. Frameworks, such as Chistopher Becker's five factors of just sustainability design (JSD), are a useful basis for such a study [207]. Regrettably, performing such a thorough analysis could demand another doctoral degree's worth of study, which I have deemed out of scope.

Chapter 6

Conclusion

A robust model with the ability to reliably infer the interactions of two proteins given only their sequences in real-world, out-of-distribution would be deeply impactful — accelerating discoveries that currently take weeks or months to potentially mere minutes or seconds. Despite three decades of developing computational methods for PPI inference, the persistent failure of these approaches to generalize to real-world use cases is a challenge the scientific community continues to grapple with. This is partly due to the many years in which researchers were unaware of or neglected this challenge.

This thesis catalogues the work I've conducted over the last half decade to identify and mitigate obstacles which are at the root of this failure, while anticipating new obstacles to generalizability. A great deal of progress is needed before these sources of bias are sufficiently addressed and PPI inference models become useful for real-world tasks. The three studies presented here represent substantive developments towards that goal.

The development of RAPPPID demonstrated that a deep learning technique could be made to generalize to PPI pairs whose members don't appear in the training dataset. To my knowledge, the study which introduced RAPPPID was the first to do so. RAPPPID out-performed all of the contemporaneous leading PPI inference methods against which it was compared.

In our subsequent study, I expanded on the generalization of RAPPPID to include PPIs between unseen proteins originating from organisms other than that which the model was trained on. To do so, we devised an "orthologous locality" task whose formulation was motivated by both the biological underpinnings of our data as much as it was by practical considerations of model building.

Finally, our third study identified challenges in the nascent area of PLM-based models. It is in this study which I identify an emerging source of data leakage, before uncharacterized, which

already implicates a number of methods. It is also here that I demonstrate ways in which PPI inference models continue to fail to generalize to common, practical use-cases.

The studies in this thesis are humble but tangible contributions to a relatively new field which still holds many mysteries to unravel and challenges to overcome. I know not what the next three decades will uncover for PPI inference, but I am confident that a focus on methodological rigour and generalization lights the way forward.

Appendix A

Supplementary Material for "GRouNdGAN: GRN-guided simulation of single-cell RNA-seq data using causal generative adversarial networks"

Table S1: Ranges of hyperparameters used for training of RAPPPID. This table shows the range of hyperparameters and their increments considered when training RAPPPID. The final values of hyperparameters were chosen based on the performance on a randomly selected subset of the training set (i.e., a validation set).

Hyperparameter	Number of LSTM Layers	Embedding Dropout Rate	LSTM Dropconnect Rate	Classifier Dropout Rate	Learning Rate
Range	2-3	0.1-0.4	0.1-0.4	0.1-0.4	10^{-3}-10^{-2}
Increment	1	0.1	0.1	0.1	0.009

Table S2: The final values of hyperparameters selected for the analysis reported in Table 1 of the manuscript for different datasets. The hyperparameters are selected based on the performance of a validation set (separate from the testing set).

Experiment	Number of LSTM Layers	Embedding Dropout Rate	LSTM Dropconnect Rate	Classifier Dropout Rate	Learning Rate
C1	3	0.1	0.1	0.1	10^{-2}
C2	2	0.3	0.3	0.2	10^{-2}
C3	2	0.3	0.3	0.2	10^{-2}

Table S3: Statistics for the STRING C1, C2, and C3 datasets.

	C1	C2	C3
Number of Proteins	9 340	9 340	9 279
Number of Edges	263 130	263 130	173 914
Training Split Size (%)	79.9%	64.1%	97.0%
Validation Split Size (%)	10.0%	19.0%	1.6%
Testing Split Size (%)	10.1%	16.9%	1.4%
Train/Validation Protein Overlap (%)	100%	83.2%	0%
Train/Test Protein Overlap (%)	100%	84.3%	0%
Train Edges w/ Test Proteins	9 137	3 758	0
Train Edges w/ Validation Proteins	9 170	3 663	0

Table S4: Repeatability of results across randomly generated C3 datasets.

Dataset #	Validation Loss	Validation AUROC	Validation AUPR	Testing AUROC	Testing AUPR
1	0.555	0.804	0.803	0.804	0.809
2	0.560	0.804	0.807	0.806	0.803
3	0.559	0.780	0.776	0.782	0.789

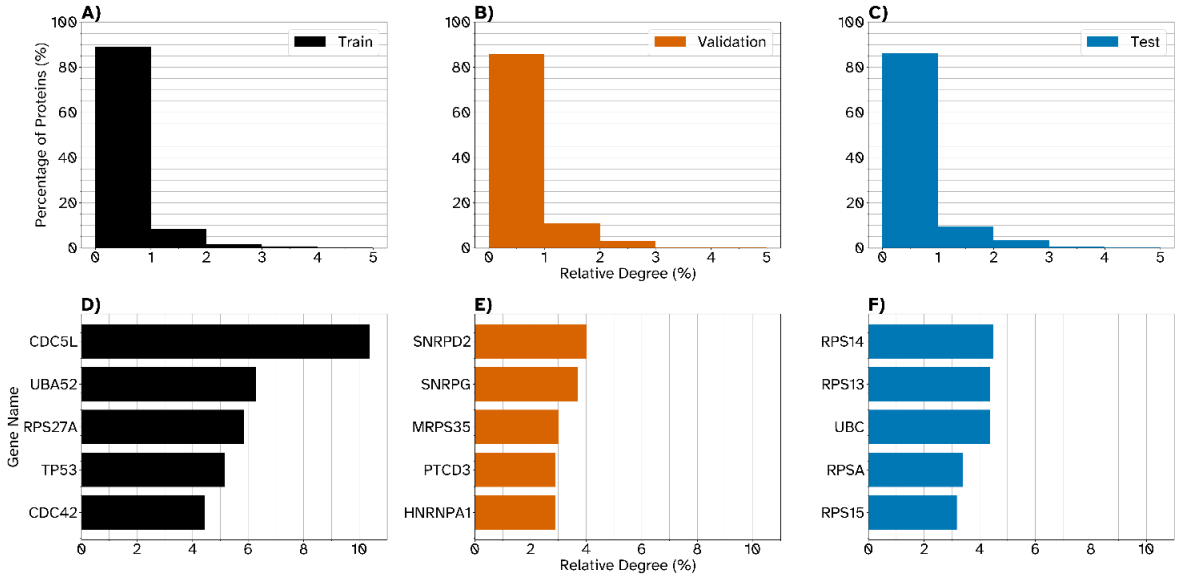


Figure S1: Relative Degree of Proteins in the *H. sapiens* STRING C3 dataset. The distribution of the relative degree of proteins between the three splits of the *H. sapiens* STRING C3 datasets reveal that the vast majority of proteins have a relative degree of less than 1% **(A-C)**. The top five proteins with the highest relative degrees are plotted in bar graphs D through F. *CDC5L* has a relative degree of 10.37%, the highest of the entire dataset. Many of the proteins of high relative degree belong to the families of ribosomal or ribonucleoproteins.

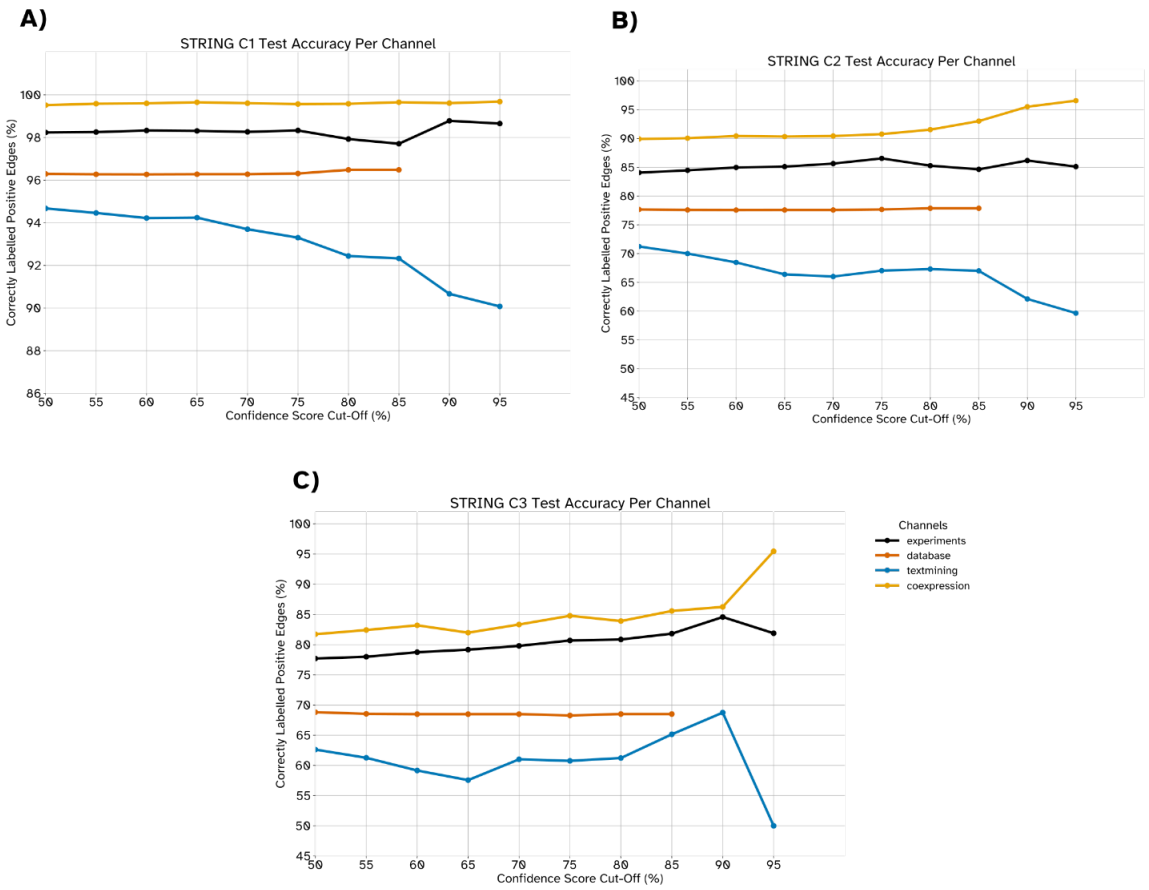


Figure S2: Accuracy of positive edges across edge confidence stratified by STRING channels. The percentage of correctly labelled positive edges are plotted for each major STRING channel in the C1 **(A)**, C2 **(B)**, and C3 **(C)** datasets. The x-axis denotes the channel edge confidence cut-off score for each curve's respective channel.

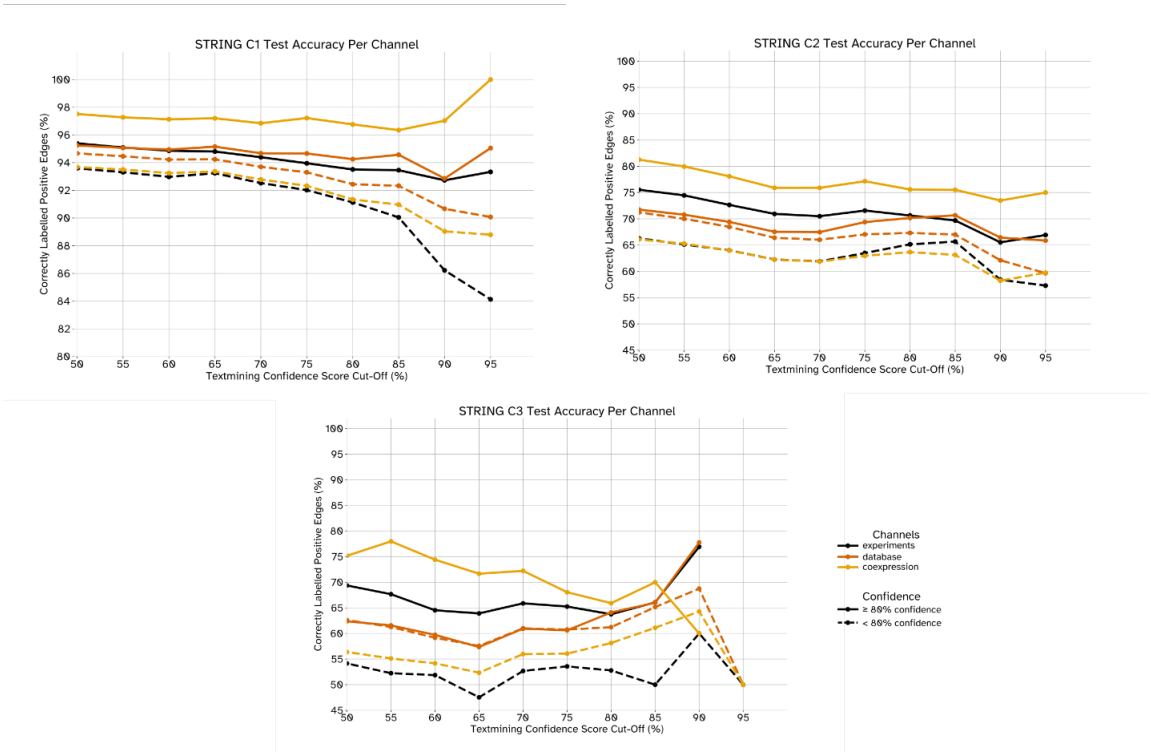


Figure S3: Accuracy of positive edges across edge confidence stratified by STRING channels. The percentage of correctly labelled positive edges are plotted for each major STRING channel, excluding text-mining. Edges are excluded according to the text-mining confidence threshold (x-axis). The solid curves include edges which have a channel confidence ≤ 80% for the channel indicated by the curve's colour. Dashed curves conversely include edges whose channel confidence is 80% for the channel indicated by the curve's colour. Data shown reflects the C1 model/dataset in panel **(A)**, the C2 model/dataset in panel **(B)**, and the C3 model/dataset in panel **(C)**.

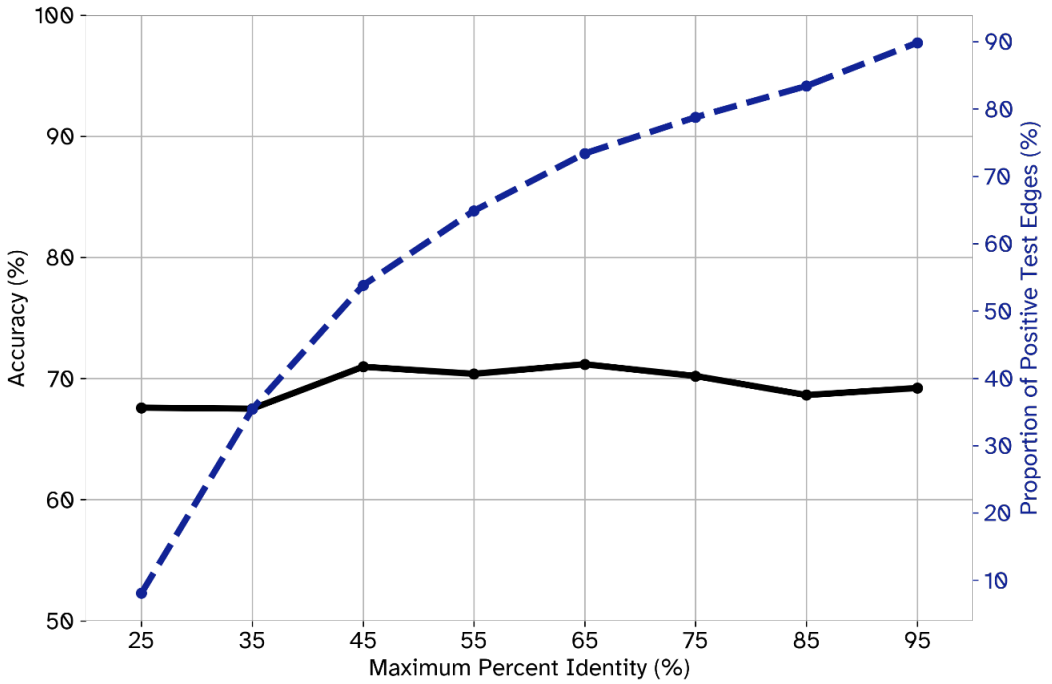


Figure S4: Accuracy of positive edges as a function of similarity between testing and training proteins in C3. The similarity between testing and training proteins was measured using their percent identity as computed by NCBI's PSI-BLAST software. The highest percent identity between any training protein and a testing protein in a testing edge was considered to be that testing edge's "maximum percent identity". The percentage of accurately labelled positive edges (black curve, left y-axis) is reported for edges with maximum percent identities lower than the threshold reported on the x-axis. The proportion of testing edges for each threshold values is reported by the dashed blue curve and the right y-axis.

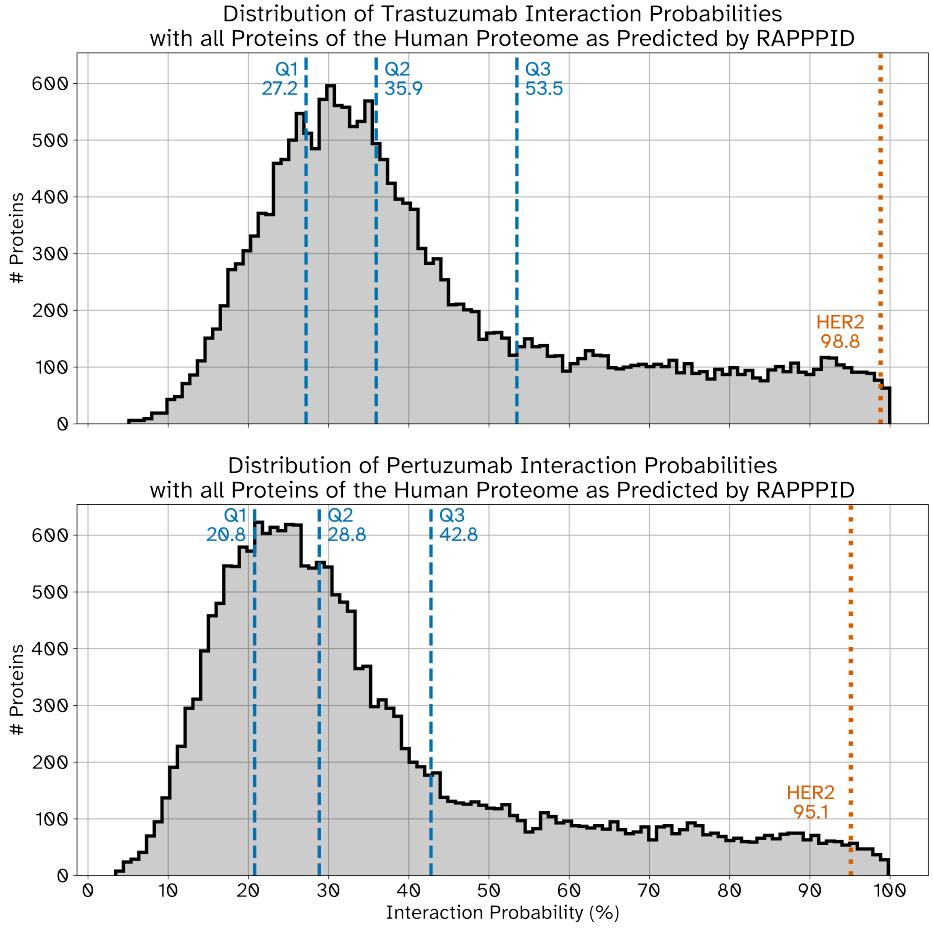


Figure S5: Distribution of Interaction Probabilities of Trastuzumab and Pertuzumab with all Proteins of the Human Proteome as Predicted by rapppid. The probability of interaction between the antibodies (Trastuzumab and Pertuzumab) and each protein in the human proteome (according to STRING) were individually predicted using RAPPPID. The distribution of these probabilities is here shown for interactions between Trastuzumab (top) and Pertuzumab (bottom). The quartiles are marked by blue dashed lines, and illustrate that the majority of proteins have a probability of interacting less than 35.9% and 28.8% for Trastuzumab and Pertuzumab, respectively. The dotted orange line demarks where HER2 is positioned in the distribution, holding a probability of interaction of 98.8% and 95.1% with Trastuzumab and Pertuzumab, respectively.

Appendix B

Supplementary Material for "Causal single-cell RNA-seq simulation, in silico perturbation, and GRN inference benchmarking using GRouNdGAN-Toolkit"

Supplementary Methods

Preparing strict PPI datasets

A PPI dataset $\mathscr{D}$ is a set of interacting protein-pairs reflecting binary interactions $D_i \in \mathscr{D}$. $\mathscr{D}$ is partitioned into three mutually disjoint subsets: a training subset ($\mathscr{D}_{\text{Tr}}$), a testing subset ($\mathscr{D}_{\text{Te}}$), and a validation subset ($\mathscr{D}_{\text{V}}$). Let P_{Tr}, P_{Te}, and P_{V} be the set of proteins present in $\mathscr{D}_{\text{Tr}}$, $\mathscr{D}_{\text{Te}}$, and $\mathscr{D}_{\text{V}}$, respectively, and let $\mathscr{P}$ be the collection of protein sets $P_{\text{Tr}}, P_{\text{Te}}, P_{\text{V}}$.

We define a strict PPI dataset as a dataset $\mathscr{D}$ (with its collection of protein sets $\mathscr{P}$) that fulfills the following two criteria.

Criterion 1 — Distinct Protein Identity
The resulting protein sets $P_{\text{Tr}}, P_{\text{Te}}, P_{\text{V}}$ of the dataset must be mutually disjoint:

$$\forall Q, R \in \mathscr{P}\ Q \cap R = \emptyset \text{ if } Q \neq R.$$

Criterion 2 — Distinct Sequence Identity
The amino acid sequence of any pair of proteins belonging to different protein subsets must have a sequence identity no greater than 90%:

$$\forall Q, R \in \mathcal{P},\ \forall q \in Q\ \forall r \in R\ f(q,r) \leq 90\% \text{ if } Q \neq R$$

where $f(\cdot,\cdot)$ is some protein sequence identity function.

For the datasets used in the experiments carried out herein, the UniRef90 identifiers provided by UniProt were used to determine sequence identity, and as such we adopt the same definition of sequence identity [1,2].

High-confidence examples of PPIs

To minimize our exposure to falsely labelled PPIs, we required a database that provided information regarding the confidence of the interactions. We selected the STRING database as the source of PPIs for our datasets as it is large, contains information for many species, and integrates confidence scores from a number of modalities [3]. We exclude all PPIs which do not have a combined confidence score greater or equal to 90%, and only include PPIs which are classified as "physical interactions". PPI information for seven species were included in our analyses.

Scientific Name	Common Name	NCBI Taxon ID	Patristic Distance from *H. sapiens*
Homo sapiens	Human	9606	0.000
Mus musculus	House mouse	10090	0.0455
Danio rerio	Zebrafish	7955	0.156
Drosophila melanogaster	Fruit fly	7227	0.358
Caenorhabditis elegans	Nematode	6239	0.455
Arabidopsis thaliana	Thale cress	3702	0.693
Saccharomyces cerevisiae	Baker's yeast	4932	0.734

Patristic distance is a measure of the divergence of two lineages. Patristic distances were computed using DendroPy and data from iTOL [4,5].

Negative examples of PPIs

As INTREPPPID is a supervised method, it requires examples of pairs of proteins which do not interact (negative examples), in addition to pairs which do (positive examples). Obtaining negative examples of PPIs is a fraught process as there exists no one agreed upon "gold standard" of proteins which do not interact with one another. While many approaches for obtaining negative PPI examples have been devised over the years, evidence has shown that many of these methods introduce a great amount of bias [6]. For this reason, we opted to sample pairs of proteins in a uniformly random manner to represent our negative examples. For the training set, negatives were randomly drawn without replacement from the set N_{Tr}, defined as:

$$N_{\text{Tr}} = \left\{\{p,q\} \,|\, p \in P_{\text{Tr}}, q \in P_{\text{Tr}}, \{p,q\} \notin \mathcal{D}\right\}$$

Negative examples for the testing and validation sets were drawn from N_{Te} and N_{V}, which were similarly defined as the set of protein-pairs of P_{Te} and P_{V}, respectively, which were not in $\mathscr{D}$ (*i.e.*, were not considered interacting in the dataset).

Random sampling of protein-pairs requires the fewest assumptions about the distribution of negative pairs, thereby reducing bias, albeit at the risk of false-negatives. Through the exclusion of known interactions from N_{Tr}, however, we've reduced the likelihood of drawing false-negatives.

Considerations for cross-species datasets

For experiments herein, it was necessary to prepare datasets suitable for testing the cross-species task. Namely, we needed to construct training sets whose *ppi*s were composed of proteins of one species, and many corresponding testing sets each belonging to a different species. All combinations of the training set and the many testing sets must satisfy both Criteria 1 and 2. The first criterion is typically met *de facto*, as protein identifiers between species normally do not overlap. In order for the second criterion to be met, positive and negative interactions of proteins that do not satisfy Criterion 2 were excluded from the datasets.

Supplementary Tables and Figures

Tables

Table S1

Table S1 is provided as a separate .xlsx spreadsheet file.

Its caption is as follows:

> Statistics of each dataset used in this study. Datasets are listed by source organism and random seed used to generate the random splits.

This file is available for download from OUP Briefings in Bioinformatics: https://doi.org/10.1093/bib/bbae405.

Table S2: Hyperparameters used to train INTREPPPID in all experiments.

Hyperparameter	Value
Batch Size	80
Embedding Size	64
Number of RNN Layers	2
RNN Drop-connect Rate	0.3
Embedding Drop-out Rate	0.3

Table S3

Table S3 is provided as a separate .xlsx spreadsheet file.

Its caption is as follows:

> Confusion matrix statistics of various methods trained and tested on *H. sapiens* data.

This file is available for download from OUP Briefings in Bioinformatics:
https://doi.org/10.1093/bib/bbae405.

Table S4: Performance of INTREPPPID trained on *H. sapiens* datasets with a 1:1 negative-to-positive interaction ratios, and tested on increasing negative-to-positive ratios.

Negative:Positive	1:1	2:1	10:1	50:1	100:1
AUROC Mean	0.752	0.723	0.733	0.728	0. 733
AUROC Std	0.017	0.004	0.006	0.011	0.004

Table S5

Table S5 is provided as a separate .xlsx spreadsheet file.

Its caption is as follows:

> Metrics of various PPI inference methods trained on *H. sapiens* data, and tested on various species. Each model was tested on a total of twenty-four dataset: four different random testing splits for each six different organism. The metrics reported here are averaged over all four datasets, with the standard deviation between the four datasets also reported. The first worksheet (AUROC) reports the area-under-the-receiver-operator-curve (AUROC). The second worksheet (AP) reports the average-precision. The third worksheet (F150) reports the F_1 measure with a 50% threshold. The fourth worksheet (MCC50) reports the Matthew's correlation coefficient (MCC) with a 50% threshold. The fifth and final worksheet (Brier) reports the Brier score. Higher values indicate better performance for all metrics but the Brier score, where the inverse is true.

This file is available for download from OUP Briefings in Bioinformatics:
https://doi.org/10.1093/bib/bbae405.

Table S6: Performance of INTREPPPID trained on *H. sapiens* datasets with scrambled and not scrambled orthologue information. Bold-face indicates the best value for each metric.

Scramble Status	Test AUROC	Test AP	Test MCC@50%	Test Brier Score
Not Scrambled	**0.761**	**0.737**	**0.378**	**0.198**
Scrambled	0.524	0.527	0.017	0.314

Table S7: Performance of INTREPPPID when trained on *H. sapiens* dataset which omits orthology data for the seven species tested below, relative to the performance of INTREPPPIED trained on dataset which includes all orthology data available.

Testing Dataset	Test AUROC	Test AP	Test MCC@50%	Test Brier Score
H. sapiens	-0.49%	-1.36%	-9.00%	2.00%
M. musculus	-13.21%	-14.79%	34.39%	21.06%
D. rerio	-13.69%	-16.16%	-38.85%	8.66%
D. melanogaster	-11.29%	-11.67%	-26.48%	5.81%
C. elegans	-12.51%	-17.58%	-13.45%	-0.73%
S. cerevisiae	-9.21%	-15.12%	-13.31%	-3.17%
A. thaliana	-9.29%	-11.55%	-30.75%	5.39%

Table S8

Table S8 is provided as a separate .xlsx spreadsheet file.

Its caption is as follows:

> Results of the statistical over-expression analysis conducted on top INTREPPPID-unique positive hits for seven species. The statistical over-expression of GO Molecular Function annotations among proteins that make up the top INTREPPPID-unique positive hits was measured relative to the proteins which make up the testing dataset for the specific species tested. A Fisher's Exact test was performed by the online PANTHER Classification System. The fold enrichment, raw p values, and corrected p values using Benjamini-Hochberg false-discover rate (FDR) correction are reported for GO Molecular Function annotations found to be over-expressed. Only results with a corrected p value of less than 0.05 are included in this table. Each worksheet in the spreadsheet is labelled with a species name, and reports the over-expression analysis for the testing dataset that belongs to that species.

This file is available for download from OUP Briefings in Bioinformatics:
https://doi.org/10.1093/bib/bbae405.

Table S9: Analysis of 22 randomly sampled articles advertising online PPI-related tools and the online availability of said tools as of January 2024. Articles were found using the SemanticScholar.org academic search engine with the search terms "protein-protein interaction website".

DOI	CorpusID	Year Published	URI	Status
10.1093/nar/gks434	9914396	2012	Link	Dead
10.1007/978-1-0716-0892-0_12	226218388	2020	Link	Alive
10.1007/978-1-4939-9873-9_12	203659921	2020	Link	Alive
10.1515/jib-2006-27	16195629	2006	Link	Dead
10.1089/adt.2013.506	12969509	2013	Link	Dead
10.1093/nar/gkx333	31083221	2017	Link	Alive
10.1371/journal.pone.0056228	2014834	2013	Link	Inaccessible[1]
10.1093/nar/gkp335	3041302	2009	Link	Inaccessible[2]
10.1093/NAR/GKG056	15003973	2003	Link	Dead
10.1109/CBMS.2008.113	19040991	2009	Link	Dead
10.1093/nar/gkx385	3268569	2017	Link	Alive
10.1093/nar/gkac1000	253479599	2022	Link	Alive
10.7726/AJBCB.2014.1003	12004796	2014	Link	Alive
10.1093/bioinformatics/btn584	206340817	2009	Link	Dead
10.1186/1471-2105-11-S10-P1	40906621	2010	Link	Alive
10.1093/bioinformatics/btx312	4420839	2017	Link	Alive
10.1155/2013/737591	5548876	2013	Link	Dead
10.1038/nmeth.1931	52853893	2012	Link	Alive
10.3389/fgene.2021.722198	229549097	2020	Link	Alive
10.1093/nar/gkv1072	8843610	2015	Link	Alive
10.1093/nar/gkt1210	560515	2013	Link	Dead

[1]Presents a login page with no context or registration option.

[2]URI resolves but the described online web server is not present.

Figures

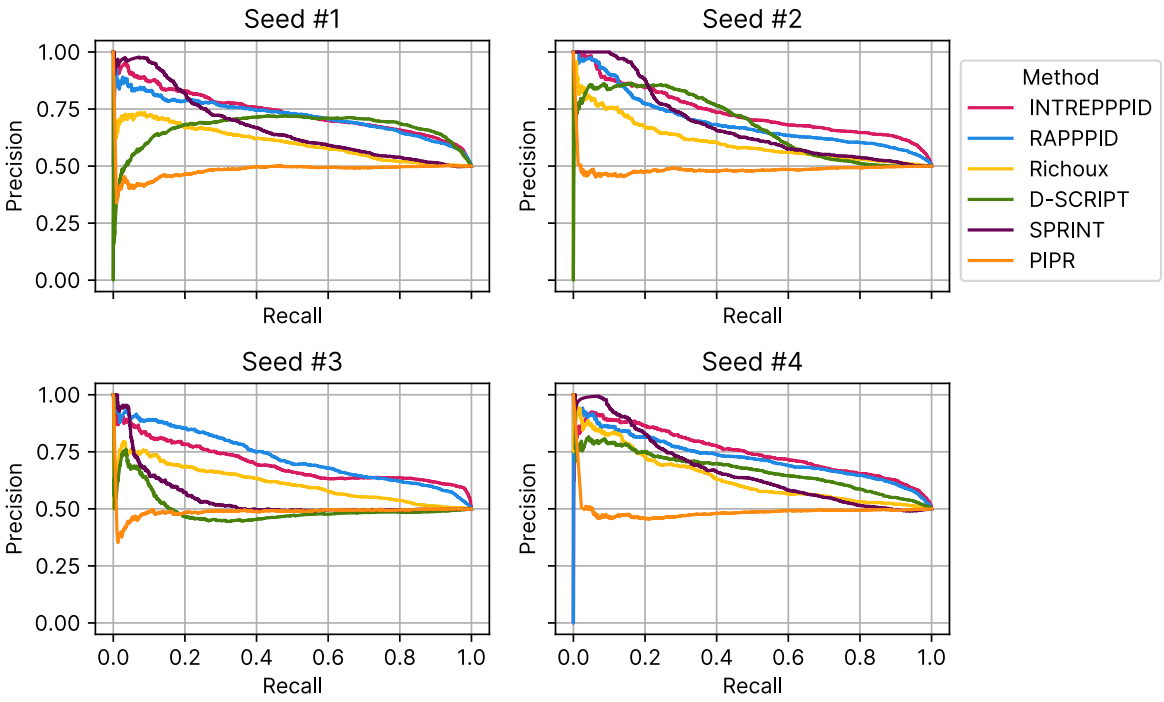


Figure S1: Precision-Recall curves of various PPI inference methods tested and trained on *H. sapiens* data. Each panel reports the Precision-Recall curve for one of four different datasets.

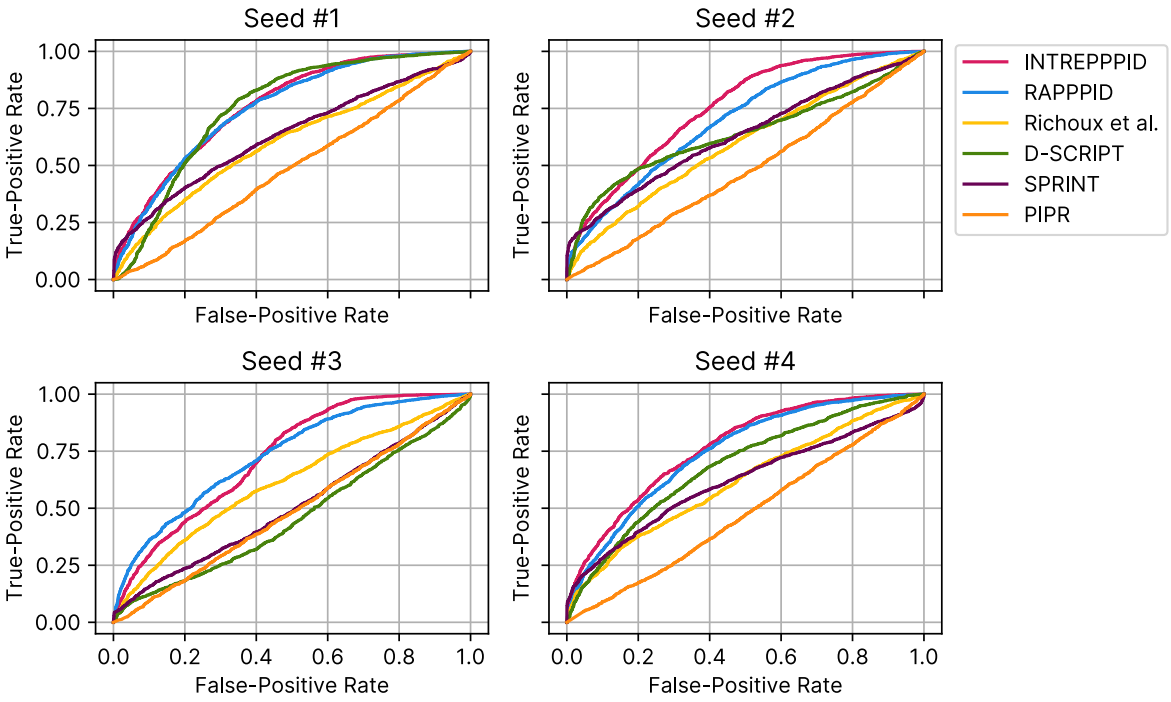


Figure S2: Receiver-Operator curves of various PPI inference methods tested and trained on *H. sapiens* data. Each panel reports the Receiver-Operator curve for one of four different datasets.

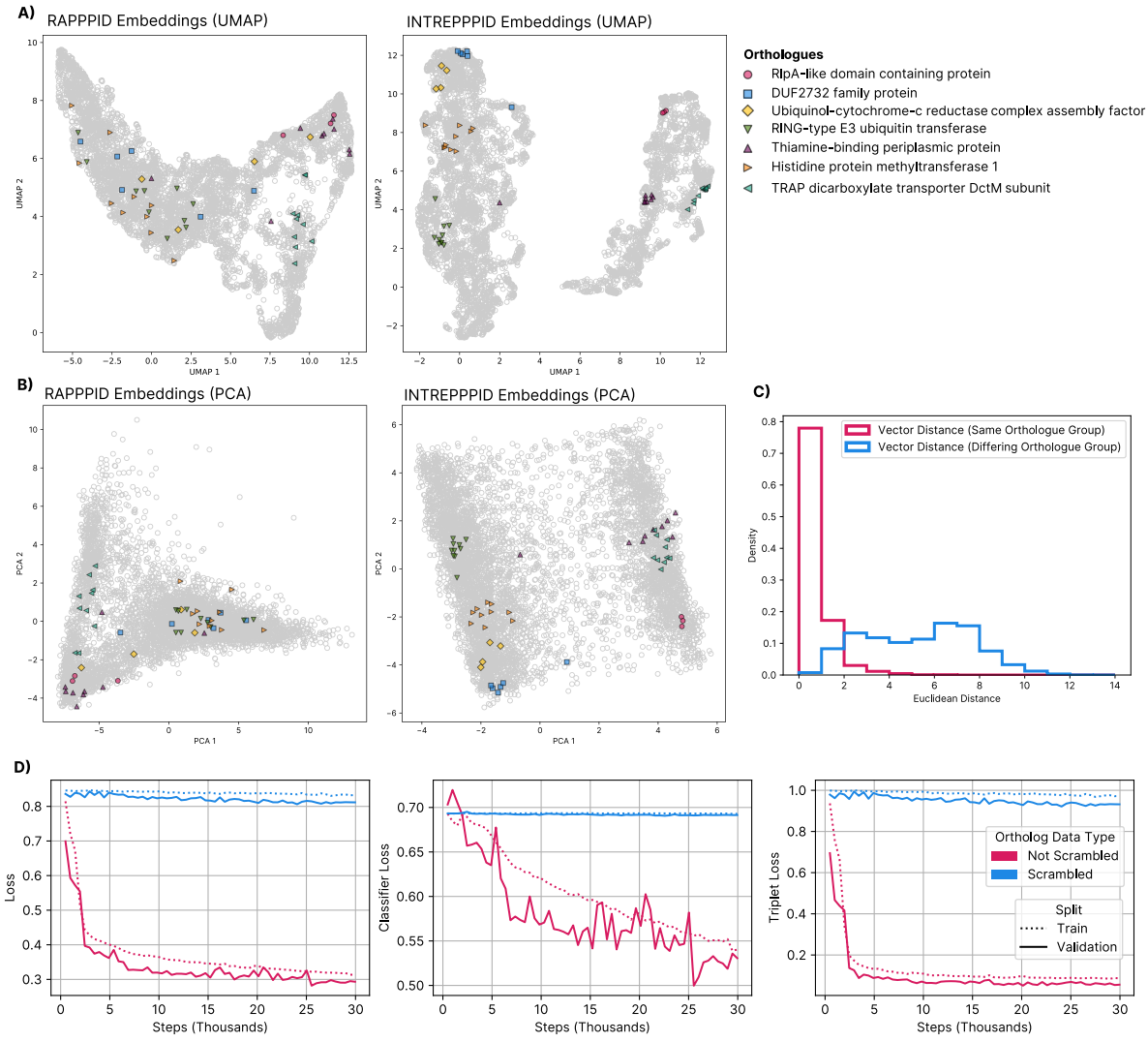


Figure S3: The effect of orthologous locality task. (A-B) A random sample of protein sequences from various species were embedded using RAPPPID (left) and INTREPPPID (right) and projected onto a 2D space using the **(A)** UMAP and **(B)** PCA algorithm. Distinct markers indicate proteins which belong to the same orthologous group according to the OMA database. The orthologous groups shown were chosen at random. **(C)** Histogram of the Euclidean distance between INTREPPPID latent vector representations of protein sequences of the same orthologous group (magenta) and of differing orthologous groups (blue). **(D)** Training (dotted line) and validation loss (solid line) of INTREPPPID trained on *H. sapiens* datasets whose orthologue data is "scrambled" (blue) and "not scrambled" (magenta). The overall loss (left), the loss of the classification network (middle), and the loss of the triplet network (right) are plotted above, demonstrating that the "scrambled" dataset fails to optimize any aspect of the loss in any appreciable way.

References

1. Suzek BE, Wang Y, Huang H, *et al.* UniRef clusters: a comprehensive and scalable alternative for improving sequence similarity searches. *Bioinformatics* 2015; 31:926–932

2. The UniProt Consortium, Bateman A, Martin M-J, *et al.* UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research* 2023; 51:D523–D531

3. Szklarczyk D, Kirsch R, Koutrouli M, *et al.* The *string* database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research* 2023; 51:D638–D646

4. Sukumaran J, Holder MT. DendroPy: a Python library for phylogenetic computing. *Bioinformatics* 2010; 26:1569–1571

5. Letunic I, Bork P. Interactive Tree Of Life (iTOL) v5: an online tool for phylogenetic tree display and annotation. *Nucleic Acids Research* 2021; 49:W293–W296

6. Ben-Hur A, Noble WS. Choosing negative examples for the prediction of protein-protein interactions. *BMC Bioinformatics* 2006; 7:S2–S2

Bibliography

[1] Sherlyn Jemimah and M. Michael Gromiha. Insights into changes in binding affinity caused by disease mutations in protein-protein complexes. *Computers in Biology and Medicine*, 123:103829, August 2020.

[2] Roger G. Ptak, William Fu, Brigitte E. Sanders-Beer, Jonathan E. Dickerson, John W. Pinney, David L. Robertson, Mikhail N. Rozanov, Kenneth S. Katz, Donna R. Maglott, Kim D. Pruitt, and Carl W. Dieffenbach. Short communication: Cataloguing the HIV type 1 human protein interaction network. *AIDS Research and Human Retroviruses*, 24(12):1497–1502, December 2008.

[3] David E. Gordon, Gwendolyn M. Jang, Mehdi Bouhaddou, Jiewei Xu, Kirsten Obernier, Kris M. White, Matthew J. O'Meara, Veronica V. Rezelj, Jeffrey Z. Guo, Danielle L. Swaney, Tia A. Tummino, Ruth Hüttenhain, Robyn M. Kaake, Alicia L. Richards, Beril Tutuncuoglu, Helene Foussard, Jyoti Batra, Kelsey Haas, Maya Modak, Minkyu Kim, Paige Haas, Benjamin J. Polacco, Hannes Braberg, Jacqueline M. Fabius, Manon Eckhardt, Margaret Soucheray, Melanie J. Bennett, Merve Cakir, Michael J. McGregor, Qiongyu Li, Bjoern Meyer, Ferdinand Roesch, Thomas Vallet, Alice Mac Kain, Lisa Miorin, Elena Moreno, Zun Zar Chi Naing, Yuan Zhou, Shiming Peng, Ying Shi, Ziyang Zhang, Wenqi Shen, Ilsa T. Kirby, James E. Melnyk, John S. Chorba, Kevin Lou, Shizhong A. Dai, Inigo Barrio-Hernandez, Danish Memon, Claudia Hernandez-Armenta, Jiankun Lyu, Christopher J. P. Mathy, Tina Perica, Kala Bharath Pilla, Sai J. Ganesan, Daniel J. Saltzberg, Ramachandran Rakesh, Xi Liu, Sara B. Rosenthal, Lorenzo Calviello, Srivats Venkataramanan, Jose Liboy-Lugo, Yizhu Lin, Xi-Ping Huang, YongFeng Liu, Stephanie A. Wankowicz, Markus Bohn, Maliheh Safari, Fatima S. Ugur, Cassandra Koh, Nastaran Sadat Savar, Quang Dinh Tran, Djoshkun Shengjuler, Sabrina J. Fletcher, Michael C. O'Neal, Yiming Cai, Jason C. J. Chang, David J. Broadhurst, Saker Klippsten, Phillip P. Sharp, Nicole A. Wenzell, Duygu Kuzuoglu-Ozturk, Hao-Yuan Wang, Raphael Trenker, Janet M. Young, Devin A. Cavero, Joseph Hiatt, Theodore L. Roth, Ujjwal Rathore, Advait Subramanian, Julia Noack, Mathieu Hubert, Robert M. Stroud, Alan D. Frankel, Oren S. Rosenberg, Kliment A. Verba, David A. Agard, Melanie Ott, Michael

Emerman, Natalia Jura, Mark von Zastrow, Eric Verdin, Alan Ashworth, Olivier Schwartz, Christophe d'Enfert, Shaeri Mukherjee, Matt Jacobson, Harmit S. Malik, Danica G. Fujimori, Trey Ideker, Charles S. Craik, Stephen N. Floor, James S. Fraser, John D. Gross, Andrej Sali, Bryan L. Roth, Davide Ruggero, Jack Taunton, Tanja Kortemme, Pedro Beltrao, Marco Vignuzzi, Adolfo García-Sastre, Kevan M. Shokat, Brian K. Shoichet, and Nevan J. Krogan. A SARS-CoV-2 protein interaction map reveals targets for drug repurposing. *Nature*, 583(7816):459–468, July 2020.

[4] Marc Vidal and Stanley Fields. The yeast two-hybrid assay: still finding connections after 25 years. *Nature Methods*, 11(12):1203–1206, Dec 2014.

[5] Jamie Snider, Max Kotlyar, Punit Saraon, Zhong Yao, Igor Jurisica, and Igor Stagljar. Fundamentals of protein interaction network mapping. *Molecular Systems Biology*, 11(12):848, Dec 2015.

[6] Teck Yew Low, Saiful Effendi Syafruddin, M. Aiman Mohtar, Adaikkalam Vellaichamy, Nisa Syakila A Rahman, Yuh-Fen Pung, and Chris Soon Heng Tan. Recent progress in mass spectrometry-based strategies for elucidating protein–protein interactions. *Cellular and Molecular Life Sciences*, 78(13):5325–5339, July 2021.

[7] Melina Altmann, Stefan Altmann, Claudia Falter, and Pascal Falter-Braun. High-quality yeast-2-hybrid interaction network mapping. *Current Protocols in Plant Biology*, 3(3):e20067, September 2018.

[8] Jitender Mehla, J. Harry Caufield, and Peter Uetz. Mapping protein-protein interactions using yeast two-hybrid assays. *Cold Spring Harbor Protocols*, 2015(5):442–452, May 2015.

[9] Joseph Szymborski and Amin Emad. INTREPPPID — an orthologue-informed quintuplet network for cross-species prediction of protein–protein interaction. *Briefings in Bioinformatics*, 25(5):bbae405, September 2024.

[10] Joseph Szymborski and Amin Emad. Rapppid: towards generalizable protein interaction prediction with awd-lstm twin networks. *Bioinformatics*, 38(16):3958–3967, August 2022.

[11] Damian Szklarczyk, Rebecca Kirsch, Mikaela Koutrouli, Katerina Nastou, Farrokh Mehryary, Radja Hachilif, Annika L. Gable, Tao Fang, Nadezhda T. Doncheva, Sampo Pyysalo, Peer Bork, Lars J. Jensen, and Christian von Mering. The string database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*, 51(D1):D638–D646, January 2023.

[12] Noemi del Toro, Anjali Shrivastava, Eliot Ragueneau, Birgit Meldal, Colin Combe, Elisabet Barrera, Livia Perfetto, Karyn How, Prashansa Ratan, Gautam Shirodkar, Odilia Lu, Bálint Mészáros, Xavier Watkins, Sangya Pundir, Luana Licata, Marta Iannuccelli, Matteo Pellegrini, Maria Jesus Martin, Simona Panni, Margaret Duesbury, Sylvain D Vallet, Juri Rappsilber, Sylvie Ricard-Blum, Gianni Cesareni, Lukasz Salwinski, Sandra Orchard, Pablo Porras, Kalpana Panneerselvam, and Henning Hermjakob. The IntAct database: efficient access to fine-grained molecular interaction data. *Nucleic Acids Research*, 50(D1):D648–D653, January 2022.

[13] Yungki Park and Edward M. Marcotte. Flaws in evaluation schemes for pair-input computational predictions. *Nature Methods*, 9(12):1134–1136, December 2012.

[14] Tobias Hamp and Burkhard Rost. More challenges for machine-learning protein interactions. *Bioinformatics (Oxford, England)*, 31(10):1521–1525, May 2015.

[15] Judith Bernett, David B Blumenthal, and Markus List. Cracking the black box of deep sequence-based protein–protein interaction prediction. *Briefings in Bioinformatics*, 25(2):bbae076, March 2024.

[16] Michael P. H. Stumpf, Thomas Thorne, Eric de Silva, Ronald Stewart, Hyeong Jun An, Michael Lappe, and Carsten Wiuf. Estimating the size of the human interactome. *Proceedings of the National Academy of Sciences of the United States of America*, 105(19):6959–6964, May 2008.

[17] Marc Vidal. How much of the human protein interactome remains to be mapped? *Science Signaling*, 9(427):eg7–eg7, May 2016.

[18] Philipp Blohm, Goar Frishman, Pawel Smialowski, Florian Goebels, Benedikt Wachinger, Andreas Ruepp, and Dmitrij Frishman. Negatome 2.0: a database of non-interacting proteins derived by literature mining, manual annotation and protein structure analysis. *Nucleic Acids Research*, 42(D1):D396–D400, January 2014.

[19] Anne-Claude Gingras and Brian Raught. Beyond hairballs: The use of quantitative mass spectrometry data to understand protein–protein interactions. *FEBS Letters*, 586(17):2723–2731, August 2012.

[20] Jonathan Flint and Trey Ideker. The great hairball gambit. *PLOS Genetics*, 15(11):e1008519, November 2019.

[21] Bijay Jassal, Lisa Matthews, Guilherme Viteri, Chuqiao Gong, Pascual Lorente, Antonio Fabregat, Konstantinos Sidiropoulos, Justin Cook, Marc Gillespie, Robin Haw, Fred Loney, Bruce May, Marija Milacic, Karen Rothfels, Cristoffer Sevilla, Veronica Shamovsky,

Solomon Shorser, Thawfeek Varusai, Joel Weiser, Guanming Wu, Lincoln Stein, Henning Hermjakob, and Peter D'Eustachio. The reactome pathway knowledgebase. *Nucleic Acids Research*, 48(D1):D498–D503, January 2020.

[22] George Tsitsiridis, Ralph Steinkamp, Madalina Giurgiu, Barbara Brauner, Gisela Fobo, Goar Frishman, Corinna Montrone, and Andreas Ruepp. CORUM: the comprehensive resource of mammalian protein complexes–2022. *Nucleic Acids Research*, 51(D1):D539–D545, January 2023.

[23] Matthias W. Hentze, Alfredo Castello, Thomas Schwarzl, and Thomas Preiss. A brave new world of RNA-binding proteins. *Nature Reviews. Molecular Cell Biology*, 19(5):327–341, May 2018.

[24] Jamie Heimburg-Molinaro, Akul Y. Mehta, Catherine A. Tilton, and Richard D. Cummings. Insights into glycobiology and the protein-glycan interactome using glycan microarray technologies. *Molecular Cellular Proteomics□: MCP*, 23(11):100844, September 2024.

[25] Kaspar P. Locher. Mechanistic diversity in ATP-binding cassette (ABC) transporters. *Nature Structural Molecular Biology*, 23(6):487–493, June 2016.

[26] Anna Kocyła, Józef Ba Tran, and Artur Krężel. Galvanization of protein–protein interactions in a dynamic zinc interactome. *Trends in Biochemical Sciences*, 46(1):64–79, January 2021.

[27] James W. Torrance, Malcolm W. MacArthur, and Janet M. Thornton. Evolution of binding sites for zinc and calcium ions playing structural roles. *Proteins: Structure, Function, and Bioinformatics*, 71(2):813–830, 2008.

[28] Setare Mostajabi Sarhangi and Dmitry V. Matyushov. Driving forces of protein diffusion. *The Journal of Physical Chemistry Letters*, 11(23):10137–10143, December 2020.

[29] Paul E. Smith and Wilfred F. van Gunsteren. Translational and rotational diffusion of proteins. *Journal of Molecular Biology*, 236(2):629–636, February 1994.

[30] Gideon Schreiber. *Protein–Protein Interactions: Kinetics*, page 2098–2101. Springer, Berlin, Heidelberg, 2013.

[31] Chongle Pan. Measuring dissociation rate constants of protein complexes through subunit exchange: Experimental design and theoretical modeling. *PLOS ONE*, 6(12):e28827, December 2011.

[32] Megan J Robinson and Melanie H Cobb. Mitogen-activated protein kinase pathways. *Current Opinion in Cell Biology*, 9(2):180–186, April 1997.

[33] Gavin Yong Quan Ng, Zachary Wai-Loon Loh, David Y. Fann, Karthik Mallilankaraman, Thiruma V. Arumugam, and M. Prakash Hande. Role of mitogen-activated protein (map) kinase pathways in metabolic diseases. *Genome Integrity*, 15, 2024.

[34] J. Nathaniel Diehl, Priya S. Hibshman, Irem Ozkan-Dagliyan, Craig M. Goodwin, Sarah V. Howard, Adrienne D. Cox, and Channing J. Der. *Chapter Four - Targeting the ERK mitogen-activated protein kinase cascade for the treatment of KRAS-mutant pancreatic cancer*, volume 153 of *RAS: Past, Present, and Future*, page 101–130. Academic Press, January 2022.

[35] Nadia Barbero, Lucia Napione, Sonja Visentin, Maria Alvaro, Andrea Veglio, Federico Bussolino, and Guido Viscardi. A transient kinetic study between signaling proteins: the case of the MEK–ERK interaction. *Chemical Science*, 2(9):1804–1809, 2011.

[36] Joshua R. Wayment and Joel M. Harris. Biotin–Avidin binding kinetics measured by single-molecule imaging. *Analytical Chemistry*, 81(1):336–342, January 2009.

[37] Matthias Blazek, Tomas Silva Santisteban, Roland Zengerle, and Matthias Meier. Analysis of fast protein phosphorylation kinetics in single cells on a microfluidic chip. *Lab on a Chip*, 15(3):726–734, January 2015.

[38] Harvey Lodish, Arnold Berk, Chris A. Kaiser, Monty Kreiger, Anthony Bretscher, Hidde Ploegh, Angelika Amon, and Kelsey C. Martin. Protein binding and enzyme catalysis. In *Molecular cell biology*, page 89–97. W H Freeman, New York, NY, USA, 8th edition, 2016.

[39] J. C. Kendrew, G. Bodo, H. M. Dintzis, R. G. Parrish, H. Wyckoff, and D. C. Phillips. A three-dimensional model of the myoglobin molecule obtained by X-Ray analysis. *Nature*, 181(4610):662–666, March 1958.

[40] M. F. Perutz, M. G. Rossmann, Ann F. Cullis, Hilary Muirhead, Georg Will, and A. C. T. North. Structure of hæmoglobin: A three-dimensional fourier synthesis at 5.5-Å. resolution, obtained by X-Ray analysis. *Nature*, 185(4711):416–422, February 1960.

[41] Alan R. Fersht. From the first protein structures to our current knowledge of protein folding: delights and scepticisms. *Nature Reviews Molecular Cell Biology*, 9(8):650–654, August 2008.

[42] C. B. Anfinsen, E. Haber, M. Sela, and F. H. White. The kinetics of formation of native ribonuclease during oxidation of the reduced polypeptide chain. *Proceedings of the National Academy of Sciences of the United States of America*, 47(9):1309–1314, September 1961.

[43] C. B. Anfinsen. Principles that govern the folding of protein chains. *Science*, 181(4096):223–230, July 1973.

[44] Jorge A. Vila. Metamorphic proteins in light of Anfinsen's dogma. *The Journal of Physical Chemistry Letters*, 11(13):4998–4999, July 2020.

[45] Prakash Kulkarni, Vitor B. P. Leite, Susmita Roy, Supriyo Bhattacharyya, Atish Mohanty, Srisairam Achuthan, Divyoj Singh, Rajeswari Appadurai, Govindan Rangarajan, Keith Weninger, John Orban, Anand Srivastava, Mohit Kumar Jolly, Jose N. Onuchic, Vladimir N. Uversky, and Ravi Salgia. Intrinsically disordered proteins: Ensembles at the limits of Anfinsen's dogma. *Biophysics Reviews*, 3(1):011306, March 2022.

[46] Francis Crick. Central dogma of molecular biology. *Nature*, 227(5258):561–563, August 1970.

[47] Kris Pauwels, Inge Van Molle, Jan Tommassen, and Patrick Van Gelder. Chaperoning Anfinsen: the steric foldases. *Molecular Microbiology*, 64(4):917–922, 2007.

[48] Ursula Jakob, Richard Kriwacki, and Vladimir N. Uversky. Conditionally and transiently disordered proteins: Awakening cryptic disorder to regulate protein function. *Chemical Reviews*, 114(13):6779–6805, 2014. PMID: 24502763.

[49] Arunachalam Vinayagam, Travis E. Gibson, Ho-Joon Lee, Bahar Yilmazel, Charles Roesel, Yanhui Hu, Young Kwon, Amitabh Sharma, Yang-Yu Liu, Norbert Perrimon, and Albert-László Barabási. Controllability analysis of the directed human protein interaction network identifies disease genes and drug targets. *Proceedings of the National Academy of Sciences*, 113(18):4976–4981, May 2016.

[50] Ian W. Taylor and Jeffrey L. Wrana. Protein interaction networks in medicine and disease. *PROTEOMICS*, 12(10):1706–1716, May 2012.

[51] Quan Zhong, Nicolas Simonis, Qian-Ru Li, Benoit Charloteaux, Fabien Heuze, Niels Klitgord, Stanley Tam, Haiyuan Yu, Kavitha Venkatesan, Danny Mou, Venus Swearingen, Muhammed A Yildirim, Han Yan, Amélie Dricot, David Szeto, Chenwei Lin, Tong Hao, Changyu Fan, Stuart Milstein, Denis Dupuy, Robert Brasseur, David E Hill, Michael E Cusick, and Marc Vidal. Edgetic perturbation models of human inherited disorders. *Molecular Systems Biology*, 5(1):321, January 2009.

[52] Yongsheng Li, Brandon Burgman, Ishaani S Khatri, Sairahul R Pentaparthi, Zhe Su, Daniel J McGrail, Yang Li, Erxi Wu, S Gail Eckhardt, Nidhi Sahni, and S Stephen Yi. e-MutPath: computational modeling reveals the functional landscape of genetic mutations rewiring interactome networks. *Nucleic Acids Research*, 49(1):e2–e2, January 2021.

[53] Minkyu Kim, Jisoo Park, Mehdi Bouhaddou, Kyumin Kim, Ajda Rojc, Maya Modak, Margaret Soucheray, Michael J. McGregor, Patrick O'Leary, Denise Wolf, Erica Stevenson, Tzeh Keong Foo, Dominique Mitchell, Kari A. Herrington, Denise P. Muñoz, Beril Tutuncuoglu, Kuei-Ho Chen, Fan Zheng, Jason F. Kreisberg, Morgan E. Diolaiti, John D. Gordan, Jean-Philippe Coppé, Danielle L. Swaney, Bing Xia, Laura Van 'T Veer, Alan Ashworth, Trey Ideker, and Nevan J. Krogan. A protein interaction landscape of breast cancer. *Science*, 374(6563):eabf3066, October 2021.

[54] Cendrine Tourette, Biao Li, Russell Bell, Shannon O'Hare, Linda S. Kaltenbach, Sean D. Mooney, and Robert E. Hughes. A large scale huntingtin protein interaction network implicates Rho GTPase signaling pathways in huntington disease. *Journal of Biological Chemistry*, 289(10):6709–6726, March 2014.

[55] Lloyd Mabonga and Abidemi Paul Kappo. Protein-protein interaction modulators: advances, successes and remaining challenges. *Biophysical Reviews*, 11(4):559–581, August 2019.

[56] Duncan E. Scott, Andrew R. Bayly, Chris Abell, and John Skidmore. Small molecules, big targets: drug discovery faces the protein–protein interaction challenge. *Nature Reviews Drug Discovery*, 15(8):533–550, August 2016.

[57] Stanley Fields and Ok-kyu Song. A novel genetic system to detect protein–protein interactions. *Nature*, 340(6230):245–246, July 1989. Publisher: Nature Publishing Group.

[58] Jamie Snider, Saranya Kittanakom, Dunja Damjanovic, Jasna Curak, Victoria Wong, and Igor Stagljar. Detecting interactions with membrane proteins using a membrane two-hybrid assay in yeast. *Nature Protocols*, 5(7):1281–1293, July 2010.

[59] Julia Petschnigg, Bella Groisman, Max Kotlyar, Mikko Taipale, Yong Zheng, Christoph F. Kurat, Azin Sayad, J. Rafael Sierra, Mojca Mattiazzi Usaj, Jamie Snider, Alex Nachman, Irina Krykbaeva, Ming-Sound Tsao, Jason Moffat, Tony Pawson, Susan Lindquist, Igor Jurisica, and Igor Stagljar. The mammalian-membrane two-hybrid assay (MaMTH) for probing membrane-protein interactions in human cells. *Nature Methods*, 11(5):585–592, May 2014. Publisher: Nature Publishing Group.

[60] Tom K. Kerppola. Bimolecular fluorescence complementation (BiFC) analysis as a probe of protein interactions in living cells. *Annual Review of Biophysics*, 37:465–487, 2008.

[61] Irma Lemmens, Sam Lievens, and Jan Tavernier. MAPPIT, a mammalian two-hybrid method for in-cell detection of protein-protein interactions. *Methods in Molecular Biology (Clifton, N.J.)*, 1278:447–455, 2015.

[62] Sam Lievens, Sarah Gerlo, Irma Lemmens, Dries J. H. De Clercq, Martijn D. P. Risseeuw, Nele Vanderroost, Anne-Sophie De Smet, Elien Ruyssinck, Eric Chevet, Serge Van Calenbergh, and Jan Tavernier. Kinase Substrate Sensor (KISS), a mammalian in situ protein interaction sensor. *Molecular & cellular proteomics: MCP*, 13(12):3332–3342, December 2014.

[63] Ian M. Evans and Ketevan Paliashvili. Co-immunoprecipitation Assays. In Lorna R. Fiedler and Caroline Pellet-Many, editors, *VEGF Signaling: Methods and Protocols*, pages 125–132. Springer US, New York, NY, 2022.

[64] Prashant Kaushal, Manisha R. Ummadi, Gwendolyn M. Jang, Yennifer Delgado, Sara K. Makanani, Kareem Alba, Declan M. Winters, Sophie F. Blanc, Jiewei Xu, Benjamin Polacco, Yuan Zhou, Erica Stevenson, Manon Eckhardt, Lorena Zuliani-Alvarez, Robyn Kaake, Danielle L. Swaney, Nevan J. Krogan, and Mehdi Bouhaddou. Protocol for mapping differential protein-protein interaction networks using affinity purification-mass spectrometry. *STAR Protocols*, 5(4):103286, November 2024.

[65] Kyle J. Roux, Dae In Kim, Brian Burke, and Danielle G. May. BioID: A Screen for Protein-Protein Interactions. *Current Protocols in Protein Science*, 91(1):19.23.1–19.23.15, 2018. _eprint: https://currentprotocols.onlinelibrary.wiley.com/doi/pdf/10.1002/cpps.51.

[66] Linlin Ma, Fan Yang, and Jie Zheng. Application of fluorescence resonance energy transfer in protein studies. *Journal of Molecular Structure*, 1077:87–100, December 2014.

[67] Chris Soon Heng Tan, Ka Diam Go, Xavier Bisteau, Lingyun Dai, Chern Han Yong, Nayana Prabhu, Mert Burak Ozturk, Yan Ting Lim, Lekshmy Sreekumar, Johan Lengqvist, Vinay Tergaonkar, Philipp Kaldis, Radoslaw M. Sobota, and Pär Nordlund. Thermal proximity coaggregation for system-wide profiling of protein complex dynamics in cells. *Science*, 359(6380):1170–1177, March 2018.

[68] R. M. Kini and H. J. Evans. Prediction of potential protein-protein interaction sites from amino acid sequence. Identification of a fibrin polymerization site. *FEBS letters*, 385(1–2):81–86, April 1996.

[69] F. Pazos, M. Helmer-Citterich, G. Ausiello, and A. Valencia. Correlated mutations contain information about protein-protein interaction. *Journal of Molecular Biology*, 271(4):511–523, August 1997.

[70] Susan Jones and Janet M Thornton. Prediction of protein-protein interaction sites using patch analysis. *Journal of Molecular Biology*, 272(1):133–143, September 1997.

[71] Hani Neuvirth, Ran Raz, and Gideon Schreiber. ProMate: A Structure Based Prediction Program to Identify the Location of Protein–Protein Binding Sites. *Journal of Molecular Biology*, 338(1):181–199, April 2004.

[72] Michael S. Chapman. Sequence similarity scores and the inference of structure—function relationships. *Bioinformatics*, 10(2):111–119, 1994.

[73] Edward M. Marcotte, Matteo Pellegrini, Ho-Leung Ng, Danny W. Rice, Todd O. Yeates, and David Eisenberg. Detecting Protein Function and Protein-Protein Interactions from Genome Sequences. *Science*, 285(5428):751–753, July 1999.

[74] Shailesh V Date and Edward M Marcotte. Discovery of uncharacterized cellular systems by genome-wide analysis of functional linkages. *Nature Biotechnology*, 21(9):1055–1062, September 2003.

[75] S. Altschul. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research*, 25(17):3389–3402, September 1997.

[76] M. Kanehisa. KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*, 28(1):27–30, January 2000.

[77] Sylvain Pitre, Frank Dehne, Albert Chan, Jim Cheetham, Alex Duong, Andrew Emili, Marinella Gebbia, Jack Greenblatt, Mathew Jessulat, Nevan Krogan, Xuemei Luo, and Ashkan Golshani. PIPE: a protein-protein interaction prediction engine based on the re-occurring short polypeptide sequences between known interacting protein pairs. *BMC Bioinformatics*, 7(1):365, July 2006.

[78] S. Pitre, C. North, M. Alamgir, M. Jessulat, A. Chan, X. Luo, J. R. Green, M. Dumontier, F. Dehne, and A. Golshani. Global investigation of protein–protein interactions in yeast Saccharomyces cerevisiae using re-occurring short polypeptide sequences. *Nucleic Acids Research*, 36(13):4286–4294, August 2008.

[79] Andrew Schoenrock. *PIPE3 : a massively parallel protein-protein interatction prediction engine*. Master of Computer Science, Carleton University, Ottawa, Ontario, 2010.

[80] Andrew Schoenrock, Frank Dehne, James R. Green, Ashkan Golshani, and Sylvain Pitre. MP-PIPE: a massively parallel protein-protein interaction prediction engine. In *Proceedings of the international conference on Supercomputing*, ICS '11, pages 327–337, New York, NY, USA, May 2011. Association for Computing Machinery.

[81] K. Dick, Bahram Samanfar, Bradley Barnes, E. Cober, B. Mimee, Le Hoa Tan, S. Molnar, K. Biggar, A. Golshani, F. Dehne, and J. Green. PIPE4: Fast PPI Predictor for Comprehensive Inter- and Cross-Species Interactomes. *Scientific Reports*, 2020.

[82] Yiwei Li and Lucian Ilie. SPRINT: ultrafast protein-protein interaction prediction of the entire human interactome. *BMC Bioinformatics*, 2017.

[83] Koki Tsukamoto, Tatsuya Yoshikawa, Kiyonobu Yokota, Yuichiro Hourai, and Kazuhiko Fukui. The development of an affinity evaluation and prediction system by using protein–protein docking simulations and parameter tuning. *Advances and Applications in Bioinformatics and Chemistry*, 2:1–15, January 2009. Publisher: Dove Press.

[84] Yuri Matsuzaki, Yusuke Matsuzaki, Toshiyuki Sato, and Yutaka Akiyama. In silico screening of protein–protein interactions with all-to-all rigid docking and clustering: an application to pathway analysis. *Journal of Bioinformatics and Computational Biology*, 07(06):991–1012, December 2009. Publisher: World Scientific Publishing Co.

[85] Rong Chen, Li Li, and Zhiping Weng. ZDOCK: an initial-stage protein-docking algorithm. *Proteins*, 52(1):80–87, July 2003.

[86] Masahito Ohue, Yuri Matsuzaki, Nobuyuki Uchikoga, Takashi Ishida, and Yutaka Akiyama. MEGADOCK: An All-to-All Protein-Protein Interaction Prediction System Using Tertiary Structure Data. *Protein & Peptide Letters*, 21(8):766–778, July 2013.

[87] Joel R. Bock and David A. Gough. Predicting protein–protein interactions from primary structure. *Bioinformatics*, 17(5):455–460, May 2001.

[88] Alexey Ya. Chervonenkis. Early History of Support Vector Machines. In Bernhard Schölkopf, Zhiyuan Luo, and Vladimir Vovk, editors, *Empirical Inference*, pages 13–20. Springer Berlin Heidelberg, Berlin, Heidelberg, 2013.

[89] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20(3):273–297, September 1995.

[90] William S Noble. What is a support vector machine? *Nature Biotechnology*, 24(12):1565–1567, December 2006.

[91] B. Scholkopf. The Kernel Trick for Distances. 2000.

[92] Shawn M. Gomez, William Stafford Noble, and Andrey Rzhetsky. Learning to predict protein-protein interactions from protein sequences. *Bioinformatics (Oxford, England)*, 19(15):1875–1881, October 2003.

[93] Shinsuke Dohkan, Asako Koike, and Toshihisa Takagi. Support Vector Machines for Predicting Protein-Protein Interactions. *Genome Informatics*, 14:502–503, 2003.

[94] Yanzhi Guo, Lezheng Yu, Zhining Wen, and Menglong Li. Using support vector machine combined with auto covariance to predict protein–protein interactions from protein sequences. *Nucleic Acids Research*, 36(9):3025–3030, May 2008.

[95] Lun Hu and Keith C. C. Chan. Discovering Variable-Length Patterns in Protein Sequences for Protein-Protein Interaction Prediction. *IEEE Transactions on NanoBioscience*, 14(4):409–416, June 2015.

[96] Shawn Martin, Diana Roe, and Jean-Loup Faulon. Predicting protein–protein interactions using signature products. *Bioinformatics*, 21(2):218–226, January 2005.

[97] A. Ben-Hur and W. S. Noble. Kernel methods for predicting protein-protein interactions. *Bioinformatics*, 21(Suppl 1):i38–i46, June 2005.

[98] Zhu-Hong You, Jian-Zhong Yu, Lin Zhu, Shuai Li, and Zhen-Kun Wen. A MapReduce based parallel SVM for large-scale predicting protein–protein interactions. *Neurocomputing*, 145:37–43, December 2014.

[99] Jeffrey Dean and Sanjay Ghemawat. MapReduce: simplified data processing on large clusters. *Commun. ACM*, 51(1):107–113, January 2008.

[100] Yanjun Qi, Judith Klein-Seetharaman, and Ziv Bar-Joseph. A mixture of feature experts approach for protein-protein interaction prediction. *BMC Bioinformatics*, 8(10):S6, December 2007.

[101] Eric W. Bell, Jacob H. Schwartz, Lydia Freddolino, and Yang Zhang. PEPPI: Whole-proteome Protein-protein Interaction Prediction through Structure and Sequence Similarity, Functional Association, and Machine Learning. *Journal of Molecular Biology*, 434(11):167530, June 2022.

[102] Leo Breiman. Random Forests. *Machine Learning*, 45(1):5–32, October 2001.

[103] Xiuquan Du, Jiaxing Cheng, Tingting Zheng, Zheng Duan, and Fulan Qian. A Novel Feature Extraction Scheme with Ensemble Coding for Protein–Protein Interaction Prediction. *International Journal of Molecular Sciences*, 15(7):12731–12749, July 2014.

[104] Zhu-Hong You, Keith C. C. Chan, and Pengwei Hu. Predicting Protein-Protein Interactions from Primary Protein Sequences Using a Novel Multi-Scale Local Feature Representation Scheme and the Random Forest. *PLOS ONE*, 10(5):e0125811, May 2015.

[105] Yijie Ding, Jijun Tang, and Fei Guo. Predicting protein-protein interactions via multivariate mutual information of protein sequences. *BMC Bioinformatics*, 17(1):398, September 2016.

[106] Bin Yu, Cheng Chen, Hongyan Zhou, Bingqiang Liu, and Qin Ma. GTB-PPI: Predict Protein–Protein Interactions Based on L1-Regularized Logistic Regression and Gradient Tree Boosting. *Genomics, Proteomics & Bioinformatics*, 18(5):582–592, October 2020.

[107] Hongli Gao, Cheng Chen, Shuangyi Li, Congjing Wang, Weifeng Zhou, and Bin Yu. Prediction of protein-protein interactions based on ensemble residual convolutional neural network. *Computers in Biology and Medicine*, 152:106471, January 2023.

[108] Mario R. Guarracino and Adriano Nebbia. Predicting Protein-Protein Interactions with K-Nearest Neighbors Classification Algorithm. In David Hutchison, Takeo Kanade, Josef Kittler, Jon M. Kleinberg, Friedemann Mattern, John C. Mitchell, Moni Naor, Oscar Nierstrasz, C. Pandu Rangan, Bernhard Steffen, Madhu Sudan, Demetri Terzopoulos, Doug Tygar, Moshe Y. Vardi, Gerhard Weikum, Francesco Masulli, Leif E. Peterson, and Roberto Tagliaferri, editors, *Computational Intelligence Methods for Bioinformatics and Biostatistics*, volume 6160, pages 139–150. Springer Berlin Heidelberg, Berlin, Heidelberg, 2010. Series Title: Lecture Notes in Computer Science.

[109] E. Fix and J. Hodges. Discriminatory Analysis - Nonparametric Discrimination: Small Sample Performance. August 1952.

[110] Friedrich Riesz. Untersuchungen über Systeme integrierbarer Funktionen. *Mathematische Annalen*, 69(4):449–497, December 1910.

[111] David D. Lewis. Naive (Bayes) at forty: The independence assumption in information retrieval. In Claire Nédellec and Céline Rouveirol, editors, *Machine Learning: ECML-98*, pages 4–15, Berlin, Heidelberg, 1998. Springer.

[112] Chia Hsin Liu, Ker-Chau Li, and Shinsheng Yuan. Human protein–protein interaction prediction by a novel sequence-based co-evolution method: co-evolutionary divergence. *Bioinformatics*, 29(1):92–98, January 2013.

[113] Piero Fariselli, Florencio Pazos, Alfonso Valencia, and Rita Casadio. Prediction of protein–protein interaction sites in heterocomplexes with neural networks. *European Journal of Biochemistry*, 269(5):1356–1361, March 2002.

[114] Frank Rosenblatt. The perceptron a perceiving and recognizing automaton. *tech. rep., Technical Report 85-460-1*, 1957.

[115] J J Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, April 1982. Publisher: Proceedings of the National Academy of Sciences.

[116] Paul J. Werbos. Applications of advances in nonlinear sensitivity analysis. In R. F. Drenick and F. Kozin, editors, *System Modeling and Optimization*, pages 762–770, Berlin, Heidelberg, 1982. Springer.

[117] David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams. Learning representations by back-propagating errors. *Nature*, 323(6088):533–536, October 1986. Publisher: Nature Publishing Group.

[118] Kunihiko Fukushima. Neocognitron: A self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. *Biological Cybernetics*, 36(4):193–202, April 1980.

[119] Jeffrey L. Elman. Finding Structure in Time. *Cognitive Science*, 14(2):179–211, 1990. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1207/s15516709cog1402_1.

[120] John D. Kelleher. A Brief History of Deep Learning. In *Deep Learning*, pages 101–159. MIT Press, Cambridge, Massachusetts, 2019.

[121] Igor N. Aizenberg, Naum N. Aizenberg, and Joos Vandewalle. *Multi-Valued and Universal Binary Neurons*. Springer US, Boston, MA, 2000.

[122] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25, 2012.

[123] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, NIPS'17, page 6000–6010, Red Hook, NY, USA, December 2017. Curran Associates Inc.

[124] Tanlin Sun, Bo Zhou, Luhua Lai, and Jianfeng Pei. Sequence-based prediction of protein protein interaction using a deep-learning algorithm. *BMC Bioinformatics*, 18(1):277, May 2017.

[125] Dor Bank, Noam Koenigstein, and Raja Giryes. Autoencoders. In Lior Rokach, Oded Maimon, and Erez Shmueli, editors, *Machine Learning for Data Science Handbook: Data Mining and Knowledge Discovery Handbook*, pages 353–374. Springer International Publishing, Cham, 2023.

[126] Hua Wang, Heng Huang, Chris Ding, and Feiping Nie. Predicting Protein–Protein Interactions from Multimodal Biological Data Sources via Nonnegative Matrix Tri-Factorization. *Journal of Computational Biology*, 20(4):344–358, April 2013.

[127] K. Fukushima. Cognitron: a self-organizing multilayered neural network. *Biological Cybernetics*, 20(3-4):121–136, November 1975.

[128] D. Hubel and T. Wiesel. Functional architecture of macaque monkey visual cortex. 1977.

[129] D. H. Hubel and T. N. Wiesel. Receptive Fields and Functional Architecture in two Nonstirate Visual Areas (18 and 19) of the Cat. *Journal of Neurophysiology*, 28:229–289, March 1965.

[130] Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel. Backpropagation Applied to Handwritten Zip Code Recognition. *Neural Computation*, 1(4):541–551, December 1989.

[131] Yann LeCun. Generalization and Network Design Strategies. Technical Report CRG-TR-89-4, University of Toronto, Toronto, ON, June 1989.

[132] Serkan Kiranyaz, Onur Avci, Osama Abdeljaber, Turker Ince, Moncef Gabbouj, and Daniel J. Inman. 1D convolutional neural networks and applications: A survey. *Mechanical Systems and Signal Processing*, 151:107398, April 2021.

[133] Somaye Hashemifar, Behnam Neyshabur, Aly A Khan, and Jinbo Xu. Predicting protein–protein interactions through sequence-based deep learning. *Bioinformatics*, 34(17):i802–i810, September 2018.

[134] Sepp Hochreiter and Jürgen Schmidhuber. Long Short-Term Memory. *Neural Computation*, 9(8):1735–1780, November 1997.

[135] Taku Kudo. Subword Regularization: Improving Neural Network Translation Models with Multiple Subword Candidates, April 2018.

[136] Taku Kudo and John Richardson. SentencePiece: A simple and language independent subword tokenizer and detokenizer for Neural Text Processing. In Eduardo Blanco and Wei Lu, editors, *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 66–71, Brussels, Belgium, November 2018. Association for Computational Linguistics.

[137] Muhao Chen, Chelsea J T Ju, Guangyu Zhou, Xuelu Chen, Tianran Zhang, Kai-Wei Chang, Carlo Zaniolo, and Wei Wang. Multifaceted protein–protein interaction prediction based on Siamese residual RCNN. *Bioinformatics*, 35(14):i305–i314, July 2019.

[138] Florian Richoux, Charlène Servantie, Cynthia Borès, and Stéphane Téletchéa. Comparing two deep learning sequence-based models for protein-protein interaction prediction. (arXiv:1901.06268), January 2019. number: arXiv:1901.06268 arXiv:1901.06268 [cs, q-bio, stat].

[139] Claude E. Shannon. Prediction and Entropy of Printed English. In *Claude E. Shannon: Collected Papers*, pages 194–208. IEEE, 1993.

[140] T. Cover and R. King. A convergent gambling estimate of the entropy of English. *IEEE Transactions on Information Theory*, 24(4):413–421, July 1978.

[141] Yoshua Bengio, Réjean Ducharme, Pascal Vincent, and Christian Jauvin. A Neural Probabilistic Language Model. *Journal of Machine Learning Research*, 3(Feb):1137–1155, 2003.

[142] Stephen Merity, Nitish Shirish Keskar, and Richard Socher. An Analysis of Neural Language Modeling at Multiple Scales, March 2018. arXiv:1803.08240 [cs].

[143] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *arXiv:1810.04805 [cs]*, May 2019. arXiv: 1810.04805.

[144] Ethan C. Alley, Grigory Khimulya, Surojit Biswas, Mohammed AlQuraishi, and George M. Church. Unified rational protein engineering with sequence-based deep representation learning. *Nature Methods*, 16(12):1315–1322, December 2019. Publisher: Nature Publishing Group.

[145] Tristan Bepler and Bonnie Berger. Learning protein sequence embeddings using information from structure, October 2019. arXiv:1902.08661 [cs, q-bio, stat].

[146] Tristan Bepler and Bonnie Berger. Learning the protein language: Evolution, structure, and function. *Cell Systems*, 12(6):654–669.e3, June 2021. Publisher: Elsevier.

[147] Nadav Brandes, Dan Ofer, Yam Peleg, Nadav Rappoport, and Michal Linial. ProteinBERT: a universal deep-learning model of protein sequence and function. *Bioinformatics*, 38(8):2102–2110, April 2022.

[148] Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, Debsindhu Bhowmik, and Burkhard Rost. ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7112–7127, October 2022.

[149] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, March 2023. Publisher: American Association for the Advancement of Science.

[150] Ahmed Elnaggar, Hazem Essam, Wafaa Salah-Eldin, Walid Moustafa, Mohamed Elkerdawy, Charlotte Rochereau, and Burkhard Rost. Ankh □: Optimized Protein Language Model Unlocks General-Purpose Modelling. January 2023. Repository: Bioinformatics.

[151] Tuoyu Liu, Han Gao, Xiaopu Ren, Guoshun Xu, Bo Liu, Ningfeng Wu, Huiying Luo, Yuan Wang, Tao Tu, Bin Yao, Feifei Guan, Yue Teng, Huoqing Huang, and Jian Tian. Protein-protein interaction and site prediction using transfer learning. *Briefings in Bioinformatics*, 24(6):bbad376, September 2023.

[152] Liuzhenghao Lv, Zongying Lin, Hao Li, Yuyang Liu, Jiaxi Cui, Calvin Yu-Chian Chen, Li Yuan, and Yonghong Tian. Prollama: A protein language model for multi-task protein language processing. 2024.

[153] Daria Frolova, Marina A. Pak, Anna Litvin, Ilya Sharov, Dmitry N. Ivankov, and Ivan Oseledets. Mulan: Multimodal protein language model for sequence and structure encoding. page 2024.05.30.596565, June 2024.

[154] Thomas Hayes, Roshan Rao, Halil Akin, Nicholas J. Sofroniew, Deniz Oktay, Zeming Lin, Robert Verkuil, Vincent Q. Tran, Jonathan Deaton, Marius Wiggert, Rohil Badkundri, Irhum Shafkat, Jun Gong, Alexander Derry, Raul S. Molina, Neil Thomas, Yousuf A. Khan, Chetan Mishra, Carolyn Kim, Liam J. Bartie, Matthew Nemeth, Patrick D. Hsu, Tom Sercu, Salvatore Candido, and Alexander Rives. Simulating 500 million years of evolution with a language model. *Science*, 387(6736):850–858, February 2025. Publisher: American Association for the Advancement of Science.

[155] Samuel Sledzieski, Rohit Singh, Lenore Cowen, and Bonnie Berger. D-SCRIPT translates genome to phenome with sequence-based, structure-aware, genome-scale predictions of protein-protein interactions. *Cell Systems*, 12(10):969–982.e6, October 2021. Publisher: Elsevier.

[156] Yanlei Kang, Arne Elofsson, Yunliang Jiang, Weihong Huang, Minzhe Yu, and Zhong Li. AFTGAN: prediction of multi-type PPI based on attention free transformer and graph attention network. *Bioinformatics*, 39(2):btad052, February 2023.

[157] Kewei Zhou, Chenping Lei, Jingyan Zheng, Yan Huang, and Ziding Zhang. Pre-trained protein language model sheds new light on the prediction of Arabidopsis protein–protein interactions. *Plant Methods*, 19(1):141, December 2023.

[158] Young Su Ko, Jonathan Parkinson, Cong Liu, and Wei Wang. TUnA: an uncertainty-aware transformer model for sequence-based protein–protein interaction prediction. *Briefings in Bioinformatics*, 25(5):bbae359, September 2024.

[159] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. The MIT Press, November 2005.

[160] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, September 2023.

[161] Ron Kohavi. A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th international joint conference on Artificial intelligence - Volume 2*, IJCAI'95, pages 1137–1143, San Francisco, CA, USA, August 1995. Morgan Kaufmann Publishers Inc.

[162] M. Stone. Cross-Validatory Choice and Assessment of Statistical Predictions. *Journal of the Royal Statistical Society: Series B (Methodological)*, 36(2):111–133, January 1974.

[163] Joseph Szymborski and Amin Emad. A flaw in using pre-trained pLLMs in protein-protein interaction inference models, April 2025. Pages: 2025.04.21.649858 Section: New Results.

[164] Stephen Merity, Nitish Shirish Keskar, and Richard Socher. Regularizing and optimizing LSTM language models. *arXiv:1708.02182 [cs]*, August 2017. arXiv: 1708.02182.

[165] Gang Fang, Nitin Bhardwaj, Rebecca Robilotto, and Mark B. Gerstein. Getting Started in Gene Orthology and Functional Analysis. *PLOS Computational Biology*, 6(3):1–8, March 2010. Publisher: Public Library of Science.

[166] Roded Sharan, Silpa Suthram, Ryan M. Kelley, Tanja Kuhn, Scott McCuine, Peter Uetz, Taylor Sittler, Richard M. Karp, and Trey Ideker. Conserved patterns of protein interaction in multiple species. *Proceedings of the National Academy of Sciences*, 102(6):1974–1979, February 2005.

[167] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A Simple Framework for Contrastive Learning of Visual Representations, June 2020. Number: arXiv:2002.05709 arXiv:2002.05709 [cs, stat].

[168] Pierre Sermanet, Corey Lynch, Yevgen Chebotar, Jasmine Hsu, Eric Jang, Stefan Schaal, and Sergey Levine. Time-Contrastive Networks: Self-Supervised Learning from Video, March 2018. Number: arXiv:1704.06888 arXiv:1704.06888 [cs].

[169] Adrian de Wynter, Xun Wang, Alex Sokolov, Qilong Gu, and Si-Qing Chen. An Evaluation on Large Language Model Outputs: Discourse and Memorization. *Natural Language Processing Journal*, 4:100024, September 2023. arXiv:2304.08637 [cs].

[170] Li Wan, Matthew Zeiler, Sixin Zhang, Yann Le Cun, and Rob Fergus. Regularization of neural networks using DropConnect. In Sanjoy Dasgupta and David McAllester, editors, *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, page 1058–1066, Atlanta, Georgia, USA, June 2013. PMLR.

[171] Nitish Srivastava, Geoffrey E. Hinton, A. Krizhevsky, Ilya Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. 15:1929–1958, 2014.

[172] Justin Bayer, Christian Osendorfer, Daniela Korhammer, Nutan Chen, Sebastian Urban, and Patrick van der Smagt. On fast dropout and its applicability to recurrent networks. (arXiv:1311.0701), December 2013. arXiv:1311.0701 [stat].

[173] Luke Taylor and Geoff Nitschke. Improving deep learning with generic data augmentation. In *2018 IEEE Symposium Series on Computational Intelligence (SSCI)*, page 1542–1547, November 2018.

[174] Rui Sun, Lirong Wu, Haitao Lin, Yufei Huang, and Stan Z. Li. Enhancing protein predictive models via proteins data augmentation: A benchmark and new directions. (arXiv:2403.00875), March 2024. arXiv:2403.00875 [q-bio].

[175] Limin Fu, Beifang Niu, Zhengwei Zhu, Sitao Wu, and Weizhong Li. CD-HIT: accelerated for clustering the next-generation sequencing data. *Bioinformatics (Oxford, England)*, 28(23):3150–3152, December 2012.

[176] Chris Sander and Reinhard Schneider. Database of homology-derived protein structures and the structural meaning of sequence alignment. 9:56–68, 1991.

[177] Wouter G. Touw, Coos Baakman, Jon Black, Tim A. H. te Beek, E. Krieger, Robbie P. Joosten, and Gert Vriend. A series of PDB-related databanks for everyday needs. *Nucleic Acids Research*, 43(D1):D364–D368, January 2015.

[178] Burkhard Rost. Twilight zone of protein sequence alignments. *Protein Engineering, Design and Selection*, 12(2):85–94, February 1999.

[179] Mustafa Umit Oner, Yi-Chih Cheng, Hwee Kuan Lee, and Wing-Kin Sung. Training machine learning models on patient level data segregation is crucial in practical clinical applications. page 2020.04.23.20076406, April 2020.

[180] Sean Whalen, Jacob Schreiber, William S. Noble, and Katherine S. Pollard. Navigating the pitfalls of applying machine learning in genomics. *Nature Reviews Genetics*, 23(3):169–181, March 2022.

[181] Joseph Szymborski and Amin Emad. A flaw in using pre-trained pllms in protein-protein interaction inference models. page 2025.04.21.649858, April 2025.

[182] Stuart Geman, Elie Bienenstock, and René Doursat. Neural networks and the bias/variance dilemma. *Neural Computation*, 4(1):1–58, January 1992.

[183] Pierre Baldi. Deep learning over-parameterization: the shallow fallacy. page 7–12. PMLR, January 2024.

[184] Brady Neal. On the bias-variance tradeoff: Textbooks need an update, 2019.

[185] TOP500. June 2025 | top500, September 2025.

[186] Jack J. Dongarra, Piotr Luszczek, and Antoine Petitet. The linpack benchmark: past, present and future. *Concurrency and Computation: Practice and Experience*, 15(9):803–820, 2003.

[187] Sara Hooker. The hardware lottery. (arXiv:2009.06489), September 2020. arXiv:2009.06489 [cs].

[188] Yi Tay, Mostafa Dehghani, Samira Abnar, Yikang Shen, Dara Bahri, Philip Pham, Jinfeng Rao, Liu Yang, Sebastian Ruder, and Donald Metzler. Long range arena: A benchmark for efficient transformers. (arXiv:2011.04006), November 2020. arXiv:2011.04006 [cs].

[189] Itamar Zimerman and Lior Wolf. On the long range abilities of transformers. (arXiv:2311.16620), November 2023. arXiv:2311.16620 [cs].

[190] Jiayu Ding, Shuming Ma, Li Dong, Xingxing Zhang, Shaohan Huang, Wenhui Wang, Nanning Zheng, and Furu Wei. Longnet: Scaling transformers to 1,000,000,000 tokens. (arXiv:2307.02486), July 2023. arXiv:2307.02486 [cs].

[191] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness. (arXiv:2205.14135), June 2022. arXiv:2205.14135 [cs].

[192] Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-attention with linear complexity. (arXiv:2006.04768), June 2020. arXiv:2006.04768 [cs].

[193] Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer. (arXiv:2004.05150), December 2020. arXiv:2004.05150 [cs].

[194] Forrest N. Iandola, Albert E. Shaw, Ravi Krishna, and Kurt W. Keutzer. Squeezebert: What can computer vision teach nlp about efficient neural networks? (arXiv:2006.11316), June 2020. arXiv:2006.11316 [cs].

[195] Roland Moore-Colyer. Nvidia GeForce RTX 3090 release date, price, specs, restocks and more, July 2022.

[196] John Keefer. Nvidia launches 3, 000*titanvpcgpubasedonteslav*100, *September*2025.

[197] The UniProt Consortium, Alex Bateman, Maria-Jesus Martin, Sandra Orchard, Michele Magrane, Shadab Ahmad, Emanuele Alpi, Emily H Bowler-Barnett, Ramona Britto, Hema Bye-A-Jee, Austra Cukura, Paul Denny, Tunca Dogan, ThankGod Ebenezer, Jun Fan, Penelope Garmiri, Leonardo Jose Da Costa Gonzales, Emma Hatton-Ellis, Abdulrahman Hussein, Alexandr Ignatchenko, Giuseppe Insana, Rizwan Ishtiaq, Vishal Joshi, Dushyanth Jyothi, Swaathi Kandasaamy, Antonia Lock, Aurelien Luciani, Marija Lugaric, Jie Luo, Yvonne Lussi, Alistair MacDougall, Fabio Madeira, Mahdi Mahmoudy, Alok Mishra, Katie Moulang, Andrew Nightingale, Sangya Pundir, Guoying Qi, Shriya Raj, Pedro Raposo, Daniel L Rice, Rabie Saidi, Rafael Santos, Elena Speretta, James Stephenson, Prabhat Totoo, Edward Turner, Nidhi Tyagi, Preethi Vasudev, Kate Warner, Xavier Watkins, Rossana Zaru, Hermann Zellner, Alan J Bridge, Lucila Aimo, Ghislaine Argoud-Puy, Andrea H Auchincloss, Kristian B Axelsen, Parit Bansal, Delphine Baratin, Teresa M Batista Neto, Marie-Claude Blatter, Jerven T Bolleman, Emmanuel Boutet, Lionel Breuza, Blanca Cabrera Gil, Cristina Casals-Casas, Kamal Chikh Echioukh, Elisabeth Coudert, Beatrice Cuche, Edouard De Castro, Anne Estreicher, Maria L Famiglietti, Marc Feuermann, Elisabeth Gasteiger, Pascale Gaudet, Sebastien Gehant, Vivienne Gerritsen, Arnaud Gos, Nadine Gruaz, Chantal Hulo, Nevila Hyka-Nouspikel, Florence Jungo, Arnaud Kerhornou, Philippe Le Mercier, Damien Lieberherr, Patrick Masson, Anne Morgat, Venkatesh Muthukrishnan, Salvo Paesano, Ivo Pedruzzi, Sandrine Pilbout, Lucille Pourcel, Sylvain Poux, Monica Pozzato, Manuela Pruess, Nicole Redaschi, Catherine Rivoire, Christian J A Sigrist, Karin Sonesson, Shyamala Sundaram, Cathy H Wu, Cecilia N Arighi, Leslie Arminski, Chuming Chen, Yongxing Chen, Hongzhan Huang, Kati Laiho, Peter McGarvey, Darren A Natale, Karen Ross, C R Vinayaka, Qinghua Wang, Yuqi Wang, and Jian Zhang. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research*, 51(D1):D523–D531, January 2023.

[198] Statistics Canada. Table 25-10-0060-01 Household energy consumption, Canada and provinces, March 2024.

[199] Canada Energy Regulator. Canada's Energy Future Data Appendices, 2021.

[200] Jennifer Gabrys. *Silicon Elephants: the transformative materiality of microchips*, page 20–44. University of Michigan Press, 2014.

[201] Sarker M Parvez, Farjana Jahan, Marie-Noel Brune, Julia F Gorman, Musarrat J Rahman, David Carpenter, Zahir Islam, Mahbubur Rahman, Nirupam Aich, Luke D Knibbs, and Peter D Sly. Health consequences of exposure to e-waste: an updated systematic review. *The Lancet Planetary Health*, 5(12):e905–e920, December 2021.

[202] Blake Alcott. Jevons' paradox. *Ecological Economics*, 54(1):9–21, July 2005.

[203] Salem Endris, Zemenu Tamir, and Abay Sisay. Medical laboratory waste generation rate, management practices and associated factors in Addis Ababa, Ethiopia. *PloS One*, 17(4):e0266888, 2022.

[204] Joana Alves, Fiona A. Sargison, Hanne Stawarz, Willow B. Fox, Samuel G. Huete, Amany Hassan, Brian McTeir, and Amy C. Pickering. A case report: insights into reducing plastic waste in a microbiology laboratory. *Access Microbiology*, 3(3), March 2021.

[205] Anjali Thakur, Tapasyapreeti Mukhopadhyay, and Ashok Kumar Ahirwar. Approaching sustainability in Laboratory Medicine. *Clinical Chemistry and Laboratory Medicine*, 62(9):1787–1794, August 2024.

[206] Mei Lan Tan, Chee Keat Ying, and Shahrul Bariyah Sahul Hamid. Plastic Pollution and Sustainable Managing of Single-Use Laboratory Plastic Waste. *Sustainability and Climate Change*, 15(1):6–16, February 2022. Publisher: Mary Ann Liebert, Inc., publishers.

[207] Christoph Becker. What is Just Sustainability Design? In *Insolvent: How to Reorient Computing for Just Sustainability*. MIT Press, Cambridge, MA, USA, June 2023.